

PART SIX - ATTACHMENTS

General

- 01_Permits Record of Decision (PROD)
- 02_Base Access Forms: DON Local Population ID Card/Base Access Pass Registration & 2023 Statement of Purpose

Architectural Drawings

- 03_A2.1 Level 1 Floor Plan (Partial)
- 04_A2.2 Level 2 Floor Plan (Partial)
- 05_A2.3 Level 3 Floor Plan (Partial)
- 06_A2.4 Level 4 Floor Plan
- 07_A2.6_Level 1 Floor Plan (Partial)
- 08_A2.7_Level 2 Floor Plan (Partial)
- 09_A2.8_Level 3 Floor Plan (Partial)

Environmental

- 10_Contractor Guidance Manual - Construction Storm Water Management 24 JAN 2023
- 10a_Refrigerant Compliance Plan Naval Station Great Lakes (January 2025)

Utility Maps

- 11_Communication Ducts
- 12_Electrical Manholes and Ducts
- 13_Electric Feeder H52
- 14_UEM Fiber
- 15_Sanitary Sewer
- 16_Abandoned Steam
- 17_Street Lights
- 18_Potable Water

Templates

- 19_Asbestos Hazard Abatement Plan
- 20_Lead, Cadmium, Chromium Compliance Plan

Note: The attached documents and drawings are provided for reference purposes only. They may not be fully accurate or comprehensive. Contractor is advised to exercise discretion and field-verify all dimensions, details, and conditions as necessary.

PERMITS RECORD OF DECISION (PROD)

PERMIT REQUIRED (Check Box)	PROJECT TITLE:		WORK ORDER NUMBER:	
	LOCATION:			
	GOVERNMENT PROJECT MANAGER:			
**Use referenced notes where additional space required.				
Air Quality	PERMIT: Construction		Date Obtained:	Date Closed:
☐	Basis of Decision (Yes/No):**			
	Issuing Agency:			
	Special Provisions and Requirements:**			
Air Quality	PERMIT: Operating		Date Obtained:	Date Closed:
☐	Basis of Decision (Yes/No):**			
	Issuing Agency:			
	Special Provisions and Requirements:**			
Air Quality	PERMIT: Asbestos Demolition & Removal		Date Obtained:	Date Closed:
☐	Basis of Decision (Yes/No):**			
	Issuing Agency:			
	Special Provisions and Requirements:**			
Air Quality	PERMIT: Other		Date Obtained:	Date Closed:
☐	Basis of Decision (Yes/No):**			
	Issuing Agency:			
	Special Provisions and Requirements:**			

PERMITS RECORD OF DECISION (PROD)

PERMIT REQUIRED (Check Box)	PROJECT TITLE:	WORK ORDER NUMBER:
	LOCATION:	
	GOVERNMENT PROJECT MANAGER:	
Water Pollution	PERMIT: Wastewater Collection System	Date Obtained: Date Closed:
☐	Basis of Decision (Yes/No):**	
	Issuing Agency:	
	Special Provisions and Requirements:**	
Water Pollution	PERMIT: Wastewater Pump Station	Date Obtained: Date Closed:
☐	Basis of Decision (Yes/No):**	
	Issuing Agency	
	Special Provisions and Requirements:	
Water Pollution	PERMIT: Wastewater Treatment Plant	Date Obtained: Date Closed:
☐	Basis of Decision (Yes/No):**	
	Issuing Agency:	
	Special Provisions and Requirements:**	
Water Pollution	PERMIT: Pretreatment, i.e. Oil/Water Separator	Date Obtained: Date Closed:
☐	Basis of Decision (Yes/No):**	
	Issuing Agency:	
	Special Provisions and Requirements:**	

PERMITS RECORD OF DECISION (PROD)

PERMIT REQUIRED (Check Box)	PROJECT TITLE:	WORK ORDER NUMBER:
	LOCATION:	
	GOVERNMENT PROJECT MANAGER:	
Water Pollution	PERMIT: Septic System	Date Obtained: Date Closed:
☐	Basis of Decision (Yes/No):**	
	Issuing Agency:	
	Special Provisions and Requirements:**	
Water Pollution	PERMIT: Erosion & Sediment Control	Date Obtained: Date Closed:
☐	Basis of Decision (Yes/No):**	
	Issuing Agency:	
	Special Provisions and Requirements:**	
Water Pollution	PERMIT: Stormwater Management	Date Obtained: Date Closed:
☐	Basis of Decision (Yes/No):**	
	Issuing Agency:	
	Special Provisions and Requirements:**	
Water Pollution	PERMIT: Other	Date Obtained: Date Closed:
☐	Basis of Decision (Yes/No):**	
	Issuing Agency:	
	Special Provisions and Requirements:**	

PERMITS RECORD OF DECISION (PROD)

PERMIT REQUIRED (Check Box)	PROJECT TITLE:		WORK ORDER NUMBER:	
	LOCATION:			
	GOVERNMENT PROJECT MANAGER:			
Discharge Permit	PERMIT: New or Increased Capacity NPDES	Date Obtained:	Date Closed:	
☐	Basis of Decision (Yes/No):**			
	Issuing Agency:			
	Special Provisions and Requirements:**			
Discharge Permit	PERMIT: General NPDES (≥1 Acre Land Disturbance)	Date Obtained:	Date Closed:	
☐	Basis of Decision (Yes/No):**			
	Issuing Agency:			
	Special Provisions and Requirements:**			
Discharge Permit	PERMIT: Other	Date Obtained:	Date Closed:	
☐	Basis of Decision (Yes/No):**			
	Issuing Agency:			
	Special Provisions and Requirements:**			
Drinking Water	PERMIT: Water Distribution System	Date Obtained:	Date Closed:	
☐	Basis of Decision (Yes/No):**			
	Issuing Agency:			
	Special Provisions and Requirements:**			

PERMITS RECORD OF DECISION (PROD)

PERMIT REQUIRED (Check Box)	PROJECT TITLE:	WORK ORDER NUMBER:
	LOCATION:	
	GOVERNMENT PROJECT MANAGER:	
Drinking Water	PERMIT: Water Treatment Plant	Date Obtained: Date Closed:
☐	Basis of Decision (Yes/No):**	
	Issuing Agency:	
	Special Provisions and Requirements:**	
Drinking Water	PERMIT: Well Construction	Date Obtained: Date Closed:
☐	Basis of Decision (Yes/No):**	
	Issuing Agency:	
	Special Provisions and Requirements:**	
Drinking Water	PERMIT: Underground Injection	Date Obtained: Date Closed:
☐	Basis of Decision (Yes/No):**	
	Issuing Agency:	
	Special Provisions and Requirements:**	
Drinking Water	PERMIT: Other	Date Obtained: Date Closed:
☐	Basis of Decision (Yes/No):**	
	Issuing Agency:	
	Special Provisions and Requirements:**	

PERMITS RECORD OF DECISION (PROD)

PERMIT REQUIRED (Check Box)	PROJECT TITLE:	WORK ORDER NUMBER:
	LOCATION:	
	GOVERNMENT PROJECT MANAGER:	
Fuel Tanks	PERMIT: Underground Storage Tank Construction	Date Obtained: Date Closed:
☐	Basis of Decision (Yes/No):**	
	Issuing Agency:	
	Special Provisions and Requirements:**	
Fuel Tanks	PERMIT: Underground Storage Tank Operating	
☐	Basis of Decision (Yes/No):**	
	Issuing Agency:	
	Special Provisions and Requirements:**	
Fuel Tanks	PERMIT: Other	Date Obtained: Date Closed:
☐	Basis of Decision (Yes/No):**	
	Issuing Agency:	
	Special Provisions, and Requirements:	
Solid and Hazardous Waste	PERMIT: Hazardous Waste Treatment, Storage, Disposal, Handling	Date Obtained: Date Closed:
☐	Basis of Decision (Yes/No):**	
	Issuing Agency:	
	Special Provisions and Requirements:**	

PERMITS RECORD OF DECISION (PROD)

PERMIT REQUIRED (Check Box)	PROJECT TITLE:	WORK ORDER NUMBER:
	LOCATION:	
	GOVERNMENT PROJECT MANAGER:	
Solid and Hazardous Waste	PERMIT: Landfill	Date Obtained: Date Closed:
☐	Basis of Decision (Yes/No):**	
	Issuing Agency:	
	Special Provisions and Requirements:**	
Solid and Hazardous Waste	PERMIT: Used Oil Collection Center, Aggregation Point, Transporter & Transfer Facility	Date Obtained: Date Closed:
☐	Basis of Decision (Yes/No):**	
	Issuing Agency:	
	Special Provisions and Requirements:**	
Solid and Hazardous Waste	PERMIT: Other	Date Obtained: Date Closed:
☐	Basis of Decision (Yes/No):**	
	Issuing Agency:	
	Special Provisions and Requirements:**	
Coastal Management Permit	PERMIT: Coastal Consistency Determination Authorization	Date Obtained: Date Closed:
☐	Basis of Decision (Yes/No):**	
	Issuing Agency:	
	Special Provisions and Requirements:**	

PERMITS RECORD OF DECISION (PROD)

PERMIT REQUIRED (Check Box)	PROJECT TITLE:	WORK ORDER NUMBER:
	LOCATION:	
	GOVERNMENT PROJECT MANAGER:	
Coastal Management Permit	PERMIT: Coastal Barrier	Date Obtained: Date Closed:
☐	Basis of Decision (Yes/No):**	
	Issuing Agency:	
	Special Provisions and Requirements:**	
Coastal Management Permit	PERMIT: Floodplain Management	Date Obtained: Date Closed:
☐	Basis of Decision (Yes/No):**	
	Issuing Agency:	
	Special Provisions and Requirements:**	
Coastal Management Permit	PERMIT: Other	Date Obtained: Date Closed:
☐	Basis of Decision (Yes/No):**	
	Issuing Agency:	
	Special Provisions and Requirements:**	
Other Permits	PERMIT: Work in Navigable Waters	Date Obtained: Date Closed:
☐	Basis of Decision (Yes/No):**	
	Issuing Agency:	
	Special Provisions and Requirements:**	

PERMITS RECORD OF DECISION (PROD)

PERMIT REQUIRED (Check Box)	PROJECT TITLE:	WORK ORDER NUMBER:
	LOCATION:	
	GOVERNMENT PROJECT MANAGER:	
Other Permits	PERMIT: Dredging	Date Obtained: Date Closed:
☐	Basis of Decision (Yes/No):**	
	Issuing Agency:	
	Special Provisions and Requirements:**	
Other Permits	PERMIT: Clearing	Date Obtained: Date Closed:
☐	Basis of Decision (Yes/No):**	
	Issuing Agency:	
	Special Provisions and Requirements:**	
Other Permits	PERMIT: Essential Fish Habitat Assessment Consultation	Date Obtained: Date Closed:
☐	Basis of Decision (Yes/No):**	
	Issuing Agency:	
	Special Provisions and Requirements:**	
Other Permits	PERMIT: Marine Mammal Protection Act	Date Obtained: Date Closed:
☐	Basis of Decision (Yes/No):**	
	Issuing Agency:	
	Special Provisions and Requirements:**	

PERMITS RECORD OF DECISION (PROD)

PERMIT REQUIRED (Check Box)	PROJECT TITLE:	WORK ORDER NUMBER:
	LOCATION:	
	GOVERNMENT PROJECT MANAGER:	
Other Permits	PERMIT: Take Permits	Date Obtained: Date Closed:
☐	Basis of Decision (Yes/No):**	
	Issuing Agency:	
	Special Provisions and Requirements:**	
Other Permits	PERMIT: Work in Wetlands	Date Obtained: Date Closed:
☐	Basis of Decision (Yes/No):**	
	Issuing Agency:	
	Special Provisions and Requirements:**	
Other Permits	PERMIT: Digging Permit	Date Obtained: Date Closed:
☐	Basis of Decision (Yes/No):**	
	Issuing Agency:	
	Special Provisions and Requirements:**	
Other Permits	PERMIT: Traffic	Date Obtained: Date Closed:
☐	Basis of Decision (Yes/No):**	
	Issuing Agency:	
	Special Provisions and Requirements:**	

PERMITS RECORD OF DECISION (PROD)

PERMIT REQUIRED (Check Box)	PROJECT TITLE:	WORK ORDER NUMBER:
	LOCATION:	
	GOVERNMENT PROJECT MANAGER:	
Other Permits	PERMIT: Airport Hazard/Airfield Safety Clearances	Date Obtained: Date Closed:
☐	Basis of Decision (Yes/No):**	
	Issuing Agency:	
	Special Provisions and Requirements:**	
Other Permits	PERMIT: Railroad Crossing	Date Obtained: Date Closed:
☐	Basis of Decision (Yes/No):**	
	Issuing Agency:	
	Special Provisions and Requirements:**	
Other Permits	PERMIT: Historic Preservation	Date Obtained: Date Closed:
☐	Basis of Decision (Yes/No):**	
	Issuing Agency:	
	Special Provisions and Requirements:**	
Other Permits	PERMIT: Noise Abatement	Date Obtained: Date Closed:
☐	Basis of Decision (Yes/No):**	
	Issuing Agency:	
	Special Provisions and Requirements:**	

PERMITS RECORD OF DECISION (PROD)			
PERMIT REQUIRED (Check Box)	PROJECT TITLE:		WORK ORDER NUMBER:
	LOCATION:		
	GOVERNMENT PROJECT MANAGER:		
Other Permits	PERMIT: Endangered/Species/Critical Habitat	Date Obtained:	Date Closed:
☐	Basis of Decision (Yes/No):**		
	Issuing Agency:		
	Special Provisions and Requirements:**		
Other Permits	PERMIT: Other	Date Obtained:	Date Closed:
☐	Basis of Decision (Yes/No):**		
	Issuing Agency:		
	Special Provisions and Requirements:**		

Prepared by: _____

Date _____

DEPARTMENT OF THE NAVY LOCAL POPULATION ID CARD/BASE ACCESS PASS REGISTRATION

PRIVACY ACT STATEMENT:

AUTHORITY: 10 U.S.C. 113, Secretary of Defense; DoD Directive 1000.25, DoD Personnel Identity Protection (PIP) Program; DoD Instruction 5200.08, Security of DoD Installations and Resources and the DoD Physical Security Review Board (PSRB); DoD 5200.08-R, Physical Security Program; DoD Directive 5200.27, Acquisition of Information Concerning Persons and Organizations not Affiliated with the Department of Defense (Exception to policy memos); Directive-Type Memorandum (DTM) 09-012, Interim Policy Guidance for DoD Physical Access Control; DTM 14-005, DoD Identity Management Capability Enterprise Services Application (IMESA) Access to FBI National Crime Information Center (NCIC) Files; and E.O. 9397 (SSN), as amended; OPNAVINST 5530.14E, Navy Physical Security and Law Enforcement Program; Marine Corps Order P5530.14, Marine Corps Physical Security Program Manual; SORN NM05512-2 Badge and Access Control System Records and DMDC 16, Identity Management Engine for Security and Analysis (IMESA): <http://dpclid.defense.gov/Privacy/SORNsindex>

PURPOSE(S): To control physical access to Department of Defense (DoD), Department of the Navy (DON) or U.S. Marine Corps Installations/Units controlled information, installations, facilities, or areas over which DoD, DON, or U.S. Marine Corps has security responsibilities by identifying or verifying an individual through the use of biometric databases and associated data processing/information services for designated populations for purposes of protecting U.S./Coalition/allied government/national security areas of responsibility and information; to issue badges, replace lost badges, and retrieve passes upon separation; to maintain visitor statistics; collect information to adjudicate access to facility; and track the entry/exit times of personnel.

ROUTINE USE(S): To designated contractors, Federal agencies, and foreign governments for the purpose of granting Navy officials access to their facility.

DISCLOSURE: Providing registration information is voluntary. Failure to provide requested information may result in denial of access to benefits, privileges, and DoD installations, facilities and buildings.

IDENTITY PROOFING AND APPLICANT INFORMATION

1. LAST NAME:	2. FIRST NAME:	3. MIDDLE NAME:	4. NAME SUFFIX: ☐ Jr. ☐ Sr. ☐ I ☐ II ☐ III ☐ IV
5. RACE (Check one or more): ☐ AMERICAN INDIAN OR ALASKA NATIVE ☐ ASIAN ☐ BLACK OR AFRICAN AMERICAN ☐ HISPANIC OR LATINO ☐ NATIVE HAWAIIAN OR OTHER PACIFIC ISLANDER ☐ WHITE			
6. GENDER (Check one): ☐ MALE ☐ FEMALE	7. DATE OF BIRTH:	8. CITY OF BIRTH:	9. STATE OF BIRTH:
11. US CITIZEN (Check): ☐ YES ☐ NO		12. DUAL CITIZENSHIP: ☐ YES ☐ NO CITIZENSHIP IF OTHER THAN US (Country) :	
10. BIRTH COUNTRY:			

U.S. Citizen Minimum Documentation Required:

By Birth - Social Security No and/or State ID/Drivers License.

Naturalized - Certification Number, Petition Number, Date, Place and Court, United States passport number, Social Security No and/or State ID/Drivers License.

Derived - Parent's certification number, Social Security No and/or State ID/Drivers License.

Alien Minimum Documentation Required:

Registration Number, Expiration date, Date of entry, Port of entry.

13. IDENTITY SOURCE DOCUMENTS PRESENTED:	14. DOCUMENT NUMBER:	15. ISSUED BY STATE/COURT:	16. ISSUED BY COUNTRY:	17. ISSUED:	18. EXPIRES:
☐ Social Security No.			United States		
☐ State ID/Drivers License			United States		
☐ Passport No.					
☐ Certification Number and Petition Number					
☐ Derived - Parent's Certification Number:			United States		
☐ Alien Registration No.			United States		
		Date of Entry:	Port of Entry:		

OTHER APPROVED IDENTITY SOURCE DOCUMENTS:

☐					
☐					
19. WEIGHT (Pounds):	20. HEIGHT (Inches):	21. HAIR COLOR (Check one): ☐ Blond ☐ Brown ☐ Black ☐ Gray ☐ Red ☐ White ☐ Silver ☐ Auburn ☐ Bald		22. EYE COLOR (Check one): ☐ Brown ☐ Green ☐ Blue ☐ Hazel ☐ Black ☐ Gray ☐ Violet ☐ Unknown	
23. HOME ADDRESS (Include city, state, zip code):				HOME PHONE (Include Area Code):	
24. BASE SPONSOR'S NAME:				SPONSOR PHONE (Include Area Code):	

EMPLOYMENT ACTIVITY INFORMATION			
25. EMPLOYER NAME AND ADDRESS (Include city/state/zip code):		EMPLOYER PHONE (Include Area Code):	
26. SUPERVISOR NAME AND ADDRESS (Include city/state/zip code):		SUPERVISOR PHONE (Include Area Code):	
27. Check the applicable box for WORK HOURS box or check the OTHER box and enter the work hours, then check the applicable for WORK DAYS: WORK HOURS: ☐ 0600-1800 ☐ 0800-1700 ☐ OTHER WORK DAYS: ☐ SN ☐ M ☐ T ☐ W ☐ TH ☐ F ☐ ST			
PRIOR FELONY CONVICTIONS			
28. Have you ever been convicted of a Felony? ☐ YES ☐ NO _____ Initial			
REQUIREMENT TO RETURN LOCAL POPULATION ID CARD			
29. I understand that I am required to return my Local Population Identification Card to the Base Pass Office when it expires or if my employment is terminated for any reason. _____ (initial)			
AUTHORIZATION AND RELEASE AND CERTIFICATION			
30. I hereby authorize the DOD/DON and other authorized Federal agencies to obtain any information required from the Federal government and/or state agencies, including but not limited to, the Federal Bureau of Investigation (FBI), the Defense Security Service (DSS), the U.S. Department of Homeland Security (DHS). I have been notified of DON right to perform minimal vetting and fitness determination as a condition of access to DON installation/facilities. I understand that I may request a record identifier; the source of the record and that I may obtain records from the State Law Enforcement Office as may be available to me under the law. I also understand that this information will be treated as privileged and confidential information. I release any individual, including records custodians, any component of the U.S. Government or the individual State Criminal History Repository supplying information, from all liability for damages that may result on account of compliance, or any attempts to comply with this authorization. This release is binding, now and in the future, on my heirs, assigns, associates, and personal representative(s) of any nature. Copies of this authorization that show my signature are as valid as the original release signed by me. FALSE STATEMENTS ARE PUNISHABLE BY LAW AND COULD RESULT IN FINES AND/OR IMPRISONMENT UP TO FIVE YEARS. BEFORE SIGNING THIS FORM, REVIEW IT CAREFULLY TO MAKE SURE YOU HAVE ANSWERED ALL QUESTIONS FULLY AND CORRECTLY. I DECLARE UNDER PENALTY OF PERJURY THAT THE STATEMENTS MADE BY ME ON THIS FORM ARE TRUE, COMPLETE AND CORECT.			
DATE _____ SIGNATURE _____			
FINAL DETERMINATION ON YOUR ACCESS: The Base Commanding Officer has final authority for determination on granting physical access to DON controlled installations/facilities under his/her jurisdiction.			
BELOW COMPLETED BY BASE REGISTRAR PERSON CONDUCTING IDENTITY PROOFING and NCIC CHECK			
31. INFORMATION VERIFIED BY:	32. ENTERED IN C/S SYSTEM BY:	33. PASS ISSUE DATE:	34. PASS EXPIRATION DATE:
35. NCIC CHECK PERFORMED BY:	36. RESULTS OF NCIC CHECK: ☐ NO RECORDS ☐ RECORD IDENTIFIER RECORD NUMBER: _____		37. RESULTS OF LOCAL RECORDS CHECK: ☐ NO RECORDS ☐ RECORD IDENTIFIER RECORD NUMBER: _____
Office of Under Secretary of Defense, DoD Manual 5200.08 VOLUME 3, PHYSICAL SECURITY PROGRAM: "ACCESS TO DOD INSTALLATIONS," January 2, 2019. The DoD Manual requires DoD installation government representatives query the National Crime Information Center (NCIC) and Terrorist Screening Database to vet the claimed identity and to determine the fitness of non-federal government and non-DoD-issued card holders (i.e. visitors) who are requesting unescorted access to a DoD installation. The minimum criteria to determine the fitness of a visitor is: 1) not on a terrorist watch list; 2) not on an DoD installation debarment list; and 3) not on a FBI National Criminal Information Center (NCIC) felony wants and warrants list. Additionally, SECNAV Memo, Policy for Sex Offender Tracking and Assignment and Access Restrictions within the Department of the Navy, of 7 Oct 08 and OPNAVINST 1752.3 established the Navy's policy on sex offenders, requiring Region Commanders (REGCOMs) and Installation Commanding Officers (COs) to prohibit sex offender access to DoN facilities and Navy owned, leased or PPV housing. This form describes the authority and purpose to collect and share the required information; and identifies the applicant/visitor and sponsor; and authorizes the DoD to perform the minimum vetting and fitness determination criteria. A favorable response on the vetting and fitness determination is required to receive access to DOD controlled installation/facilities.			

Instruction for completing the Local Population Access Registration Form

INSTRUCTIONS: Please complete all information in black ink (printed) or by typing. By voluntarily providing your Personal Information, you agree to the following terms and restrictions:

RESTRICTIONS: Local Population Identification Card/Base Access Pass may only be used by person to whom they are issued and for the specific business/purpose issued. Applicants are reminded that soliciting (i.e., door-to-door sales) is prohibited on the base, and that such activity is grounds for cancellation of the Pass. Additionally, such action may result in debarment from the base and legal action. The Base Commanding Officer has discretion over specifying the period of validity for any Local Population ID Cards/Base Access Passes that are issued under his/her jurisdiction. Review the Privacy At Statement that is printed at the top of the form

Block 1: Enter the Last Name. Block 2: Enter the First Name. Block 3: Enter the Middle Name. Block 4: If applicable, check the box for Name Suffix. Block 5: Check the applicable box for Race. Block 6: Check the applicable box for Gender. Block 7: Enter Date of Birth. Block 8: Enter City of Birth. Block 9: Enter State of Birth. Block 10: Enter Country of Birth. Block 11: Check the applicable box for US Citizenship. Block 12: If not a US Citizen, enter the name of the Country of Citizenship. Block 13: Two forms of identity source documents from the list of acceptable documents listed below must be presented to the base registrar with this completed form. Check the box for the type of Documents that will be presented for identity proofing. If the document type is not listed, use the two rows under Other Approved Identity Source Documents to enter the type of document(s) that you will present. Block 14: Enter the Document Number located on the Identity Proofing Source document that was checked in Block 13. Block 15: Enter the State that issued the Identity Source Document. Block 16: Enter the Country that issued the Identity Source Document.	Block 17: Enter the Date that the Identity Source Document was issued. Block 18: Enter the Date that the Identity Source Document will expire. Block 19: Enter Weight in pounds. Block 20: Enter Height in inches. Block 21: Check the applicable box for Hair Color. Block 22: Check the applicable box for Eye Color. Block 23: Enter Home Address Including City, State, Zip Code, and Home Telephone Number. Block 24: Enter Name of Registrant's Base Sponsor and Base Sponsor's Telephone Number. Block 25: Enter Employer Name and address including City, State, Zip Code, and Employer's Telephone Number. Block 26: Enter Supervisor's Name including City, State, Zip Code, and Supervisor's Telephone Number. Block 27: Check the applicable box for Work Hours box or check the OTHER box and enter the work hours, then check applicable boxes for Work Days. Block 28: Check the applicable answer if you have been convicted of Felony and enter initials. Block 28: Check the applicable box for felony conviction. Block 29: Enter initials to accept terms for returning Local Population Identification Card. Block 30: Sign and date the form to attest that the foregoing information is true and complete to best of your knowledge.
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LIST OF ACCEPTABLE DOCUMENTS - All documents must not be expired.

Must present one selection from List A or a combination of one selection from List B and one selection from List C.

List A - Documents that Establish Identity and Employment Authorization	OR	List B - Documents that Establish Identity	AND	List C - Documents that Establish Employment Authorization
1. U.S. Passport or U.S. Passport Card. 2. Permanent Resident Card or Alien Registration Receipt Card (Form I-551). 3. Foreign passport that contains a temporary I-551 stamp or temporary I-551 printed notation on a machine-readable immigrant visa. 4. Employment Authorization Document that contains a photograph (Form I-766). 5. For a nonimmigrant alien authorized to work for a specific employer because of his or her status: a. Foreign Passport; and b. Form I-94 or Form I-94A that has the following: (1) The same name as the passport; and (2) An endorsement of the alien's nonimmigrant status as long as that period of endorsement has not yet expired and the proposed employment is not in conflict with and restrictions or limitations identified on form. 6. Passport from the Federal States of Micronesia (FSM) or the Republic of the Marshal Islands (RM) with Form I-94 or Form I-94A indicating nonimmigrant admission under the Compact of Free Association Between the United States and FSM or RM.		1. Driver's license or ID card issued by a State or outlying possession of the United States provided it contains a photograph or information such as name, date of birth, gender, height, eye color, and address. 2. ID card issued by federal, state or local government agencies or entities, provided it contains a photograph or information such as name, date of birth, gender, height, eye color, and address. 3. School ID card with a photograph 4. Voter's registration card. 5. U.S. Military card or draft record. 6. Military dependent's ID card. 7. U.S. Coast Guard Merchant Mariner Card. 8. Native American tribal document. 9. Driver's license issued by a Canadian government authority. For persons under age 18 who are unable to present a document listed above: 10. School record or report card. 11. Clinic, doctor, or hospital record. 12. Day-care or nursery school record.		1. A Social Security Account Number card, unless the card includes one of the following restrictions: (1) NOT VALID FOR EMPLOYMENT (2) VALID FOR WORK ONLY WITH INS AUTHORIZATION. (3) VALID FOR WORK ONLY WITH DHS AUTHORIZATION. 2. Certification of Birth Abroad issued by the Department of State (Form FS-545). 3. Certification of Birth issued by the Department of State (Form DS-1360). 4. Original or certified copy of birth certificate issued by a State, county, municipal authority or territory of the United States bearing an official seal. 5. Native American tribal document. 6. U.S. Citizen ID Card (Form I-197). 7. Identification Card for Use of Resident Citizen in the United States (Form I-179). 8. Employment authorization document issued by the Department of Homeland Security.

The remainder of the form will be completed by the Base Registrar Person conducting Identify Proofing process and NCIC check.

AGENCY DISCLOSURE STATEMENT:

The public reporting burden for this collection of information, OMB 0703-0061, is estimated to average ten (10) minutes per response, including the time for reviewing instructions, searching existing data sources, gathering and maintaining the data needed, and completing and reviewing the collection of information. Send comments regarding the burden estimate or burden reduction suggestions to the Department of Defense, Washington Headquarters Services, Executive Services, at whs.mc-alex.esd.mbx.dd-dod-information-collections@mail.mil. Respondents should be aware that notwithstanding any other provision of law, no person shall be subject to any penalty for failing to comply with a collection of information if it does not display a currently valid OMB control number.

PLEASE DO NOT RETURN RESPONSE TO THE ABOVE ADDRESS.

Responses should be sent to the Base Registrar.

5000-82
Memo

MEMORANDUM

From (Trusted Agent, Command):

To: Commanding Officer, Naval Station Great Lakes (N34/VCC)

Subj: INSTALLATION ACCESS REQUEST STATEMENT OF PURPOSE

☐ 1. _____ (Company Name) has a legitimate need to access the installation due to contracted or service agreement.

Or

☐ 1. _____ (Individual's name) has a legitimate need to access the installation for the purpose of:

2. The following information is provided:

a. Government Sponsor (Trusted Agent):

Name (First MI. Initial):

Phone Number:

Email Address:

b. Company Information:

Address:

Point of Contact Name:

Phone Number:

Email Address:

Contract Number/Project Number:

Contract Period of Performance:

3. DBIDS Credential Information:

a. DBIDS Registration: NEW; RENEWAL

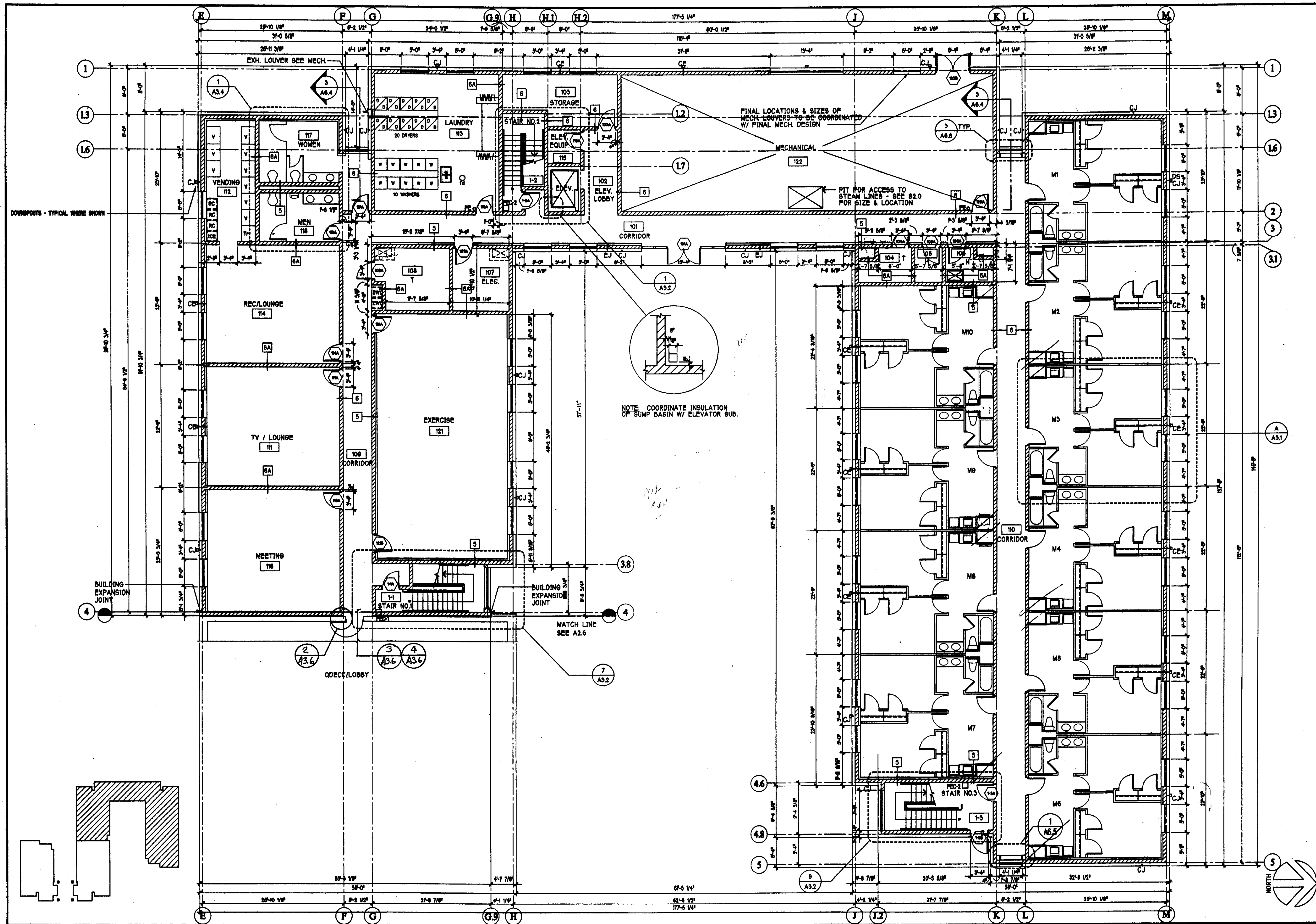
b. Length (Provide specific dates):

c. Access Days/Times Required (E.g. Mon-Fri, 0500-1800). 24/7 access requires justification:

4. Access to Recruit Training Complex Required: YES (justification required); NO

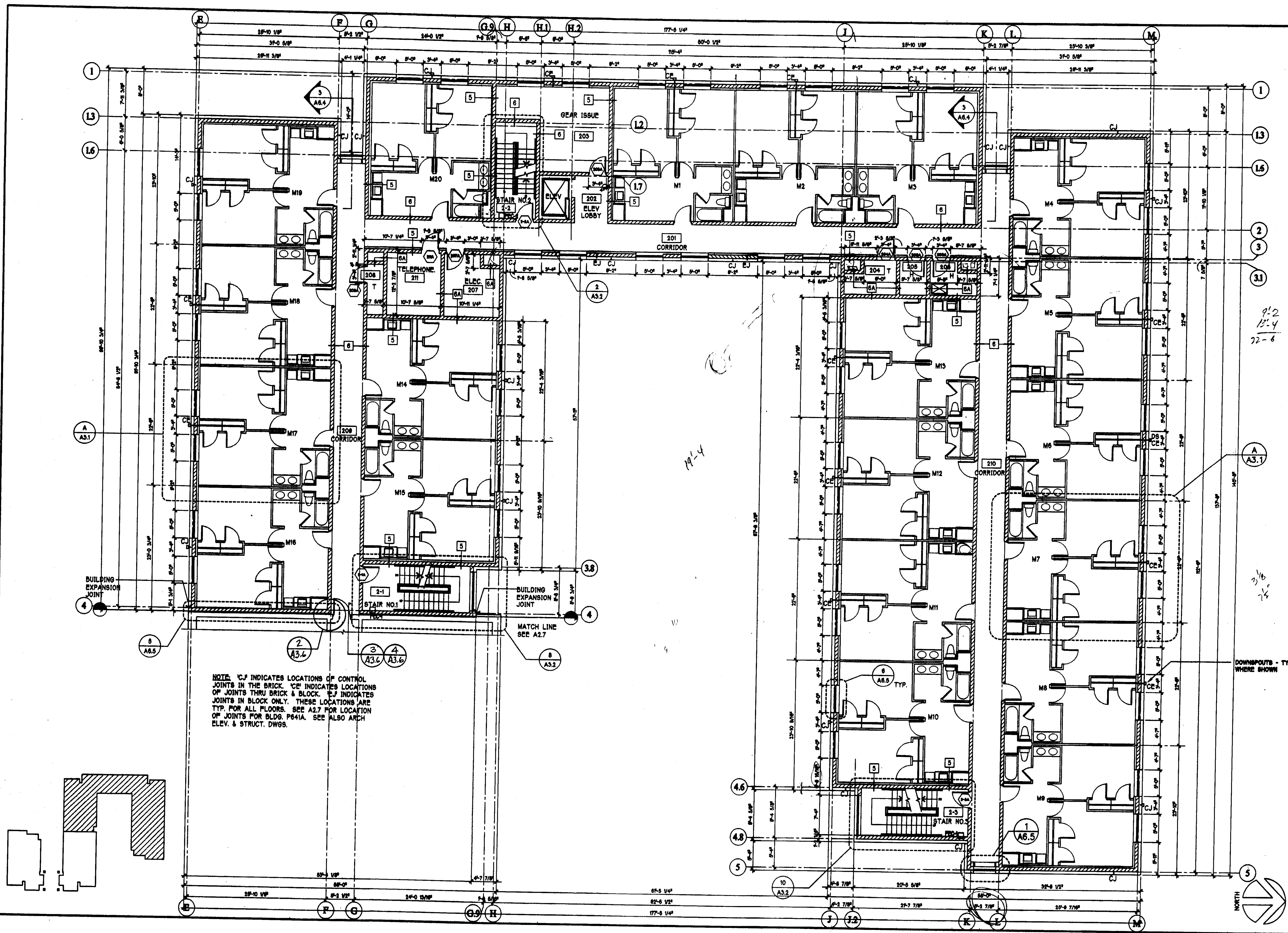
5. Company List of Employees Name (Last, First MI):

GOVERNMENT SPONSOR'S SIGNATURE



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ISSUED FOR STRUCTURES/EXTERIOR ENVELOPE	2 FEBRUARY 98	2 FEBRUARY 98
ARCHITECTURAL INTERIORS	3 MARCH 98	3 MARCH 98
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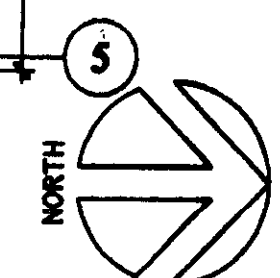
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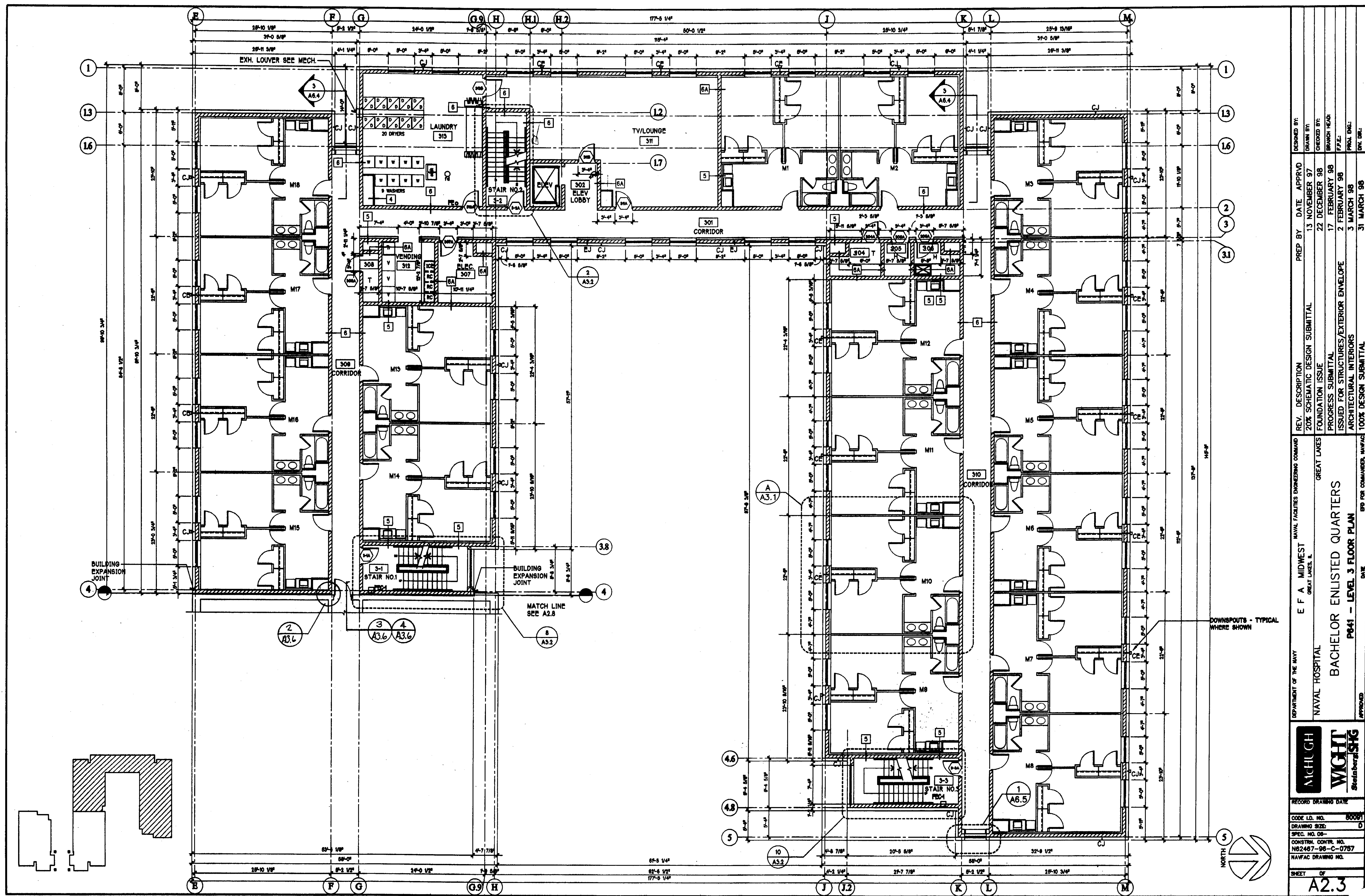
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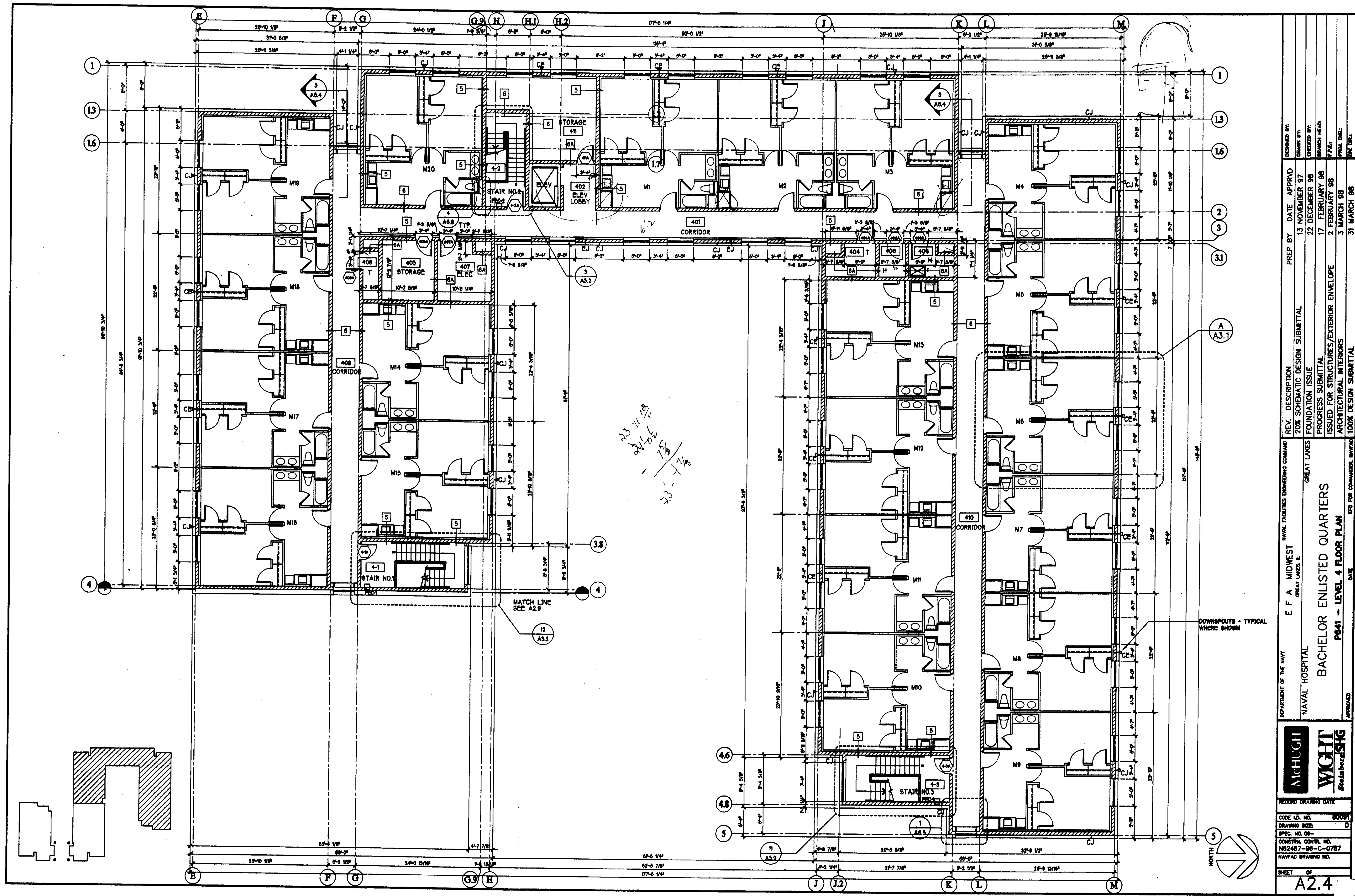
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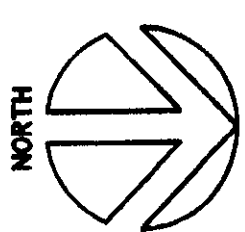
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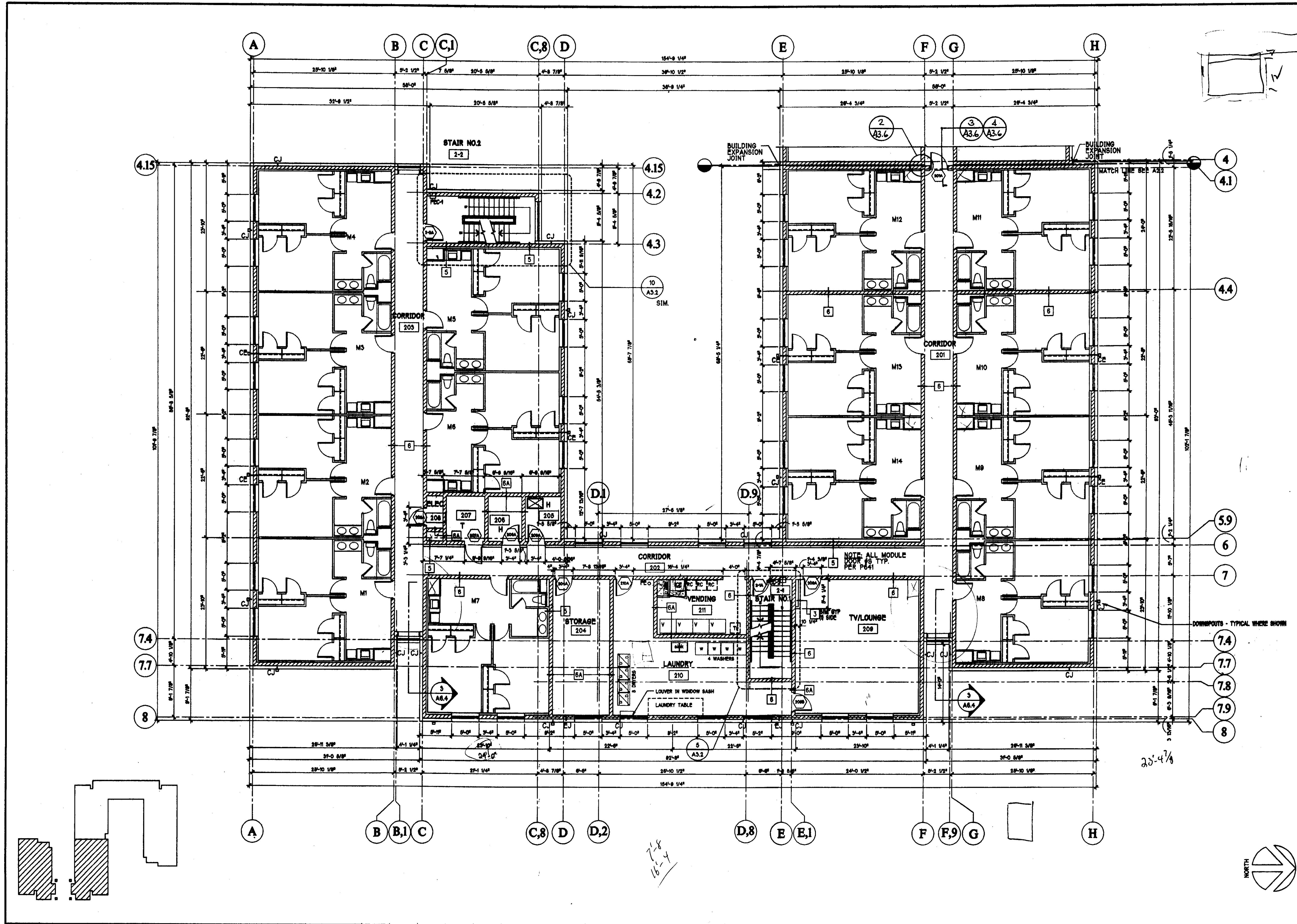
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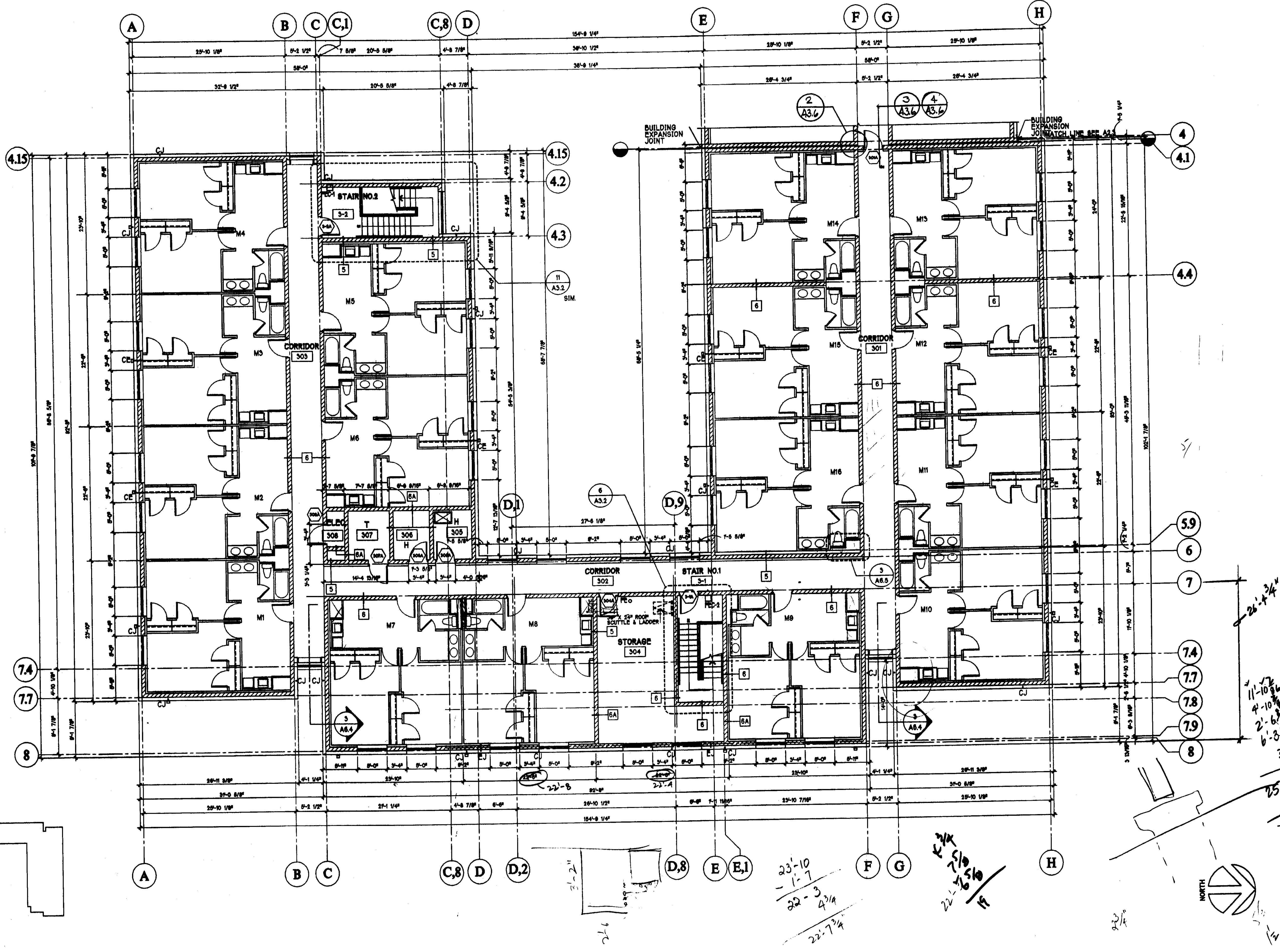


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PROJECT:	ARCHITECTURAL INTERIORS	3 MARCH 98	
PROJECT:	100% DESIGN SUBMITTAL	31 MARCH 98	

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BACHELOR ENLISTED QUARTERS		
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Contractor Guidance Manual

Construction Stormwater Management

Naval Station Great Lakes Public Works Department, Environmental Operations
January 24, 2023



This guidance manual provides contractors working at Naval Station Great Lakes (NSGL) with expectations regarding the need for strict compliance with the Federal Clean Water Act (CWA), the National Pollution Discharge Elimination System (NPDES) permit requirements, and similar federal, state, and local laws regulating stormwater run-off and erosion and sediment control on construction job-sites. The content of this manual represents the minimum requirements for compliance with construction stormwater discharge regulations.

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NAVAL STATION GREAT LAKES CONSTRUCTION STORMWATER MANAGEMENT CONTRACTOR GUIDANCE MANUAL

TABLE OF CONTENTS

Part I	Overview of NSGL Policy, Stormwater Requirements, and Responsibilities.....	1
1-1	Scope.....	1
1-2	Navy Environmental Policy	2
1-3	Summary of Applicable Regulatory Requirements.....	2
1-4	Responsibilities	3
1-4.1	General Contractor Responsibilities	3
1-4.2	Subcontractor Responsibilities.....	5
Part II	Construction Stormwater Requirements – NPDES Permitted Projects	7
2-1	Pre-Construction Requirements	7
2-1.1	Procedures and Paperwork.....	7
2-1.2	Borrow and Spoil Conditions.....	8
2-2	Active Construction Requirements	8
2-2.1	General Contractor Inspection Procedures.....	8
2-2.2	Inspection, Paperwork, and Training Requirements	9
2-2.3	Spill Prevention and Response Procedures	10
2-2.4	Regulatory Inspection Procedures.....	11
2-3	Post Construction Requirements	12
2-3.1	Notice of Termination (NOT) and Close Out	12
2-3.2	Records Turn-Over.....	13
Part III	– NSGL Erosion and Sedimentation Control Standards	15
3-1	Structural Controls	15
3-2	Non Structural Controls	16
3-3	Materials Management Plan.....	16
3-4	Other Stormwater Pollution Prevention Requirements	20
Part IV	– NSGL Water Resources.....	22
4-1	Additional Requirements for Construction in or near Station Water Bodies (Pettibone Creek, Skokie River, and Lake Michigan).....	22
4-2	Total Maximum Daily Load (TMDL) for Station Water Resources.....	22

SWPPP FORMS

1. General Contractor Certification (NSGL)
2. Construction Site SWPPP Notice (NSGL)
3. 2022 CGP Site Inspection Report (EPA – *Found on the EPA website)
4. 2022 CGP Corrective Action Log (EPA – *Found on the EPA website)
5. SWPPP Template (EPA – *Found on the EPA website)

*See details to download in the SWPPP Forms section

List of Appendices

- A** Not Used
- B** NSGL Watershed Boundary and Drainage Areas Map

Part I Overview of NSGL Policy, Stormwater Requirements, and Responsibilities

1-1 Scope

The purpose of this manual is to provide contractors working at Naval Station Great Lakes (NSGL) with the expectations and requirements for strict compliance with the Federal Clean Water Act (CWA), the National Pollution Discharge Elimination System (NPDES) permit requirements, and similar federal, state and local laws regulating stormwater runoff and erosion and sediment control on construction job-sites. This manual specifically pertains to contractors at NSGL performing construction activities requiring NPDES permitting. This is defined as construction activities disturbing greater than one acre of land.

In general terms, NSGL expects its contractors:

- To independently investigate, understand and comply with all such laws, both as their independent obligation under the law and as well as their affirmative obligation under their construction contracts with NSGL;
- To assign only personnel to NSGL projects trained in understanding and complying with such laws;
- To have a system of monitoring and quality control on NSGL construction job-sites to ensure compliance with such laws and to prevent the discharge of any impermissible substances from NSGL construction job-sites;
- To promptly report any impermissible discharges to the appropriate NSGL representative and, where required by law, to allow NSGL to report such discharges to appropriate public officials, and to work with NSGL and any governmental agencies with jurisdiction to correct the problem which caused the impermissible discharges and to correct and mitigate any harm from the impermissible discharges; and
- To maintain at all times accurate and timely records required under such laws and required under this manual. NSGL expects its contractors to be proactive in meeting all of these requirements and to communicate on a regular basis with the Navy Construction Manager and/or Environmental Engineer assigned to a NSGL construction project on stormwater compliance related matters.

This manual will form part of the Contract Documents on any NSGL construction project. Although this manual is intended to provide certain basic guidance on NSGL expectations, this manual does not relieve a contractor of its independent obligation to investigate, understand and comply with all laws on a NSGL construction project.

1-2 Navy Environmental Policy

Navy policy for the management of environmental, natural, and cultural resources for all Navy ship and shore activities is stated in Navy Operations Instruction (OPNAVINST) 5090.1E. As required by OPNAVINST 5090.1E (9-4), Navy facilities will comply with all substantive and procedural requirements applicable to point and non-point sources of pollution as required by E.O. 12088 and the CWA. These requirements include Federal, State, interstate, and local laws and regulations regarding the control and abatement of water pollution such as load reduction requirements resulting from the development of total maximum daily limits (TMDLs) for impaired water bodies. Navy facilities must comply in the same manner and to the same extent as any nongovernmental entity, including the payment of reasonable service charges.

1-3 Summary of Applicable Regulatory Requirements

The Illinois Environmental Protection Agency (IEPA) has authority to implement the federal stormwater regulations under the National Pollutant Discharge Elimination System (NPDES) program (40 CFR § 122.26). IEPA NPDES permits requirements are codified under 35 Ill. Adm. Code 309 which regulate discharges into navigable waters. This authority to implement the federal NPDES requirements is codified by IEPA under Sections 13 and 13.3 and authorized by Section 27 of the Illinois Environmental Protection Act [415 ILCS 5/13, 13.3 and 27].

Person or persons with either operational control of construction project plans and specifications, or day-to-day operational control of activities necessary to ensure compliance with stormwater NPDES permit conditions are required to obtain permit coverage for construction activity that results in a total land disturbance of 1 acre or more or common plan of development or sale. Below is a link to the IEPA General NPDES Permit (ILR10) For Stormwater Discharges from Construction Site Activities.

<https://www2.illinois.gov/epa/topics/forms/water-permits/storm-water/Documents/General-NPDES-Permit-ILR10-Construction.pdf>

The Lake County Stormwater Management Commission (LCSMC) has adopted a Watershed Development Ordinance (WDO) as part of the county's Comprehensive Stormwater Management Plan (adopted by ordinance June 11, 1990). A memorandum of understanding (MOU) exists between LCSMC and NSGL dated April 1995 regarding coordination between the two government entities over stormwater management procedures within a shared watershed. The MOU generally requires NSGL to submit for comment and approval specific construction plans for compliance with the WDO. Below is a link to the LCSMC WDO.

<https://www.lakecountyil.gov/2358/Watershed-Development-Ordinance>

1-4 Responsibilities

1-4.1 General Contractor Responsibilities

The following list provides responsibilities and NSGL expectations of the General Contractor (GC) working on an NSGL project under an approved NPDES construction stormwater discharge permit:

1. One of the keys to success on a construction project is NSGL's relationship with its contractors. NSGL contractors will be responsible for effectively communicating with NSGL personnel, and its consultants, inspectors and other occupants and contractors present on the site. Examples of common communications include:
 - a. Results of stormwater permit required inspections;
 - b. Major modifications or revisions to project Stormwater Pollution Prevention Plan (SWPPP);
 - c. Major modification to site Best Management Practices (BMPs); and
 - d. Monthly SWPPP training of personnel and subcontractors.
2. NSGL policy for all projects disturbing greater than one acre requires the contractor to file a Notice of Intent (NOI) for coverage under the IEPA construction stormwater discharge permit (ILR10). Department of Navy (DON) shall be listed as the owner and the Naval Facilities Engineering Command (NAVFAC) Mid-Atlantic NSGL Public Works Officer (PWO) or authorized designee shall sign as Owner.
3. GC prepares a site specific SWPPP and NOI that are reviewed as part of contract submittals. The NOI will be completed online through the EPA's NPDES eReporting Tool, NeT, found at the following link: https://usepa.servicenowservices.com/oeca_icis?id=net_homepage. The GC will input information for the project profile and create the NOI as a "Preparer". When complete, the form will be assigned to a "Signatory" (PWO) to be certified before EPA review. As before, operators will be required to provide the following information:
 - Approval to use a paper NOI form if granted a waiver from electronic reporting by an EPA Regional Office;
 - Permit information;
 - Operator information;
 - Project/site information;
 - Discharge information;
 - Chemical treatment information, if applicable;
 - SWPPP information;
 - Endangered species protection information;
 - Historic preservation information;

- Certification of NOI; and
 - Contact information for NOI preparer.
- a. Unless notified by the Agency with a Notice of Incompleteness letter, coverage under the Stormwater General NPDES permit is automatic, and operators are authorized to discharge stormwater from construction sites under the terms and conditions of the permit **30 days after the date the NOI is received by the Agency**, provided the project has received sign-off from Illinois Department of Natural Resources (IDNR) and Illinois Historic Preservation Act (IHPA) that the project complies with endangered species and historic preservation laws and appropriate application fee has been received by the Agency.
 - b. IEPA will issue a Notice of Coverage (NOC) which is typically sent to the owner and contact listed on the NOI. The NOC will be forwarded to the GC for inclusion in the site SWPPP binder and posting at the site.
 - c. A copy of the NOI submittal shall be included with the on-site SWPPP. Proof of submission, such as a notification of receipt, with the NOI in the on-site SWPPP. This will serve as the basis for permit coverage in absence of the NOC. As stated above, coverage under the Stormwater General NPDES permit is automatic after 30 days of receipt by the Agency of the NOI. **UNDER NO CIRCUMSTANCES SHALL CONSTRUCTION ACTIVITIES BEGIN WITHOUT A FULLY EXECUTED STORMWATER PERMIT NOI AND A NOC FROM THE AGENCY or 30 DAYS HAS PASSED SINCE AGENCY RECEIPT OF THE NOI.**
 4. GC is responsible for regulatory stormwater inspections and providing qualified individual(s) to conduct such inspections. The IEPA construction stormwater discharge general permit specifies requirements for qualifications of stormwater inspectors. The qualification requirement as stated in Part IV.D.4 of the ILR10 general permit states:

.....Qualified personnel means a person knowledgeable in the principles and practices of erosion and sediment controls measures, such as a licensed Professional Engineer (P.E.), a Certified Professional in Erosion and Sediment Control (CPESC), a Certified Erosion Sediment and Stormwater Inspector (CESSWI) or other knowledgeable person who possesses the skills to assess conditions at the construction site that could impact stormwater quality and to assess the effectiveness of any sediment and erosion control measures selected to control the quality of stormwater discharges from the construction activities.

IEPA's phrase "...or other knowledgeable person who possesses the skills to assess..." is not interpreted to mean superintendents or other contract labor are

qualified based solely on years of experience on construction projects.

NSGL requires that the GC provide qualified personnel for stormwater inspection that have clear demonstration of knowledge in the principles and practices of erosion and sediment controls measures. NSGL retains the right to review and reject a contractor's stormwater inspector based on the lack of applicable experience and demonstration of knowledge.

The identification and qualification of the GC's stormwater inspector shall be provided in the contract submittals for approval prior to submittal of the NOI.

5. The GC is to hold a SWPPP preconstruction meeting prior to starting any work on a construction project. This can be added as an agenda item to the general pre-construction meeting. This starts the communication process and lays out the stormwater management expectations for the project team members. All applicable members of the project team are required to attend including the Contractor's Superintendent, Project Manager, Site Work Subcontractor, Stormwater Inspector, Navy Construction Manager, Navy PWD Environmental Representative, and other subcontractors as necessary.
6. GC will be required to provide to NSGL copies of all project SWPPP documentation including site inspection reports for NSGL files. Inspection reports are to be completed in a timely manner as required by the IEPA general permit. All reports are to be transmitted to the Navy Environmental Representative via email as they are completed and must also be retained in the project log book at the site.
7. Outside utility companies are not fully exempt from adhering to the SWPPP. It is the GC responsibility to inform them of the approved SWPPP project requirements and include inspections as part of their work to ensure appropriate BMPs are implemented.

1-4.2 Subcontractor Responsibilities

NSGL and its GC are ultimately responsible for compliance with the CWA and NPDES/SWPPP. The subcontractors shall be held accountable for their actions. The following are the requirements of subcontractors on all NSGL projects:

1. All subcontractors must abide by the SWPPP. A subcontractor and GC will be held liable jointly for all costs NSGL incurs as a result of a violation of permit caused by the subcontractor.
2. They are required to have qualified people who are trained on the SWPPP/BMPs.
3. **The maintenance and adherence to the SWPPP is the subcontractor's**

responsibility and must be made enforceable by GC's subcontract.

4. All staging areas must be located more than 150 feet away from the waterways and discharge points (if possible).
5. The subcontractor shall provide the GC with a notification that their off-site disposal of trash, debris, and soil is in accordance with the federal, state, and local requirements.

Part II Construction Stormwater Requirements – NPDES Permitted Projects

2-1 Pre-Construction Requirements

This section outlines specific pre-construction requirements that need to be met prior to the start of any earth disturbing activities.

2-1.1 Procedures and Paperwork

The key to successfully managing a construction site SWPPP program is to conduct regular inspections and maintain the required records. This is specifically required as a condition of the general permit. The following lists the basic steps for the minimum pre-construction requirements on a NSGL NPDES permitted project.

1. This NSGL Construction Stormwater Management – Contractor Guidance Manual is to be provided or made accessible to all GC Superintendents and Project Managers. The manual is to be used as a reference and shall not replace the site specific SWPPP.
2. No project that disturbs greater than one (1) acre of land is allowed to start without an executed Notice of Intent (NOI) (**No exceptions**).
3. SWPPP Pre-construction Meeting – A pre-construction meeting must be held prior to starting any project. The GC Superintendent and GC Project Manager must attend this meeting.
4. The following forms must be executed and signed prior to starting work:
 - a. Illinois Environmental Protection Agency (IEPA) – Notice of Intent (NOI) for coverage under the General NPDES Permit for Stormwater Discharges from Construction Site Activities. Note that this includes a 30 day period for automatic coverage under ILR10 if the IEPA has not provided notice of coverage earlier than the 30 days. This must be completed online, through the EPA's NeT portal.
 - b. General Contractor Certification - The general contractor certification form must be completed by the GC in accordance with Part VI.G of the general permit.
5. The following paperwork and forms must be posted prior to starting work:
 - a. Notice Of Intent (NOI) – On the job trailer wall and at the main entry to the site for public viewing.
 - b. Construction Site SWPPP Notice – This must be posted at the main entrance to the project and in the project job trailer. Emergency contact

numbers must be included along with a schedule where and when the SWPPP can be viewed by the public.

- c. Working As Built SWPPP Drawing – On the job trailer wall.
- d. Spill Notification Procedures (Appendix B) – On the job trailer wall and material storage area.

2-1.2 Borrow and Spoil Conditions

All offsite borrow or spoil, including topsoil haul-off, must be to a licensed/permitted site. This needs to be reviewed in the SWPPP Pre-Construction meeting for the project and tracked on site by the Superintendent to ensure adherence.

Project spoils, construction & demolition debris, excess dirt, etc. shall be removed from NSGL within 30-days of project completion. Soil shall be verified free of contamination before use at an alternate location on NSGL. Alternate sites must be approved in advance by the PWD-Environmental Department Director.

2-2 Active Construction Requirements

2-2.1 General Contractor Inspection Procedures

General

Inspect disturbed areas of the construction site, areas that have not been finally stabilized, areas used for storage of materials exposed to precipitation, stabilization practices, structural practices, other controls, and areas where

vehicles exit the site at least once every seven (7) calendar days and within 24 hours of the end of any storm that produces 0.5 inches or more rainfall at the site.

Inspections Details

Inspect disturbed areas [and areas used for material storage that are exposed to precipitation] for evidence of, or the potential for, pollutants entering the drainage system. Observe erosion and sediment control measures identified in the SWPPP to ensure that they are installed and operating correctly. Inspect discharge locations or points to ascertain whether erosion control measures are effective in preventing significant impacts to receiving waters. Inspect locations where vehicles exit the site for evidence of offsite sediment tracking.

Inspection Reports

For each inspection conducted, prepare a report (SWPPP FORM 3) summarizing the scope of the inspection, name(s) and qualifications of personnel making the inspection, the date(s) of the inspection, major observations relating to the implementation of the SWPPP, maintenance performed, and actions taken. Furnish the report to the Environmental Department within 24 hours of the inspection. The original inspection report shall be maintained on the job site in the site specific SWPPP. The inspection report shall be certified by signature of the inspector.

2-2.2 Inspection, Paperwork, and Training Requirements

The following are the active construction inspection, paperwork, and training requirements for all NSGL projects:

SWPPP FORM No.	NAME	INSTRUCTION	FREQUENCY
3	2022 CGP Site Inspection Report	All SWPPP/Best Management Practice (BMP) measures must be inspected on this form.	Every seven (7) days & within 24 hours any rain event of ½" or more.
4	2022 CGP Corrective Action Log	This form is used to document corrective actions taken as a result of deficiencies noted during stormwater inspections.	As Needed
NA	Working As-Built SWPPP Drawings	The SWPPP drawings, which are to be posted in the job trailer wall, are to be kept up to date by field mark-up to reflect current configuration of stormwater BMPs at the project site.	Whenever changes are made to site BMPs.

2-2.3 Spill Prevention and Response Procedures

The Contractor will train all personnel in the proper handling and cleanup of spilled materials. It shall be the responsibility of the job site superintendent to properly train all personnel in spill prevention and clean up procedures. No spilled hazardous materials or hazardous wastes will be allowed to come in contact with stormwater discharges. If such contact occurs, the stormwater discharge will be contained on site until appropriate measures in compliance with State and Federal regulations are taken to dispose of such contaminated stormwater.

In order to minimize the potential for a spill of hazardous materials to come into contact with stormwater, the following steps will be implemented:

1. All materials with hazardous properties (such as pesticides, petroleum products, fertilizers, detergents, construction chemicals, acids, paints, paint solvents, cleaning solvents, additives for soil stabilization, concrete curing compounds and additives, etc.) will be stored in a secure location, with their lids on, preferably under cover, when not in use. Storage locations shall be approved by a NSGL PWD Environmental Division Representative.
2. Special approval is required for use and storage of pesticides. Contact the NSGL PWD Environmental Division for information.
3. The minimum practical quantity of all such materials will be kept on the job site.
4. A spill control and containment kit (containing, for example, absorbent materials, acid neutralizing powder, brooms, dust pans, mops, rags, gloves, goggles, plastic and metal trash containers, etc.) will be provided at the storage site. This is also a specific requirement of the spill prevention, control, and countermeasures (SPCC) rule if fuels or oils will be kept on site in containers greater than 55 gals in size.
5. Manufacturer's recommended methods for spill cleanup will be clearly posted and site personnel will be trained regarding these procedures and the location of the information and cleanup supplies.

In the event of a spill, the following procedures should be followed:

1. Notify NSGL Emergency Dispatch Center (847) 688-3333
2. All spills will be cleaned up immediately after discovery.
3. The spill area will be kept well ventilated and personnel will wear appropriate protective clothing to prevent injury from contact with the hazardous substances.

4. The Facilities Engineering and Acquisitions Division (FEAD) Construction Manager will be notified immediately. **A NSGL Spill Notification Procedure flyer has been provided in Appendix B for posting at the job site.**
5. **Spills Outside of NSGL Property:** Spills of toxic or hazardous materials will be reported to the appropriate Federal, State, and/or local government agency. Spills of amounts that exceed Reportable Quantities of certain substances specifically mentioned in federal regulations (40 CFR 110, 40 CFR 117, and 40 CFR 302) must be immediately reported to the EPA National Response Center, telephone 1- 800-424-8802.
6. If the spill exceeds a Reportable Quantity, the SWPPP must be modified within seven (7) calendar days of knowledge of the discharge to provide a description of the release, the circumstances leading to the release, and the date of the release. The plan must identify measures to prevent the recurrence of such releases and to respond to such releases.

The job site superintendent will be the spill prevention and response coordinator. He will designate the individuals who will receive spill prevention and response training. These individuals will each become responsible for a particular phase of prevention and response. The spill notification flyer (Appendix B) will be posted in the material storage area and in the office trailer onsite.

2-2.4 Regulatory Inspection Procedures

Federal, State, and local inspections for environmental regulatory compliance and enforcement **are** authorized for NSGL projects. While outside agency regulatory inspections of projects within the secured portion of the station are not conducted without advance notice, projects occurring outside the secured area may be inspected without prior notification. The following are guidelines provided for contractors working on NSGL projects for scheduled or unscheduled inspection by an outside agency.

Overview

Under the CWA, the Federal (EPA), State (EPA), or Local permitting agencies are authorized to conduct workplace inspections to determine whether permit holders are complying with standards issued by the agency for NPDES/SWPPP. Inspections are usually conducted without advance notice.

The following bullets outline NSGL expectations for the contractor superintendent to observe during a SWPPP inspection.

Your Behavior

- **Notify FEAD Construction Manager of unannounced inspection.**
- Be polite, respectful, and cooperative.
- **Remember that you are a professional and are representing your company**

and that your conduct could affect the Navy.

- Do not resent the presence of an inspector.
- The atmosphere shall be that of cooperation.
- Only answer direct questions, preferably "yes" or "no".
- Never argue with the inspector.
- Take notes.
- Never guess at an answer to a question. If you don't know, say so, and promise to investigate the question and respond later by mail or phone.
- Make no admissions of liability.

Credentials

- The superintendent has the right to and shall ask for the inspector's credentials before letting him inspect the job.
- In order to ensure that this is an authorized local, state or federal official, you may record the inspector's name and number and ask the inspector to wait until his or her credentials are verified. They can be verified by contacting the nearest local office.
- Once the credentials are verified, do not delay the inspector or deliberately waste his/her time.

Inspection Process

The inspector may request to review the records and documentation. This includes the following:

- Jobsite posting (office wall and at the entrance);
- The written SWPPP;
- BMPs in use;
- SWPPP inspector's training and qualifications;
- Rain fall and stabilization reports; and
- Inspection reports.

A "walk-around" inspection may also be performed. The superintendent must escort the inspector at all times. You must be thoroughly familiar with all SWPPP aspects of operations and must be skilled in answering questions and explaining any and all aspects of the work to defend any ambiguous points. You must be able to conduct yourself in a courteous and business-like manner at all times.

The superintendent and/or project manager shall **notify the FEAD Construction Manager** of any inspection by Federal, State or Local Officials.

2-3 Post Construction Requirements

2-3.1 Notice of Termination (NOT) and Close Out

A site can be considered finally stabilized when:

1. All soil disturbing activities have been completed;

2. A uniform perennial vegetative cover with a density of 70% for the unpaved areas has been established;
3. Areas not covered by permanent structures have equivalent permanent stabilization measures established;
4. The activity no longer discharges stormwater associated with construction activities;
5. And a Notice of Termination (NOT) form has been filed through the EPA's NeT portal.

When the site is fully stabilized, the NOT is to be submitted to the IEPA. The NOT process is as follows:

1. To initiate the process the GC must notify the FEAD Construction Manager and Environmental Department that the site meets the final stabilization criteria and is ready for permit termination.
2. Prior to filing the NOT a NSGL Environmental Department Representative shall visit the site and perform a final inspection for compliance with standards and authorize signing of NOT.
3. The NOT will be completed online through the EPA's NPDES eReporting Tool, NeT, found at the following link: https://usepa.servicenowservices.com/oeca_icis?id=net_homepage. When complete, the form will be assigned to a "Signatory" (PWO) to be certified before EPA review.
4. Filing the NOT terminates coverage under the General Permit and terminates the Contractor's responsibility to implement the SWPPP (**Note: requirements of the SWPPP, including periodic inspections, must be continued until the NOT is filed**).

2-3.2 Records Turn-Over

SWPPP BINDER – The 3 ring binder SWPPP book is the basis of the NSGL SWPPP program. The paperwork can only be signed by authorized and trained employees of the GC. All paperwork must be signed and dated. The key to the program is the timely tracking and documenting of the required records. At the end of the project, the SWPPP book is to be turned into the NSGL Environmental Department as part of the close out submittals.

SITE MAPS – The site map and working drawing needs to be kept up to date as the job progresses and changes to the SWPPP are made. The sequence of the SWPPP is always noted on the drawing and must be followed. Site map and working drawing shall be kept as part of the final record to be turned over to NSGL.

SWPPP FORMS – The SWPPP forms completed for the project, including inspections, delegations of authority, revision log, precipitation log, etc. are to be

kept with the SWPPP when not posted. Like the site maps, these documents will be retained as part of the final SWPPP book for project records.

Part III – NSGL Erosion and Sedimentation Control Standards

All erosion and sediment controls used on NSGL projects are to adhere to the Best Management Practice (BMP) standards specified in the current edition of the Illinois Urban Manual (IUM). The IUM can be found at the following link: <https://www.aiswcd.org/illinois-urban-manual/>. **Specific controls that will be used for the project should be referenced in the SWPPP.**

- **Erosion Control** – Erosion control measures are used to stop the dislodging and movement of sediment by water, air or human means. This can be through the use of ground cover, paving, concrete, structures, grass/sod, soil stabilization measures, mulch, or building structures.
- **Sediment Control** – Sediment control is a means of filtering or settling out the sediment prior to discharge into the waterways of the United States. This can be accomplished through the use of inline storm structures, silt fence, check dams, sediment ponds, inlet protection, walk-off mats, etc.

The following three sections (3-1, 3-2, & 3-3) provide NSGL minimum standards for structural, non-structural, and miscellaneous erosion and sediment control requirements.

3-1 Structural Controls

The following structural controls are a minimum requirement on most NSGL construction project. The IUM practice standards and details for these BMPs shall be used as a reference. While not all NSGL projects will require the use of other structural controls such as temporary sediment traps, basins, and stream crossings, all should strictly adhere to the practice standards and details in the IUM when employed.

Silt Fence (IUM Practice Standard 920)

A silt fence is designed to retain sediment-laden water and allow settlement of the suspended solids before the stormwater flows through the fabric for discharge downstream. They are good in low velocity sheet draining areas. Walk the perimeter silt fence looking for failures, tears, knock downs, sediment accumulation. If sediment accumulation is over 50% of the way up the silt fence it must be repaired.

Stabilized Construction Entrance (IUM Practice Standard 930)

Track out is not allowed on any NSGL site and should be closely monitored. The timely clean-up of streets is imperative. Do not use water to wash streets down into an unprotected storm sewer system as this is a violation of the SWPPP. The site contractor needs to have dry sweeping capabilities of the streets with equipment as part of their overall package throughout the duration of the project. The bigger the rock the more the sediment will vibrate off of the vehicles. The construction entrance needs to be replenished/replaced as it chokes full of sediment.

Inlet Protection (IUM Practice Standards 860 & 861)

There are many different types of inlet protection for paved and non-paved area inlets. The use of silt sacks and other fabrics are only as good as the maintenance program used by

the subcontractor. The use of hay bales is not recommended. If hay bales are proposed consult the design engineer for alternatives.

Temporary Concrete Washout Facility (IUM Practice Standard 954)

The concrete and masonry wash out/mix areas must be shown on the working as built drawing. The area must protect against spills outside the washout area. This area should, at a minimum, be ringed on the low side with silt fence or berm. The wash out area cannot be located in close proximity to storm structures where an inadvertent release could drain into the storm sewer system.

3-2 Non Structural Controls

Good Housekeeping

A clean site is a safe site. The porta johns cannot be located in close proximity to storm inlets where an inadvertent release could drain into the storm sewer system. Dumpsters must never be overflowing with trash and debris. On dusty jobsites, the use of water or approved chemical additives is required as dust control and is a requirement of the SWPPP.

Proper Material Storage

Avoidance of oversight on the part of subcontractors on this issue is imperative. The improper storage and disposal of chemicals, fuel, paints concrete washouts, and masonry washouts/mix areas can create serious clean up/ remediation issues for any project. The material storage areas need to be noted on the SWPPP working drawing along with a notation of the location of the spill cleanup materials/equipment (See Section 3-3 Material Management Plan for further direction).

Proper Location of Equipment Wash and Maintenance

The equipment washout and maintenance area must be shown on the working as built plan. It is best to keep the staging, maintenance and storage trailers in the same area. The area may require a berm or swale around the outside perimeter.

Paint Washout Prohibition

Absolutely no paint washout is allowed on a NSGL project. Paint contractors are to make provisions to clean and dispose of paint in accordance with state, local and federal guidelines.

3-3 Materials Management Plan

The following are the material management practices that will be used to reduce the risk of spills or other accidental exposure of materials and substances to stormwater runoff. The job site superintendent will be responsible for ensuring that these procedures are followed.

Good Housekeeping Practices

The following good housekeeping practices will be followed onsite during the construction project.

- An effort will be made to store only enough products required to do the job.

- All materials stored onsite will be stored in a neat, orderly manner and, if possible, under a roof or in a containment area. At a minimum, all containers will be stored with their lids on when not in use.
- Drip pans shall be provided under all dispensers.
- Products will be kept in their original containers with the original manufacturer's label in legible condition.
- Substances will not be mixed with one another unless recommended by the manufacturer.
- Whenever possible, all of a product will be used up before disposing of the container.
- Manufacturer's recommendations for proper use and disposal will be followed.
- The job site superintendent will be responsible for daily inspections to ensure proper use and disposal of materials.

Hazardous Products Practices

These practices will be used to reduce the risks associated with hazardous products:

- Safety Data Sheets (SDSs) for each substance with hazardous properties that is used on the job site will be obtained and used for the proper management of potential wastes that may result from these products.
- SDSs must be submitted for approval to the Contract Officer, prior to use on site.
- An SDS will be posted in the immediate area where such product is stored and/or used and another copy of each SDS will be maintained in the SWPPP file at the job site construction trailer office.
- Each employee who must handle a substance with hazardous properties will be instructed on the use of SDS sheets and the specific information in the applicable SDS for the product he/she is using, particularly regarding spill control techniques.
- Products will be kept in original containers with the original labels in legible condition.
- Original labels and SDS will be procured and used for each material.
- If surplus product must be disposed of, manufacturers or local/state/federal recommended methods for proper disposal will be followed.

Hazardous Waste Provisions

Store all hazardous and special waste in approved containers in accordance with all applicable Federal, State and Local regulations and statutes, Department of the Navy Instructions and environmental provisions contained in the contract documents rules regarding generation, storage, and disposal of hazardous waste can be found in the station's Hazardous Waste Management Plan. Contact the PWD Environmental Department for this information. All hazardous waste materials will be disposed of by the Contractor in the manner specified by local, State, and/or Federal regulations and by the manufacturer of such products. Site personnel will be instructed in these practices by the job site superintendent, who will also be responsible for seeing that these practices are followed.

Contaminated Soil Provisions

All spills or release of petroleum products, hazardous substances, regulated waste, regulated gases or sewage occurring on Navy property shall be reported immediately to:

Great Lakes Emergency Dispatch Center (HAZMAT, Fire, Police, Ambulance)	911 (847) 688-3333
NAVSTA GL PWD 24-Hour Service Desk	(847) 688-4820
NAVSTA GL Environmental Division	(847) 688-5395 x270, Matt Wollert

Any contaminated soils (resulting from spills of materials with hazardous properties) which may result from construction activities will be contained and cleaned up immediately in accordance with the procedures given in the Materials Management Plan and in accordance with all applicable Federal, State and Local regulations and statutes, Department of the Navy Instructions and environmental provisions contained in the contract documents

Material Specific Practices

The following material specific practices will be followed on the job site.

Petroleum Products

- All onsite vehicles will be monitored for leaks and receive regular preventative maintenance to reduce the chance of leakage.
- Petroleum products will be stored in tightly sealed containers which are clearly labeled.
- Any petroleum storage tanks stored onsite will be located within a containment area that is designed with an impervious surface between the tank and the ground. Secondary containment for petroleum storage tanks must follow NSGL SPCC requirements as applicable.
- Drip pans shall be provided for all dispensers.
- Any asphalt substances used onsite will be applied according to the manufacturer's recommendations.
- The location of any fuel tanks and/or equipment storage areas must be identified on the SWPPP working drawing by the contractor once the locations have been determined.
- Temporary fuel storage tanks shall not exceed 500 gallons total capacity.
- Temporary fuel storage tanks shall be double-walled and/or provided with secondary containment as approved by NSGL.

Fertilizers

- Fertilizers will be applied only in the minimum amounts recommended by the manufacturer. Once applied, fertilizer will be worked in the soil to limit exposure to stormwater.
- The contents of any partially used bags of fertilizer will be transferred to a sealable plastic bin to avoid spills.
- Note that products such as Weed & Feed are categorized as pesticide for purpose of material management.

Paints, Paint Solvents, and Cleaning Solvents

- All containers will be tightly sealed and stored when not in use. Follow all OSHA requirements for material storage, handling and disposal.
- Excess paint and solvents will not be discharged to the storm sewer system or water ways. Excess paint and solvents will be properly disposed of according to manufacturer's instructions or State and Federal regulations.
- **NSGL specifically prohibits paint wash out on any NSGL construction site.**

Concrete Wastes

- Concrete trucks will be allowed to wash out or discharge surplus concrete or drum wash water on the site, but only in either (1) specifically designated diked areas which have been prepared to prevent contact between the concrete and/or wash out and stormwater which will be discharged from the site or (2) in locations where waste concrete can be poured into forms to make rip rap or other useful concrete products.
- The hardened residue from the concrete wash out diked areas will be disposed of in the same manner as other non-hazardous construction waste materials or may be broken up and used on site as deemed appropriate by the contractor.
- The job site superintendent will be responsible for seeing that these procedures are followed. The project may require the use of multiple concrete wash out areas. All concrete wash out areas will be located in an area where the likelihood of the area contributing to stormwater discharges is negligible. If required, additional BMPs must be implemented to prevent concrete wastes from contributing to stormwater discharges.
- The location of concrete wash out area(s) must be identified on the SWPPP working drawing by the contractor once the locations have been determined.

Solid and Construction Wastes

- All waste materials will be collected and stored in an appropriately covered container and/or securely lidded metal dumpster rented from a local, licensed solid waste management company.
- Recycling is mandatory aboard NSGL. This includes, but is not limited to, construction and demolition debris, ferrous and non-ferrous metals, paper, and cardboard. **All recycling efforts will be coordinated with the Contracting Officer and NSGL Public Works Department.** Unless specifically stated otherwise in the Contract Documents, the Government has first right of refusal for the rights to recyclables generated by the project. If exercised, the Contracting Officers or their representative shall direct the disposition of such recyclables materials.
- Dumpsters will comply with all local and state solid waste management regulations.
- All trash and construction debris from the site will be deposited in the dumpster.
- Dumpsters will be emptied a minimum of twice per week or more often if necessary, and the trash will be hauled to an approved landfill. No construction waste materials will be buried on site.
- All personnel will be instructed regarding the correct procedures for waste

disposal. All waste dumpsters and roll-off containers will be located in an area where the likelihood of the containers contributing to stormwater discharges is negligible. If required, additional BMPs must be implemented, such as sandbags around the base, to prevent wastes from contributing to stormwater discharges. The location of waste dumpsters and roll-off containers must be identified on the SWPPP working drawing by the contractor once the locations have been determined.

Sanitary Wastes

- All sanitary waste will be collected from portable units at a minimum of three times per week by a licensed portable facility provider in complete compliance with all applicable Federal, State and Local regulations and statutes, Department of the Navy Instructions and environmental provisions contained in the contract documents.
- All sanitary waste units will be located in an area where the likelihood of the unit contributing to stormwater discharges is negligible. If required, additional BMPs must be implemented, such as sandbags around the base, to prevent wastes from contributing to stormwater discharges.
- The location of sanitary waste units must be identified on the SWPPP working drawing by the contractor once the locations have been determined.

3-4 Other Stormwater Pollution Prevention Requirements

Stabilization

As a minimum, any disturbed area that will not have work on going within 14 days must be stabilized in accordance with the SWPPP. This includes topsoil piles. This is one of the most critical ways to maintain a jobsite and minimize the impact of rain and wind. As areas are stabilized, they need to be documented on the jobsite stabilization form. The use of blankets, fabric, hydro seed, mulch and other techniques can improve your jobsite maintenance program.

Historical and Natural Resources

The nature of new construction in general leads the Navy into the demolition of old buildings and natural flora. The Navy, or their designated contractors cannot begin any new construction project without first coordinating Historical Preservation Requirements with the NSGL Cultural Resources Manager (CRM) and Natural Resources Protection Requirements with the NSGL Natural Resources Manager (NRM). Construction projects near threatened and endangered species and their habitat may require consultation with the U.S. Fish and Wildlife Service (USFWS) and Illinois Department of Natural Resources (IDNR). Contractor must avoid impact to wetland areas. Work near wetlands and waters of the U.S. may require permitting through the U.S. Army Corps of Engineers (USACE) and Illinois Environmental Protection Agency (IEPA). Staging areas must be at minimum 200 ft. from wetlands and waters of the U.S. Locations of protected species, wetlands, and natural areas can be found in the NSGL Integrated Natural Resources Management Plan.

IEPA will be the agency to review projects for meeting requirements of Illinois Environmental laws through the NOI. IEPA will request a review from the Illinois Historic Preservation Agency (IHPA) and Illinois Department of Natural Resources (IDNR) for projects indicating on the NOI the review has not been completed. No contractor shall communicate directly with regulatory agencies. All communication requests shall be submitted to the Navy, and then routed to the proper state agency by the appropriate Navy representative.

Control of non-Stormwater Discharges

Certain types of discharges are allowable under the IEPA Construction General Permit for Construction Activity, and it is the intent of NSGL to allow such discharges. These types of discharges will be allowed under the conditions that no pollutants will be allowed to come in contact with the water prior to or after its discharge. The control measures which have been outlined previously in the SWPPP will be strictly followed to ensure that no contamination of these non-stormwater discharges takes place.

Discharge Point

A discharge point is the last resort location to observe how well the SWPPP/BMPs are working. Sediment laden water at the outfall discharge point should not be present from a NSGL jobsite. This area usually has rip rap to slow the movement of water and allow sediment to fall out. The outfall may need to be cleaned out from time to time.

Part IV – NSGL Water Resources

Naval Station Great Lakes has several water resources and sensitive areas that need to be taken into consideration for construction stormwater management practices. These include:

- Lake Michigan,
- Pettibone Creek (including South & West Branches of Pettibone Creek), and
- Skokie Creek.

A station map showing the drainage areas for these water resources is included with Appendix C.

The Pettibone Creek, including the South & West Branches of Pettibone Creek are tributary to Lake Michigan. Construction activities whose stormwater discharges to the Pettibone Creek should be listed as such.

The Skokie River is tributary to the North Chicago River and therefore the greater Mississippi drainage basin as a result of the Chicago Sanitary & Shipping Canal Project.

4.1 Additional Requirements for Construction in or near Station Water Bodies (Pettibone Creek, Skokie River, and Lake Michigan)

If activities are to construct, operate or maintain any dam, dock, pier, wharf, sluice, levee, dike, building, utility and road crossings, piling, wall, fence or other structure which dredge, fill or otherwise alter the bed or banks of any stream, lake, wetland, floodplain or floodway subject to State or Federal regulatory jurisdiction, then a Joint Application is required. The Joint Application is a simplified approval process for an applicant seeking project authorizations from the U.S. Army Corps of Engineers (USACE), The Illinois Department of Natural Resources/Office of Water Resources (IDNR/OWR) and the Illinois Environmental Protection Agency (IEPA).

Additionally, any work within Illinois' Coastal Zone (east of Green Bay Rd.) may require consultation with IDNR, pursuant to the Coastal Zone Management Act, to assess potential impacts to coastal uses and resources. Work cannot begin until project is coordinated with the NSGL NRM and concurrence is received from IDNR.

4-2 Total Maximum Daily Load (TMDL) for Station Water Resources

The IEPA has prepared a list of impaired waters to fulfill the requirements set forth in Section 303(d) (<https://www2.illinois.gov/epa/topics/water-quality/watershed-management/tmdls/Pages/303d-list.aspx>) of the federal CWA and the Water Quality Planning and Management regulation of 40 CFR Part 130. There are currently no TMDLs planned for NSGL receiving waters cited above according to the Illinois 303(d) list dated 2020/2022.

SWPPP FORMS

1. General Contractor Certification (NSGL - Attached)
2. Construction Site SWPPP Notice (NSGL - Attached)
3. 2022 CGP Site Inspection Report (EPA - Found on the EPA website)
4. 2022 CGP Corrective Action Log (EPA - Found on the EPA website)
5. SWPPP Template (EPA - Found on the EPA website)

All EPA SWPPP forms can be downloaded on the EPA webpage for Construction General Permit Resources, Tools, and Templates:

<https://www.epa.gov/npdes/construction-general-permit-resources-tools-and-templates>

A screenshot of the EPA website with the appropriate documents highlighted is included as an attachment.

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Naval Station Great Lakes – Form 1
GENERAL CONTRACTOR CERTIFICATION

Project Information	
Project Name: _____	
Location: _____	
Contract No: _____	NPDES No. _____
General Contractor: _____	Contact information: _____

“I certify under penalty of law that I understand the terms and conditions of the general National Pollution Discharge Elimination System (NPDES) permit (ILR10) that authorizes the stormwater discharges associated with industrial activity from the construction site identified as part of this certification”

“Further, by my signature, I understand that I am fully responsible, along with all other contractors and subcontractors signing such certifications who are performing work activities under this contract, to comply with all provisions and requirements of the NPDES stormwater permit and this Stormwater Pollution Prevention Plan (SWPPP). I understand that I, and my company, are legally required under the Clean Water Act, to ensure compliance with the terms and conditions of the NPDES stormwater permit and SWPPP developed under the NPDES stormwater permit as it pertains to the scope of my services. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment for knowing violations.”

Name: _____

Company Name: _____

Signature: _____

Address: _____

Title: _____

Telephone Number: _____

Date: _____

Scope of Services:

--

General Contractor Project Manager Signature:

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Naval Station Great Lakes – Form 2
CONSTRUCTION SITE SWPPP NOTICE

STORMWATER POLLUTION PREVENTION PLAN (SWPPP)	
Project Name:	Date:
Contractor Firm:	
Contractor Address:	
Contact Name:	Contact No:
Project Description:	Disturbed Area (Acres):
SWPPP Viewing Location:	SWPPP Viewing Time:

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National Pollutant Discharge Elimination System (NPDES)

CONTACT US

- NPDES Home
- About NPDES
- All NPDES Program Areas
- Animal Feeding Operations
- Aquaculture
- Forest Roads
- Industrial Wastewater
- Municipal Wastewater
- National Pretreatment Program
- Pesticide Permitting
- Stormwater
- Water Quality Trading
- NPDES Technical Resources

Construction General Permit Resources, Tools, and Templates

On this page:

- [Tools and Maps](#)
- [SWPPP Templates](#)
- [Determine 2-Year, 24-Hour Storm Event for Sediment Basin Design](#)
- [Inspection and Corrective Action Report Templates and Other Guidance](#)
- [Monitoring and Inspection Guide for Construction Dewatering](#)
- [Professional Resources](#)

Construction Stormwater Permitting Topics

- [Overview](#)
- [2022 CGP](#)
- [Threatened and Endangered Species](#)
- [Turbidity Benchmark Monitoring \(Dewatering\)](#)
- [Getting Permit Coverage / Net CGP Waivers](#)
- [Submitting CGP Forms](#)
- [CGP Inspector Training](#)
- [Resources, Tools, and Templates](#)
- [Report Non-Compliance or Violations](#)

EPA's Permit Mapping Tools

These resources help you determine whether you discharge to an impaired water and if your construction activities will occur during a seasonally dry period in an arid or semi-arid climate.

- [Discharge Mapping Tool](#)
Helps identify whether any receiving waters to which you discharge are listed as impaired (and the pollutant for which it is impaired) and whether an approved total maximum daily load (TMDL) exists for that waterbody.
- [Seasonally Dry Period for Arid and Semi-Arid Areas](#)
 - [Seasonally Dry Period Locator Tool \(xlsx\)](#) (3.76 MB) - This tool helps you determine if your site is in an arid or semi-arid climate and the seasonally dry months for your site location. To use the tool, download the Excel file and enter your site's zip code. The tool will return text indicating if your site is located in an arid or semi-arid climate and, if applicable, the seasonally dry months for your site.
 - [Seasonally Dry Period Locator Tool-Supporting Figures \(pdf\)](#) (1.08 MB) - These maps show areas of the U.S. that have an arid or semi-arid climate and the months that those areas are considered seasonally dry.

SWPPP Templates

How to develop a Stormwater Pollution Prevention Plan (SWPPP)

These resources help operators develop effective SWPPPs that are required under construction stormwater permits. Note that construction site SWPPPs must follow the requirements of the applicable State or EPA CGP and must describe the specific conditions of the site and plans for development.

SWPPP Form 5

- [Construction Stormwater Pollution Prevention Plan \(SWPPP\) Template \(docx\)](#) (327.22 KB) - Helps construction site operators develop a SWPPP that is compliant with the minimum requirements of EPA's 2022 CGP.
- [Plantilla del Plan de Prevención de la Contaminación de Aguas Pluviales en la Construcción \(PPCAP\) \(docx\)](#) (309.33 KB) - Ayuda a los operadores de proyectos de construcción a desarrollar un PPCAP que cumpla con los requisitos mínimos del Permiso General de Construcción 2017 de la EPA.
- [Small Residential Lot SWPPP Template \(docx\)](#) (9.14 MB) - Helps small residential lot builders develop a streamlined SWPPP that is compliant with the minimum requirements of

- [Stormwater Pollution Prevention for Small Residential Construction Sites \(pdf\)](#) (3.42 MB) - An educational brochure that provides basic erosion and sediment control and stormwater pollution prevention guidance common to all small residential lot projects.

Determine 2-Year, 24-Hour Storm Event for Sediment Basin Design

Part 2.2.12 of EPA's 2022 CGP indicates that if you install a sediment basin, one of the two alternative design requirements is to provide storage for the calculated volume of runoff from a 2-year, 24-hour storm.

- [2022 CGP - 2-Year, 24-Hour Storm Frequencies \(pdf\)](#) (382.32 KB) - EPA compiled this information to help determine the volume of precipitation associated with the local 2-year, 24-hour storm event.

Inspection and Corrective Action Report Templates and Other Guidance

EPA developed the following templates to help you document your findings for site inspections, dewatering inspections, and corrective actions as required by the 2022 CGP. These templates are formatted to be filled out electronically or manually and include text fields that direct you to populate the forms with your site-specific information.

SWPPP Form 3

• [Site Inspection Report Template \(docx\)](#) (91.85 KB) - This inspection report template is designed to assist you in preparing inspection reports that meet the requirements in Part 4.7 of the 2022 CGP.

- [Dewatering Inspection Report Template \(docx\)](#) (64.37 KB) - This dewatering inspection report template is designed to assist you in preparing inspection reports related to dewatering activities required by Part 4.6.3 of the 2022 CGP.

SWPPP Form 4

• [Corrective Action Log Template \(docx\)](#) (53.29 KB) - This corrective action log form is designed to assist you in recording the required corrective action documentation that meet the requirements in Part 5.4 of the 2022 CGP.

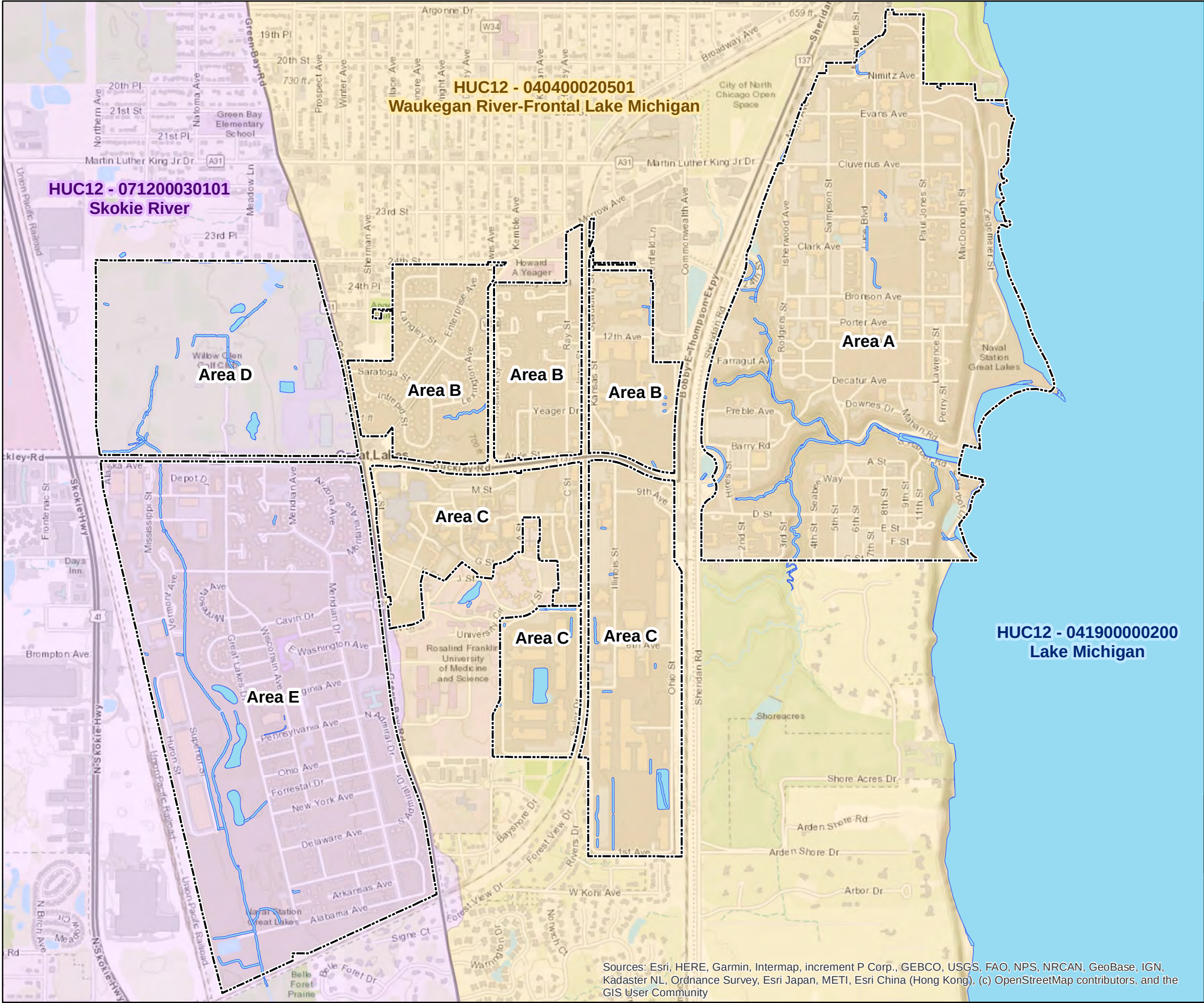
- [Routine Maintenance/Corrective Action Determination Guidelines \(pdf\)](#) (139.56 KB) - This guidance will assist you in determining whether problems found on the construction site related to stormwater controls require routine maintenance or corrective action.
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APPENDIX B

NSGL Watershed Boundary and Drainage Areas Map

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Watershed Boundary and Drainage Areas Map

Naval Station Great Lakes (N00128)

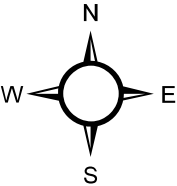
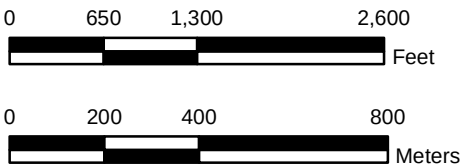


ENVIRONMENTAL BUSINESS LINE, NAVY REGION MID-ATLANTIC

Legend

- Installation Boundary
- Water Body
- HUC 12 Waukegan River - Frontal Lake Michigan Watershed
- HUC 12 Skokie River Watershed

Coordinate System: NAD_1983_StatePlane_Illinois_East_FIPS_1201_Feet
Projection: Transverse_Mercator
Datum: D_North_American_1983



Sources: Esri, HERE, Garmin, Intermap, increment P Corp., GEBCO, USGS, FAO, NPS, NRCAN, GeoBase, IGN, Kadaster NL, Ordnance Survey, Esri Japan, METI, Esri China (Hong Kong), (c) OpenStreetMap contributors, and the GIS User Community



REFRIGERANT COMPLIANCE PLAN NAVAL STATION GREAT LAKES

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TABLE OF CONTENTS

1	INTRODUCTION.....	1
2	NSGL COMPLIANCE SUMMARY	1
2.1	COMPLIANCE STRATEGY.....	1
2.1.1	General Compliance.....	1
2.1.2	Equipment Specific Compliance	1
2.1.3	Compliance Tools.....	2
3	BACKGROUND.....	2
3.1	REFRIGERANT BACKGROUND.....	2
3.2	ENVIRONMENTAL IMPACT	2
3.3	REFRIGERANT CLASSIFICATION	3
4	VENTING PROHIBITION	4
4.1	NON-COMPLIANCE PENALTIES.....	4
4.2	EXAMPLES OF EPA ENFORCEMENT ACTIONS.....	5
4.2.1	February 19, 2019: U.S. Settles with Trident Seafoods.....	5
4.2.2	August 23, 2019: U.S. Settles with Southeastern Grocers.....	6
4.2.3	May 6, 2015: Navy Settles Environmental Violations at Naval Station Norfolk.....	6
5	PREVENTIVE ACTION TIMELINE	7
6	REGULATORY REQUIREMENTS	7
6.1	APPLICABILITY.....	7
6.2	TECHNICIAN CERTIFICATION	8
6.3	VENTING PROHIBITIONS.....	8
6.4	EVACUTATION	8
6.5	EXCEPTIONS – EVACUATION REQUIREMENTS	9
6.6	RECLAMATION.....	9
6.7	SAFE DISPOSAL	10
6.8	RECOVERY & RECYCLING EQUIPMENT	10
6.9	SALES RESTRICTIONS	10
6.10	CHANGING REFRIGERANT OIL	10
6.11	REFRIGERATION LEAK RATES	10
6.11.1	Leak Rate Threshold.....	10
6.11.2	Leak Rate Calculation.....	11
6.12	REPAIR, RETROFIT OR RETIRE?	11
6.12.1	Repair Verification.....	12
6.12.2	Initial Verification Test (IVT).....	12
6.12.3	Follow-up Verification Test (FVT).....	12
6.12.4	Verification Test Methods	13
6.13	RETROFIT / RETIREMENT PLAN REQUIREMENTS.....	13
6.13.1	Retrofit / Retirement Plan – Required Information.....	13
6.13.2	Relief From Retrofit / Retirement Requirements.....	13
6.13.3	Retrofit / Retirement Plan Extensions.....	14
6.14	SYSTEM MOTHBALLING	14
7	REFRIGERANT RECOVERY & RECYCLING EQUIPMENT	15

8	DISPOSAL REQUIREMENTS	15
8.1	Disposal of Equipment.....	15
8.2	Refrigerant Contamination and Disposal	15
9	HANDLING OF REFRIGERANT AT ON-SITE STORAGE LOCATIONS.....	16
9.1	Filling Refrigerant Cylinders	16
9.2	Emission Reduction	16
10	RECORD KEEPING.....	16
10.1	Technician Certifications.....	17
10.2	Equipment Inventory	17
10.3	Responding to (Service) Work Orders	17
10.3.1	CAA 40-82F-1 Service Invoice for 50+ # Units	17
10.3.2	CAA 40-82F-2 Service and Inventory Tracking – All Appliances	18
10.3.3	CAA 40-82F-3 Disposal Invoice – All Appliances.....	18
10.4	Refrigerant Tracking.....	18
11	ROLES & RESPONSIBILITIES	18
11.1	HVAC SUPERVISOR / SME	18
11.2	ENVIRONMENTAL PROGRAM MANAGER.....	19
11.3	FACILITIES MANAGEMENT DIVISION	19
11.4	TENANT COMMANDS	20
12	ADDITIONAL REQUIREMENTS.....	20
12.1	LABELING	20
12.2	SITE CONTACTS	20
12.3	REVIEW FREQUENCY.....	21
12.4	REFERENCES.....	21

LIST OF TABLES

Table 3.1	Key Definitions of Environmental Impacts of Refrigerants.....	2
Table 3.2	Refrigerant Classification Criteria	3
Table 3.3	Types of Refrigerants used in Stationary Equipment & Air Conditioning.....	4
Table 3.4	U.S. Action to Meet the Montreal Protocol Phase-out Schedule	4
Table 6.1	US EPA Technician Certification Types.....	8
Table 6.2	Required Level of Evacuation of Appliances	9
Table 6.3	Applicable Trigger Rates (Effective 1 January 2019).....	11
Table 6.4	Rolling Average Method for Determining Leak Rates.....	11
Table 12.1	Contacts	20

LIST OF APPENDICES

Appendix A. NAVFAC Mid-Lant process # ML-B-15.2.2 Refrigerant Compliance Management

Appendix B. "CAA 40-82F" Forms

Appendix C. Retrofit / Retirement Plan Form

Appendix D. Technician Certification Levels

Appendix E. Class I Refrigerant Phase Out List

Appendix F. Class II Refrigerant Phase Out List

Appendix G. Regulatory Timeline & Checklist for Retrofit & Retirement Activities

Appendix H. Decision guide for Leaking Refrigeration Systems

Appendix I. Glossary

Appendix J. List of Commonly Used Acronyms & Abbreviations

1 INTRODUCTION

Naval Station Great Lakes (NSGL) uses stationary refrigeration and air-conditioning equipment to support its mission of national defense. The equipment at NSGL is subject to United States Environmental Protection Agency (EPA) regulation for the management of Ozone Depleting Substances (ODS), Section 608 of the Clean Air Act, that went into effect in 1990, and was subsequently updated in 2016 and 2020.

The NSGL Public Works Department – Environmental (PWD EV) oversees the ODS and refrigerant management program in accordance with 40 Code of Federal Regulations (CFR) Part 82 Protection of Stratospheric Ozone.

The purpose of this Refrigerant Compliance Plan is intended to assist NSGL owners and operators of stationary heating, ventilation, air-conditioning and refrigeration (HVAC&R) equipment to maintain compliance with applicable regulatory requirements and reduce the environmental impact of operating such equipment.

This document provides some review on refrigerants, chlorofluorocarbons, hydrofluorocarbons and Federal rules & regulations to ensure all users have enough background to implement the compliance and operating procedures discussed, and to be aware of their importance and need.

2 NSGL COMPLIANCE SUMMARY

2.1 COMPLIANCE STRATEGY

To ensure compliance with federal, state and Navy environmental policy, stationary refrigeration and air-conditioning technicians (HVAC Technicians) must follow the guidelines for all equipment regardless of classification. This includes Class I and II ODS refrigerants as well as non-DOS refrigerants (substitute refrigerants).

2.1.1 General Compliance

The following eight actions apply to all equipment regardless of refrigerant charge or classification, with details in the referenced sections:

1. Require all technicians to be EPA-certified. See section 6.2.
2. Follow the venting prohibition requirements. See section 6.3.
3. Follow refrigerant evacuation requirements. See section 6.4.
4. Follow reclamation requirements. See section 6.6.
5. Follow safe disposal requirements. See section 6.7.
6. Use only certified recovery and recycling equipment. See section 6.8.
7. Follow sales restrictions. See section 6.9.
8. Maintain records of all refrigerant transactions such as purchase, sale, disposal, recovery, reclamation and recycling. See section 10.

Per the Navy's Environmental Readiness Program Manual (OPNAVINST 5090.1E), all new shore-based, non-critical HVAC&R equipment must use an EPA-SNAP-program-approved refrigerant with an ozone depletion potential (ODP) of zero. Furthermore, new HVAC&R equipment (both mission-critical and non-mission-critical) must not contain Class II ODS refrigerants.

2.1.2 Equipment Specific Compliance

In addition to the General Compliance Actions outlined above, the following actions apply to comfort cooling equipment with 50 pounds or more (50+#) of any refrigerant, including Class I ODS, Class II ODS, and substitutes.

Comfort Cooling Equipment must complete:

1. **Leak Repair Requirements** – See section 6.12, including:
 - a. Repair leaks within 30 days
 - b. Initial and follow-up verification tests
 - c. Keep and complete retrofit and retirement plans
2. **Leak Rate Calculation** - See section 6.11.2.
3. **System Mothballing** - See section 6.14.
4. **Recordkeeping** - See section 10.

FMD is encouraged to develop a plan of specific compliance actions required for each individual refrigerant containing system at NSGL. Personnel could use these plans to identify all required compliance actions for impacted units during the transition to zero ODP refrigerant use.

2.1.3 Compliance Tools

To outline operational and compliance recordkeeping requirements, Navy Region Mid-Lant issued BMS number 15.2.2 Refrigeration Compliance Management, and 3 forms to document compliance with 40 CFR 82F. These are included in Appendices A & B.

3 BACKGROUND

3.1 REFRIGERANT BACKGROUND

A refrigerant is the fluid used for heat transfer in a refrigeration system. The refrigerant absorbs heat during evaporation at low temperature and pressure, and releases heat during condensation at a higher temperature and pressure. Refrigerants can absorb heat by undergoing a change of state (for example from liquid to vapor), or they can absorb heat without undergoing a change of state (for example they may absorb heat as a liquid and remain a liquid, albeit a hotter one).

Fluorocarbon compounds have a demonstrated ability to effectively absorb/remove heat and thus they have commonly been used as refrigerants. It has also been determined that fluorocarbon compounds that contain chlorine are harmful to the stratospheric ozone layer and thus they are also known as ODSs.

3.2 ENVIRONMENTAL IMPACT

The CAA prohibits intentional release of ODS and substitute refrigerants during maintenance, repair or disposal of appliances and equipment containing regulated refrigerants. The EPA has identified two types of environmental harm from the release of refrigerants into the atmosphere: ozone depletion and global warming. These environmental impacts are measured for each refrigerant using ozone depleting potential (ODP) and global warming potential (GWP), as explained in Table 3.1.

Table 3.1 Key Definitions of Environmental Impacts of Refrigerants

Environmental Characteristic	Definition	Basis of Numerical Assignment
Ozone Depletion Potential (ODP)	Relative amount of degradation to the stratospheric ozone layer compared to R-11; which has a fixed value of 1.0	ODP (R-11) = 1.0
Global Warming Potential (GWP)	The measure of how much a given mass of greenhouse gas contributes to global warming relative to carbon dioxide (CO ₂)	GWP (CO ₂) = 1.0

Relevant definitions are provided in Appendix J and further federal regulatory background on ODS phaseout is provided in Appendices E & F.

3.3 REFRIGERANT CLASSIFICATION

The most common refrigerants are chlorofluorocarbons (CFCs), hydrochlorofluorocarbons (HCFCs), hydrofluorocarbons (HFCs) and perfluorocarbons (PFCs). Refrigerants are also classified as Class I, Class II, or non-ODS refrigerants, as summarized in Table 3-2 below. Table 3-3 lists common refrigerants, including the classification, OWP, and GWP of each refrigerant. Refrigerants shown in blue are currently used at NSGL.

Table 3.2 Refrigerant Classification Criteria

Type of ODS	Refrigerants in this Category	Notes
Class I	CFCs (R-11)	Higher ozone depletion potential. Production prohibited after 1 January 2000.
Class II	HCFCs (R-22, R-123)	Transitional substances for Class I ODS. Reductions in production by 2010 and production prohibition in 2030 for R-123 and R-124. Ban on production and import of R-22 in 2020
Non-ODS	HFCs (R-134a, R-404a, R-410a, R-407a, R-507a)	No Ozone Depletion Potential; high GWP.

Class I ODSs include chlorinated solvents (carbon tetrachloride and methyl chloroform), compounds that contain fluorine, chlorine and carbon (CFCs), methyl bromide, and halons (compounds consisting of bromine, fluorine and carbon).

Class II ODSs are compounds consisting of hydrogen, chlorine, fluorine, and carbon and are known as HCFCs. Both classes have been rated based on their ozone depletion potential (ODP). Class I ODSs have ODPs of 0.2 or higher. Class II substances have an ODP of less than 0.2. Within the Class I substances halons have the highest ODPs because they contain bromine. Bromine is many times more effective at destroying ozone than chlorine.

HCFCs are one class of chemicals that have been used to replace the Class I CFCs. Since they do contain some chlorine, they do deplete stratospheric ozone, but to a much lesser extent than CFCs. Thus, their phase out is on a much longer timeline than what was required for Class I ODSs. Another class of fluorocarbon compounds that do not contain chlorine, and thus don't destroy ozone (ODP is equal to zero), is hydrofluorocarbons (HFCs). This makes them a viable replacement for both CFC and HCFCs.

Perfluorocarbon (PFC), a compound consisting of carbon and fluorine, also does not deplete ozone but has the disadvantage of being a very significant contributor to global warming. HCFCs and HFCs also contribute to global warming to varying extents. Also note, even though phase out requirements don't apply to HFC and PFC refrigerant/ODS substitutes, recent changes to EPA rules that regulate the use of ODSs have extended some of the provisions to HFCs and PFCs.

A list of **exempt substitutes** is below. All other refrigerants are covered by this compliance plan.

- Carbon dioxide in any application
- Nitrogen in any application
- Water in any application
- Ammonia in commercial or industrial process refrigeration or in absorption units
- Chlorine in industrial process refrigeration
- Hydrocarbons in industrial process refrigeration
- Ethane (R-170) in very low temperature refrigeration equipment and equipment for non-mechanical heat

transfer

- Propane (R-290) in certain units
- Isobutane (R600a) in certain units
- R-441A in certain units

Table 3.3 Types of Refrigerants used in Stationary Equipment & Air Conditioning*

Refrigerant Name	Refrigerant Classification	Class I, Class II, or Non-ODS	ODP	GWP	Common Uses
R-22	HCFC	Class II	0.055	1,810	A/C, Heat Pumps
R-134a	HFC	Non-ODS	0	1,430	A/C, Heat Pumps
R-404a	HFC	Non-ODS	0	3,900	Commercial Refrigeration
R-410a	HFC	Non-ODS	0	2,088	Industrial A/C
R-123	HCFC	Class II	0.02	77	Industrial A/C
R-123a	HCFC	Class II	0.02	77	Industrial A/C
R-407a	HFC	Non-ODS	0	2,107	Refrigeration
R-507a	HFC	Non-ODS	0	3,985	Commercial Refrigeration

*Note: Refrigerants in Table 3.3 highlighted in blue text are currently in use at NSGL.

In addition to the EPA rules that govern the use of ODSs, the United States follows the international mandate (known as the Montreal Protocol) to phase out the production of both Class I and Class II ODSs. For Class I ODSs the production phase out in the U.S. is already complete. The phase out of Class II ODSs will occur in stages up through 2030.

Table 3.4 U.S. Action to Meet the Montreal Protocol Phase-out Schedule

Year to Be Implemented	Implementation of HCFC Phase-out through Clean Air Act Regulations	% Reduction in HCFC Consumption & Production from Baseline
2003	No production or import of HCFC-141b	35.0%
2010	No production or import of HCFC-142b and HCFC-22, except for use in equipment manufactured before January 1, 2010	75.0%
2015	No production or import of any other HCFCs, except as refrigerants in equipment manufactured before January 1, 2020	90.0%
2020	No production or import of HCFC-142b and HCFC-22	99.5%
2030	No production or import of any HCFCs	100.0%

4 VENTING PROHIBITION

4.1 NON-COMPLIANCE PENALTIES

The enforcement tools available to the EPA to deal with ODS/refrigerant rule violations include civil injunctions, civil penalties, imprisonment, and criminal fines. Failure to comply can result in fines up to \$57,617 per day/per violation (as of 2024). Intentional violations can result in criminal penalties of up to five years imprisonment. Submitting false or misleading information or failure to submit required records also could incur criminal penalties, including prison terms of up to two years. To determine noncompliance the EPA conducts surprise inspections.

When assessing penalties the seriousness of the violation plays an important role. The seriousness of a specific violation is determined by examining the potential environmental harm, the risk of (or actual) refrigerant loss, the extent of deviation from the regulations, and the duration of the deviation. Also, factors not directly related to the violation can affect the penalty assessment. These include good faith efforts to comply, previous violation history, and payment of any previously assessed penalties for earlier violations.

In addition, there are several factors the EPA may consider that may lead them to reduce or eliminate penalties. These include, but are not limited to, the following:

- The voluntary and complete disclosure by the violator of such violation in a timely fashion after discovery of the non-compliance;
- Full and prompt cooperation by the violator following disclosure of the violation including, when appropriate, entering into a legally enforceable commitment to undertake compliance and remedial actions;
- The existence and scope of a comprehensive environmental compliance program.

In terms of the potential of a violation to do serious harm the EPA generally views non-compliance in terms of the following schemes:

Major (substantial adverse effect)

- Knowingly venting a regulated substance.
- Not using recycling /recovery equipment.
- Not repairing leaks (for equipment 50 lbs. charge or over).
- Accepting signed statement pursuant to 40 CFR 82.156(f)(2) if the person knew or had reason to know that such a signed statement is false.
- Failure to follow required practices in 40 CFR 82.156.

Moderate (significant adverse effect)

- Technicians not properly trained and certified.
- Recovery/Recycling equipment not properly maintained/does not pull specified vacuum.
- Not using equipment certified for the type of appliance.
- Altering design of certified refrigerant recycling or recovery equipment
- Sale and distribution of refrigerants to persons who are not certified technicians.

Minor (minor adverse effect)

- Recordkeeping requirements not properly followed.
- Training certificate not available on request.
- Sale of un-reclaimed refrigerant.
- Sale of refrigerant reclaimed by uncertified re-claimer.
- Failure of owner or re-claimer to certify.

4.2 EXAMPLES OF EPA ENFORCEMENT ACTIONS

4.2.1 February 19, 2019: U.S. Settles with Trident Seafoods

Trident Seafoods Corporation has agreed to reduce emissions of ozone-depleting substances from refrigeration equipment on its vessels, under a proposed settlement with the U.S. Environmental Protection Agency and U.S. Department of Justice to resolve alleged violations of the Clean Air Act. Under the settlement, Trident will spend up to \$23 million to reduce coolant leaks from refrigerators and other equipment, use alternative refrigerants, and improve company-wide compliance. The company will also pay a \$900,000 civil penalty.

Trident is one of the largest seafood processing companies in Alaska and the Pacific Northwest. Within its numerous fish catching and processing vessels, Trident uses thousands of pounds of ozone depleting substances

as refrigerants. EPA alleged that Trident violated the Clean Air Act by failing to promptly repair leaks of the refrigerant R-22, an ozone-depleting hydrochlorofluorocarbon (HCFC). Trident's failures allowed its appliances to leak refrigerant at high rates for thousands of days, causing over 200,000 pounds of harmful refrigerant to be released into the atmosphere.

Trident will retrofit or retire 23 refrigeration appliances used on 14 marine vessels to use an alternative refrigerant that does not harm the ozone layer compared to typical refrigerants. Trident agreed to retrofit nine of these appliances as part of a Supplemental Environmental Project. Because of these retrofits, nearly 100,000 pounds of harmful refrigerant will be removed from use, and future leaks will not damage the ozone layer.

Trident will also conduct routine leak inspections of all appliances, promptly repair leaks, install leak detectors to monitor appliances for leaks, add fluorescent dye into appliances to assist staff in detecting leaks, compile information to assist in identifying common failure points on appliances, and train employees to properly manage the appliances. In addition, the settlement sets a corporate-wide refrigerant leak cap and requires Trident to retain a 3rd-party auditor to review the company's compliance with the consent decree and regulations.

4.2.2 August 23, 2019: U.S. Settles with Southeastern Grocers.

Southeastern Grocers Inc. and its subsidiaries BI-LO LLC and Winn-Dixie Stores Inc. (together, "SEG"), owners and operators of regional grocery store chains BI-LO LLC, Winn-Dixie Stores Inc., Fresco y Más and Harveys Supermarket, have agreed to reduce emissions of potent ozone depleting gases from refrigeration equipment at 576 stores under a proposed settlement with the U.S. Department of Justice and the U.S. Environmental Protection Agency to resolve alleged violations of the Clean Air Act. Under the settlement, SEG will spend an estimated \$4.2 million over the next three years to reduce coolant leaks from refrigerators and other equipment and improve company-wide compliance. SEG will also pay a \$300,000 civil penalty.

The United States alleged that SEG violated the Clean Air Act by failing to promptly repair leaks of class I and class II refrigerants, ozone-depleting substances used as coolants in refrigerators. SEG also failed to keep adequate servicing records of its refrigeration equipment and failed to provide information about its compliance record.

SEG will now implement a corporate refrigerant compliance management system to comply with federal stratospheric ozone regulations and to detect and repair leaks through a new bi-monthly leak monitoring program. In addition, SEG will achieve and maintain an annual corporate-wide average leak rate of 17.0 percent through 2022, well below the grocery store sector average of 25 percent. SEG must also use non-ozone depleting advanced refrigerants at all new stores, and an additional 15 existing, non-advanced refrigerant stores.

4.2.3 May 6, 2015: Navy Settles Environmental Violations at Naval Station Norfolk

A settlement between the U.S. Environmental Protection Agency and U.S. Navy will help reduce potentially harmful discharges of ozone-depleting substances and ensure the safe storage and disposal of hazardous waste at the Naval Station Norfolk in Virginia, where EPA alleges environmental violations occurred.

Under the settlement, the Navy paid an \$83,900 penalty for violations of the Clean Air Act and the Resource Conservation and Recovery Act (RCRA). The Clean Air Act violations pertained to regulations designed to reduce discharges of ozone-depleting substances used as coolants in air conditioning units. EPA alleged that the facility did not perform leak rate calculations when it serviced the units. The Navy has implemented improved training and recordkeeping to help ensure proper servicing of equipment. In addition, EPA alleged that the facility violated emissions limits on an industrial device used for abrasive blasting and also had recordkeeping violations. The Navy has addressed both issues. The RCRA violations pertained to regulations that require the safe, environmentally-sound storage and disposal of hazardous waste. The Navy took prompt action to address the hazardous waste violations, which will help ensure that those wastes are managed properly.

5 PREVENTIVE ACTION TIMELINE

NSGL will institute a preventative maintenance program that requires periodic leak testing to ensure refrigerant system integrity. Each affected work area shall develop standard operating procedures (SOPs) for conducting leak testing relative to their equipment and refrigerant systems maintained. Leak tests shall:

- For commercial refrigeration and industrial process refrigeration appliances with a full charge of ≥ 500 pounds, leak inspections must be conducted once every three months until the owner or operator can demonstrate through the leak rate calculations that the appliance has not leaked in excess of the applicable trigger rate for four quarters in a row.
- For commercial refrigeration and industrial process refrigeration appliances with a full charge of 50-499 pounds, leak inspections must be conducted once per calendar year until NSGL can demonstrate through the leak rate calculations that the appliance has not leaked in excess of the applicable trigger rate for one year.
- For comfort cooling appliances and other appliances, leak inspections must be conducted once per calendar year until NSGL can demonstrate through leak rate calculations that the appliance has not leaked in excess of the applicable trigger rate for one year. *The vast majority of HVAC units at NSGL fall into this category.*

System leaks that exceed the trigger rate threshold and cannot be repaired within 30 days shall be documented and reports submitted electronically to 608reports@epa.gov (40 CFR §82.157(m)) by Environmental staff. If calculated leak rates are greater than allowed, repairs or a retrofit/retirement plan shall be made within 30 days from the identification of the leak.

6 REGULATORY REQUIREMENTS

6.1 APPLICABILITY

Title VI Section 608 of the CAA applies to owners and operators of stationary refrigeration and air-conditioning equipment, 50+#, containing Class I (CFCs) and Class II (HCFCs) ODS refrigerants and substitute non-ODS refrigerants, such as HFCs.

Under 415 ILCS5, Title II: Air Pollution, the state of Illinois has determined that federal Clean Air Act regulations are such that Illinois will avoid the existence of duplicative, overlapping or conflicting state and federal systems. It therefore deems federal Clean Air Act regulations and emission standards for hazardous air pollutants are applicable and enforceable in Illinois. This applies to ODS & refrigerant management.

In addition to federal and state regulations, the Department of Navy reiterates requirements and outlines environmental instructions in the Environmental Readiness Program Manual, dated June 25, 2021. Refrigerant management requirements are outlined in Chapter 22 of OPNAV M-5090.1E, which extends similar procedures for both ODS and its substitutes. Section 22-3.6, Management of Ozone Depleting Substances, states:

“Navy facilities must follow similar procedures for HFCs and PFCs as are required for ODSs in subparagraph 22-3.6e of this manual, including technician certification, recovery and recycling, and leak monitoring and repair to reduce emissions to help achieve GHG-emission reduction targets.”

6.2 TECHNICIAN CERTIFICATION

US EPA regulations require technicians that maintain, service, repair, or dispose of equipment with the potential to release ozone-depleting refrigerants or non-exempt substitutes into the atmosphere be certified.

Technicians are required to complete and pass an US EPA-approved test specific to the type of equipment maintained. Section 608 Technician Certifications are for stationary equipment and do not expire. Section 609 Technician Certification is for MVAC (Motor Vehicle Air Conditioning) and MVAC-like appliances and is good for life. Tests must be administered by an US EPA-approved certifying organization.

Only US EPA-certified technicians can purchase ozone-depleting substances (ODS) used as refrigerants. As per the CAA Amendments provided in Section 608 and Section 609, the five types of certifications for all equipment suspected of releasing ozone-depleting refrigerants are as follows:

Table 6.1 US EPA Technician Certification Types

Type of equipment serviced	Level of required certification
Small equipment (<5 pounds)	Type I
High and very-high pressure equipment	Type II
Low pressure equipment	Type III
All types	Type IV Universal*
Small appliances, MVACs, and MVAC-like appliances	609 MVAC Technician

6.3 VENTING PROHIBITIONS

US EPA regulations prohibit intentionally venting ozone-depleting substances or their non-exempt substitutes while maintaining, servicing, repairing and/or disposing of air-conditioning or refrigeration equipment.

NSGL compliance is intended to prohibit the intentional release of Class I or Class II ozone depleting substances and non-exempt substitutes. Therefore, only allowing only "de-minimis" releases associated with good faith efforts to reduce, recapture, recycle and/or safely dispose of such substances shall occur.

Only three types of releases are allowed:

- Small "de-minimis" quantities of refrigerant released during good faith attempts to recapture and recycle or safely dispose of refrigerant including releases that occur when connecting or disconnecting hoses to charge or service appliances.
- Refrigerant emitted during the normal operation of air-conditioning and refrigeration equipment (excluding maintenance, servicing, repair or disposal of this equipment), such as from mechanical purging and leaks. However, the US EPA requires that leaks above the applicable trigger rate be repaired in equipment that contains 50+lb of refrigerant.
- Releases of substitute refrigerants that US EPA has determined do not pose a threat to the environment.

6.4 EVACUTATION

Technicians are required to evacuate stationary refrigeration and air-conditioning equipment to established vacuum levels described in Table 5.1 when opening the equipment for maintenance, service, repair, or disposal (40 CFR §82.156(a)). Technicians repairing small appliances, such as household refrigerators, window AC units, and water coolers, must recover 90% of the refrigerant when the compressor is functioning and 80% of the refrigerant when the compressor is not functioning (40 CFR §82.156(b)(2)).

Table 6.2 Required Level of Evacuation of Appliances

Type of Appliance	Inches of Mercury (Hg) Vacuum (relative to standard atmospheric pressure of 29.9 inches Hg)	
	Manufactured or imported before 15 Nov 1993	Manufactured or imported on or after 15 Nov 1993
HCFC-22 appliances, or isolated component of such appliances, with a full charge of less than 200 pounds of refrigerant	0	0
HCFC-22 appliances, or isolated component of such appliances, with a full charge of 200 pounds or more of refrigerant	4	10
Very high-pressure appliances	0	0
Other high-pressure appliances, or isolated component of such appliances, with a full charge of less than 200 pounds of refrigerant	4	10
Other high-pressure appliances, or isolated component of such appliances, with a full charge of 200 pounds or more of refrigerant	4	15
Medium-pressure appliances, or isolated component of such appliances, with a full charge of less than 200 pounds of refrigerant	4	10
Medium-pressure appliances, or isolated component of such appliances, with a full charge of 200 pounds or more of refrigerant	4	15
Low-pressure appliances	25 mm Hg absolute	25 mm Hg absolute.

6.5 EXCEPTIONS – EVACUATION REQUIREMENTS

If technicians cannot evacuate systems to the specified levels because of refrigerant leaks, or it would substantially contaminate the refrigerant being recovered, the following is required:

- Isolate leaking components from non-leaking components where possible;
- Evacuate non-leaking components to the specified levels; and
- Evacuate leaking components to the lowest level that can be attained without substantially contaminating the refrigerant. This level cannot exceed zero pounds per square inch gauge (psig).

If a technician is not evacuating the equipment to the environment after a repair is completed, and if the repair is not major, then the following requirements must be met:

- For high or very high-pressure appliances, the equipment shall be evacuated to zero psig before it is opened.
- For low-pressure appliances, the equipment must be pressurized to zero psig before it is opened. Methods that require subsequent purging (e.g., nitrogen) cannot be used except with appliances containing R-113.

6.6 RECLAMATION

Recovered refrigerant can be returned to the same system or other systems owned by NSGL and/or tenants without restriction (40 CFR §82.156(h)). If recovered refrigerant changes ownership, it must be reclaimed (i.e., cleaned to the AHRI Standard of purity) by an US EPA-certified refrigerant reclaimer (40 CFR §82.164.(a)). In addition, per the Navy's Environmental Readiness Program Manual, any Class I or Class II ODS that is captured and not intended to be reused or stored on-site must be turned into the Defense Logistics Agency (DLA) ODS reserve. **Class I or II ODS refrigerants should not be retained by a 3rd party or contractor.**

To initiate ODS turn-in to DLA, contact the NSGL Hazardous Waste Program Manager (PWD-EV Division) for further instructions.

6.7 SAFE DISPOSAL

Stationary refrigeration and air-conditioning equipment that is dismantled on-site before disposal must have refrigerant recovered in accordance with EPA's requirements for servicing prior to disposal (40 CFR §82.155). This includes Class I and II refrigerants and any non-exempt substitutes.

6.8 RECOVERY & RECYCLING EQUIPMENT

The recovery or recycling of refrigerant must be conducted using equipment that is certified by an EPA-approved equipment testing organization. EPA has approved the Air conditioning, Heating, and Refrigeration Institute (AHRI) and Underwriters laboratories (UL) to certify recycling and recovery equipment. Certified equipment can be identified by a label placed by the manufacturer that states:

"This equipment has been certified by AHRI/UL to meet EPA's minimum requirements for recycling and/or recovery equipment intended for use with [appropriate category of appliance]."

6.9 SALES RESTRICTIONS

The sale of ODS and substitute refrigerants is restricted to EPA-certified technicians (40 CFR §82.154(c)); however, refrigerant owned by the Navy can only be maintained on-site or sent to DLA ODS Reserve. Sale or transfer of any Class I ODSs or HCFC-22 outside the Navy or DLA may occur **only** with written permission from the Navy chain of command and OPNAV N45 (OPNAV 5090.1E Ch22.3-6.h). The sales restriction covers ozone-depleting and substitute refrigerants contained in bulk containers, such as cans, cylinders, or drums. EPA restricts the resale of used refrigerant to a new owner unless it has been reclaimed by an EPA-certified refrigerant reclaimer (40 CFR §82.154(d)).

6.10 CHANGING REFRIGERANT OIL

The oil in a refrigeration appliance can contain large amounts of dissolved refrigerant. The US EPA requires a reduction in the pressure prior to an oil change to ensure that most of the refrigerant contained in the oil is recovered. It is a violation to change oil at higher than 5-psig. There are two acceptable procedures for recovering refrigerant contained in oil:

- Evacuate (or pressurize) the refrigeration appliance, or isolated portion, to a pressure no greater than 5-psig and then remove the oil; or
- Drain the oil into a system receiver to be evacuated (or pressurized) to a pressure no greater than 5-psig.

6.11 REFRIGERATION LEAK RATES

6.11.1 Leak Rate Threshold

Owners or operators of stationary refrigeration and air-conditioning equipment containing 50+# of refrigerant per circuit are required to repair leaks within 30 days of discovery when the calculated leak rate exceeds those presented in Table 5.2 or develop a retrofit or retirement plan. **(Note: Leak rate thresholds do not apply to non-ODS or substitute refrigerants).**

The information collected by the HVAC Technicians on Form "CFR 40-82F-1" has the information used by FMD to calculate leak rates. This form is in Appendix B.

Table 6.3 Applicable Trigger Rates (Effective 1 January 2019)

Appliance Type	Leak Rate
Industrial process refrigeration	30%
Commercial refrigeration	20%
Comfort cooling	10%
Other appliances	10%

Source: US EPA Refrigerant Leak Repair Requirements <https://www.epa.gov/section608/stationaryrefrigeration-leak-repair-requirements>.

The leak repair requirements do not apply to equipment with less than 50 lbs. of refrigerants. However, **smaller equipment is not exempt from the refrigerant venting prohibitions** outlined in Section 5.3 of this plan.

6.11.2 Leak Rate Calculation

To track leak rates, owners or operators of air-conditioning and refrigeration equipment with 50+# of charge must keep records of the quantity of refrigerant added to their equipment during servicing and maintenance procedures (40 CFR §82.157(l)). The leak rate must be calculated every time a refrigerant is added to this type of equipment.

As outlined in BMS 15-2.2 Refrigerant Compliance Management, the Facilities Management Division (FMD) of the NSGL Public Works Department (PWD) is responsible for tracking and calculating leak rates. FMD uses information from all Form CAA 40-82F-1 and any service records obtained from tenants or contracted technicians to calculate the annualized leak rate each time refrigerant is added to 50+# equipment.

The leak rate outlined in the BMS is a rolling average, expressed in terms of the percentage of the appliance's full charge that would be lost over a 12-month period if the current rate of loss were to continue over that period.

Table 6.4 Rolling Average Method for Determining Leak Rates

$$\text{Leak rate (\%)} = \frac{\begin{array}{c} \text{lbs of refrigerant added over past 365 days} \\ \text{(or since leaks were last repaired,} \\ \text{if that period is less than one year)} \end{array}}{\text{lbs of refrigerant in full charge}} \times 100\%$$

The trigger for repair requirements is the current leak rate projected over a consecutive 12-month period, rather than the total quantity of refrigerant lost.

Refrigerants purged and destroyed at a verifiable 98% efficiency are not to be included in the leak rate calculation (40 CFR §82.157(k)). When excluding purged and destroyed refrigerants from the annual leak rate calculations, owners and operators must notify EPA within 60 days after the first time the exclusion is used (40 CFR §82.157(m)(5)). However, refrigerant purging may not be applicable to NSGL operations.

6.12 REPAIR, RETROFIT OR RETIRE?

If refrigerant is leaking at a rate that would exceed the applicable trigger rate for that type of equipment (e.g., 10% for comfort cooling), then corrective action must be taken. If it is leaking at a rate that would cause it to release 125% or more of its charge over a year, thereby making it "chronically leaking" equipment, it must be repaired within 30 days of leak discovery. **Thus, the trigger for repair requirements is the current leak rate rather than the total quantity of refrigerant lost.**

FMD will calculate the leak rate determination for faulty equipment, but it is the responsibility of the service technician to repair any leak as soon as it is discovered or as quickly as possible. When a technician reports a leak to their supervisor, the supervisor should immediately alert FMD, provide the appropriate CAA 40-82F forms, and

request an initial repair ticket to occur within 10 days. **FM&S should not wait for FMD leak rate calculation to begin repairs.**

Additional leak repair time beyond the 30 days may be permitted if it has been determined that the leak repair cannot be made within 30 days. In this event, NSGL must document all repair efforts, **and notify EPA** of the inability to comply within the 30-day repair requirement window, and the reason for the inability must be submitted to the EPA in accordance with 40 CFR 82.166(n). **Such notification must be made within 30 days of discovering that leak repair efforts cannot be completed (which should be no more than 30 days from Leak Exceedance).**

NSGL is not required to repair leaks if a plan is made to retrofit or retire the leaking appliance.

- **Retrofit** means to convert an appliance from one refrigerant to another refrigerant. Retrofitting includes the conversion of the appliance to achieve system compatibility with the new refrigerant and may include, but is not limited to, changes in lubricants, gaskets, filters, driers, valves, o-rings or appliance components.
- **Retire**, when referring to an appliance, means the removal of the refrigerant and the disassembly or impairment of the refrigerant circuit such that the appliance as a whole is rendered unusable by any person in the future.

A copy of the plan must be kept on-site, **does not need to be submitted to the EPA**, but must be kept on site at the appliance, another copy in FMD/EV, and the original plan must be made available to US EPA upon request. Activities under the plan must be completed within 1 year from the date of the plan. Section 6.13 of this document describes Retrofit / Retire Plan requirements.

6.12.1 Repair Verification

The primary purpose of the Initial Verification Test (IVT) is to certify that a leak or leaks have been repaired before refrigerant is added back to the system. Even when the system is not evacuated to complete repairs, initial and follow-up verification tests are required. The primary purpose of the Follow-Up Verification Test (FVT) is to re-certify that repairs continue to hold after the system is returned to its normal operating characteristics or conditions.

Typically, IVT and FVT are not required for systems other than industrial process refrigeration, unless these other systems are federally owned. This is why comfort and other chillers at NSGL that have 50+# of refrigerant must have IVTs and FVTs.

FM&S should not wait for FMD leak rate calculation to schedule IVTs or FVTs.

6.12.2 Initial Verification Test (IVT)

An IVT **must** be conducted following any leak repairs and **must be performed within 30 days of an appliance exceeding the applicable leak rate**. A refrigeration system may not be brought back online (if it has been taken offline), unless a) an IVT indicates that repairs have been successfully completed or b) a decision to retrofit or retire is made. The IVT must be conducted before the replacement of the full charge of refrigerant in systems (or isolated portions of systems) from which the refrigerant has been evacuated. In systems that have not been evacuated, the IVT must be conducted as soon as practicable after the repair work is completed.

6.12.3 Follow-up Verification Test (FVT)

A FVT must be conducted no later than 10 days after an IVT or return to normal operation. The HVAC Technician must notify the supervisor immediately of failed verification testing. The work order will remain open until successful FVT is completed.

Exception: In certain cases, the results of a FVT may be less reliable if the test is conducted at normal operating characteristics or conditions. For example, repairs made to leaks inside heat exchangers would not normally be accessible after the system is operating. In this kind of situation, based on sound professional judgment, the FVT may be conducted prior to returning to normal operating characteristics or conditions. However, the FVT must be conducted at or near the normal operating pressure where practicable, and at or near the normal operating temperature where practicable.

If the FVT fails, and the leak repair could not be completed in subsequent attempts, or if the leak is deemed to be “un-repairable,” then a retrofit or retirement plan **must** be developed within:

- 30 days if retrofit or retire is intended rather than repair;
- 30 days of the failed FVT; or
- 30 days of detecting the leak if no leak repair attempts are conducted.

6.12.4 Verification Test Methods

Initial and Follow-up verification tests may use any method that meets sound professional judgement. Test examples include, but are not limited to:

- A soap bubble test; or
- Electronic leak detectors; or
- Ultrasonic leak detectors; or
- A pressure test; or
- A vacuum test; or
- A fluorescent dye and black light test; or
- An infrared test; or
- A near infrared (back scatter absorption gas imaging) test; or
- Halon refrigerant gas detection methods.

6.13 RETROFIT / RETIREMENT PLAN REQUIREMENTS

6.13.1 Retrofit / Retirement Plan – Required Information

A retrofit or retirement plan must, at a minimum, contain the following information:

- Identification and location of the appliance
- Type and full charge of the refrigerant used in the appliance;
- Type and full charge of the refrigerant to which the appliance will be converted, if retrofitted;
- Itemized procedure for converting the appliance to a different refrigerant, including changes required for compatibility with the new substitute, if retrofitted;
- Plan for the disposition of the appliance, if retired; and
- A schedule, not to exceed one-year, for completion of the appliance retrofit or retirement.

A copy of the retrofit or retirement plan must be signed by an authorized official, kept on site, and made available to EPA upon request (40 CFR §82.157(h)(3)). Activities under the plan must be completed within 12 months from the date of the plan (40 CFR §82.157(h)(5)), unless otherwise allowed or approved by the EPA (40 CFR §82.157(f)).

6.13.2 Relief From Retrofit / Retirement Requirements

The owners or operators of industrial process refrigeration equipment, **federally owned chillers**, commercial refrigeration and commercial comfort cooling appliances may be relieved from the retrofit or repair requirements if:

- Second efforts to repair the same leaks that were subject to the first repair efforts are successful;

- Within 180 days of the failed FVT, NSGL determines the leak rate is below the trigger rate. In this case, NSGL must notify US EPA as to how this determination will be made and must submit the information within 30 days of the failed verification test; and/or
- For appliances subject to the leak repair requirements, the timelines may be suspended if the appliance has been taken out of service and stored (i.e., mothballed).

In these cases, EV will communicate to the EPA what actions were taken, and the unit is brought below the applicable leak rate.

For reference, a Regulatory Timeline & Checklist for Retrofit & Retirement Activities, and a Decision Guide (flowchart) for Leaking Refrigeration Systems are included Appendices G and H of this plan.

6.13.3 Retrofit / Retirement Plan Extensions

US EPA will permit additional time to retrofit or retire industrial process refrigeration equipment and **federally owned appliances** as follows:

- The owners or operators must make a request within six months from the expiration of the initial 30-day period (before Leak Exceedance Date plus 120 days).
- US EPA will permit additional time to complete a retrofit or retirement plan to the extent reasonably necessary if a delay is caused by the requirements of other applicable federal, state and/or local regulations, or where components that must be replaced are not available within 30 days (40 CFR §82.157(f)). However, these time extensions require additional and stringent recordkeeping and (EPA) notification requirements.
- US EPA will permit one additional 12-month period if the appliance is custom-built and the supplier of the appliance or a critical component has quoted a delivery time of more than 30 weeks from when the order was placed (assuming the order was placed in a timely manner). In some cases, US EPA may provide additional time beyond this 12-month extension period if a request is made by the end of the ninth month of the extension.

6.14 SYSTEM MOTHBALLING

System mothballing, for the purposes of this plan, is the intentional shutdown of a refrigeration system for an extended period of time, where the refrigerant has been evacuated from the system to at least atmospheric pressure. If the intent is to make repairs to the system that require opening the system, the refrigerant must be removed to the required evacuation levels for that refrigerant before the system is opened.

When a refrigeration system is in a mothballed condition, refrigerant leaks that existed when the system was operating have effectively been stopped because the system has been evacuated. Therefore, the time-related requirements for repairing, retrofitting, or retiring a leaking system are temporarily suspended while the system is mothballed. The time-related requirements resume on the day the system is brought back online and is no longer considered mothballed.

Leak repairs may be made while a system is mothballed, and there is no time limit to make the repairs. Leak repairs following mothballing are still subject to IVT and FVT. While the system is mothballed, IVT can happen at any time. However, the repairs must pass the IVT before the system is brought online, unless there is a plan to retrofit or retire the system. When the system is brought out of a mothballed state, the deadline for performing FVT would begin. In this situation, NSGL would have 30 days to perform FVT.

System mothballing does not require a report to the EPA. However, NSGL will need to report to EPA if the decision to temporarily mothball a system results in delaying an estimated completion date that has previously been submitted to EPA for repairing, retrofitting, or retiring the system.

7 REFRIGERANT RECOVERY & RECYCLING EQUIPMENT

US EPA regulations require that refrigerant recovery and recycling equipment be tested to ensure it meets US EPA requirements. These requirements can be found in Appendix B2, B3 and B4 to Subpart F of Part 82 Performance of Refrigeration Recovery, Recycling, and/or Reclaim Equipment.

Recovery and recycling equipment used with most air-conditioning and refrigeration equipment shall meet the standards identified in Table 1 in Section 4.

Small appliance recovery equipment must be capable of recovering either 90% of the refrigerant in the small appliance when the small appliance compressor is functional, or 80% of the refrigerant in the small appliance when the compressor is not functional.

US EPA has approved the Air-Conditioning, Heating, and Refrigeration Institute (AHRI) and Underwriters Laboratories (UL) to certify recycling and recovery equipment. Certified equipment can be identified by a label that states: "This equipment has been certified by AHRI/UL to meet EPA's minimum requirements for recycling and/or recovery equipment intended for use with [appropriate category of appliance]."

8 DISPOSAL REQUIREMENTS

8.1 Disposal of Equipment

Refrigeration and air-conditioning equipment that is typically dismantled on-site before disposal (i.e., retail food refrigeration, central air conditioning, chillers and industrial process refrigeration) must have refrigerant recovered in accordance with US EPA's requirements for servicing prior to their disposal.

Disposal of appliances with full charge >5 pounds requires recordkeeping under 40 CFR 82.156(a)(3).

Disposal of small (≤ 5 pounds) appliances 40 CFR 82.155(a) requires the following:

- Evacuate and recover refrigerant as discussed in Section 10.
- Verify that refrigerant has been evacuated previously through signed statements or contracts.

Activities transferring HVAC&R equipment to DLA Disposition Services for **reuse** must label the equipment to indicate that it contains an ODS.

Activities transferring HVAC&R equipment to DLA Disposition Services for disposal as **scrap** must recover the ODS prior to disposal and label the units as having had all refrigerants and oils removed per EPA regulations. Additional information on HVAC&R equipment disposal can be found in OPNAV 5090.1E Ch. 10, reference (o). (OPNAV 5090.1E Ch. 22-3.6.h).

8.2 Refrigerant Contamination and Disposal

Refrigerant management shall include the following:

- Separate, clean and evacuated recovery vessels shall be provided for each refrigerant.
- Each refrigerant cylinder shall be labeled and marked in accordance with US EPA and U.S. Department of Transportation (DOT) regulations.
- All excess Class I refrigerants, HCFC-22 refrigerant, and Halons should continue to be recovered and returned to the DLA ODS reserve (OPNAV 5090.1E Ch. 22-3.6.c.4.b).

- Oils extracted from refrigerant, recovery machines, or vacuum pumps shall be kept separate from other oils and stored in specifically identified drum.
- Refrigerants shall not be mixed. Mixed refrigerants shall be considered unusable and resulting product shall be considered hazardous waste unless otherwise directed by DLA Disposal Services.
- Disposal of waste and/or oil shall be completed through Hazardous Waste Management Department.

9 HANDLING OF REFRIGERANT AT ON-SITE STORAGE LOCATIONS

9.1 Filling Refrigerant Cylinders

Refrigerant cylinders should not be filled in excess of 80% of the fluid capacity. It is considered overfilling if a cylinder is filled more than 80% of capacity. Overfilling may result in serious safety issues such as a buildup in hydrostatic pressure that can cause a significant explosion.

9.2 Emission Reduction

Emission reduction shall be accomplished through servicing practices that include but are not limited to the following:

- Adhering to best equipment use practices and manufacturers' instructions and procedures in combination with the diligent use of forms, reporting structure, and training outlined in this document.
- Establishing retrofitting and retirement programs for refrigerant systems.
- Maintaining efficient system performance and instituting operations to help conserve energy consumption.
- Recovering refrigerants from appliances for reuse, reclamation or proper destruction
- Using recycled refrigerant that meets or exceeds ARI requirements before reusing it in equipment.
- Using only approved, clean, and leak free containment vessels and recovery equipment.
- The official version of this information will only be maintained in an on-line web format. Review the material on-line prior to placing reliance on a dated printed version.
- Using trace gases with a lower environmental impact. For example, virgin HCFC-22 presents a lower environmental risk than a fully halogenated gas mixed with dry nitrogen.
- Use of dry nitrogen to break vacuums created by recovery actions.

10 RECORD KEEPING

As regards record keeping, NSGL follows steps outlined in the NAVFAC Mid-Lant process # ML-B-15.2.2 Refrigerant Compliance Management, last updated 28 February 2020. A copy of this process is available in the Appendix A of this plan.

All records outlined in this section shall be maintained for a period of five years.

10.1 Technician Certifications

Technicians shall keep a copy of their proof of certification at NSGL. HVAC Supervisors must keep copies of technician certifications where work occurs. They must keep these records for at least 5 years after the technician no longer works as an HVAC technician.

10.2 Equipment Inventory

The following items shall be recorded and serve as a minimum checklist:

- Facility Management Division (FMD) personnel shall keep detailed technical information in Maximo on every appliance which contains 50+# of refrigerant including:
 - Equipment description;
 - Location (building number); asset number;
 - Manufacturer; model; serial no.;
 - Refrigerant capacity at full charge (per circuit if independent;
 - How full charge was determined: if using established range, include full range, midpoint, and how range was determined);
 - How any revision to full charge was determined, along with revision date;
 - Equipment duty type (comfort cooling, commercial refrigeration, industrial process refrigeration, other application).
- FMD must flag “50+#” units in Maximo by putting “ODS” in Assets module comment field.
- FMD must update the installation-wide 50+# inventory as equipment is added or removed, including the reason for removal and the date removed from inventory.

10.3 Responding to (Service) Work Orders

FMD will generate work orders, including preventive maintenance; repairs, refrigerant removal for appliance disposal; initial leak repair verification tests; follow-up leak repair verification tests; and recurring equipment leak inspections.

When the HVAC technician responds to and completes work orders, they will use recycling/recovery equipment that meets EPA standards. They will operate recycling/recovery equipment per manufacturer instructions. They will not intentionally release refrigerant. When the work order has been responded to or completed, the HVAC technician will use the forms described in the BMS to document this activity. These forms must be kept for 5 years.

10.3.1 CAA 40-82F-1 Service Invoice for 50+# Units

This “Service Invoice” (form) is used to document:

- Any maintenance or service where refrigerant is added to or removed from an appliance (or circuit, if there is more than one circuit in the unit) containing 50+# of refrigerant;
- Results of leak inspections;
- Results of IVTs and FVTs
- Actions taken if a 50+# unit (circuit) is taken offline (mothballed) or put back online into service again.

The form itself should include the date and type of service, the Maximo Work Order number (WO#), the quantity of refrigerant added or removed. If 50#+ appliances with refrigerants, recovery equipment, recycling equipment and/or refrigerant cylinders are not functioning or are not in good working condition, or need to be repaired, it is the responsibility of the technician to report it on the Service Invoice for this Maximo WO#.

The technician must provide duplicate copies of this form to the HVAC supervisor / SME daily. They must notify their HVAC supervisor / SME immediately if a leak is not repaired.

The HVAC supervisor / SME will submit completed CAA 40-82F-1 forms to FMD immediately upon receipt so that FMD may calculate an appliance's leak rate.

10.3.2 CAA 40-82F-2 Service and Inventory Tracking – All Appliances

HVAC technicians use this form to document work on any appliances, regardless of size, and to track all refrigerant transfers (added and/or removed) for these appliances. Details of refrigerant is to include date of transfer, quantity and type of refrigerant transferred, the entity or person to whom it was transferred.

At the bottom of the form, the technician should calculate the total amount of each type of refrigerant disposed from appliances in the period tracked on this form.

Technicians should provide a copy of this form to their supervisor weekly. When no refrigerant work is performed, they should log “no work” and dates covered.

10.3.3 CAA 40-82F-3 Disposal Invoice – All Appliances

HVAC technicians use this form to track disposal of all appliances, and how much refrigerant was reclaimed or disposed. A typical example of this type of appliance is a window AC unit or a small refrigerator. Copies of this form should be provided to the HVAC supervisor / SME weekly.

10.4 Refrigerant Tracking

The purchase, use, recovery, recycling, reuse, and disposal of refrigerants shall be tracked by each work area affected by this program. In addition to the three CAA 40-82F forms, FM&S shall maintain a Refrigerant Inventory Log (in MS Excel) on a monthly basis. This log inventories pounds refrigerant on-hand at beginning of month, purchased during the month, reclaimed from equipment (added to shop stock), removed from shop stock (charged into equipment), transferred from inventory to disposal, disposed refrigerant, and shop-to-shop transfers. These are all used to then determine refrigerant on-hand at end of month.

FM&S shall provide this Refrigerant Inventory Log to FMD and EV upon request.

11 ROLES & RESPONSIBILITIES

11.1 HVAC SUPERVISOR / SME

It is the role of the HVAC supervisor / SME to make sure the technicians are properly and legibly filling out the three CAA 40-82F forms mentioned in Section 10. There can be instances where one Maximo WO# can and should be documented on all three forms for the same service event.

In the event of a leak, or suspected leak, the supervisor / SME must immediately provide the CAA 40-82F-1 form to FMD for leak rate calculation. The supervisor / SME should also request a follow-up leak repair ticket from FMD to occur as soon as possible.

All other, non-leak-related, CAA 40-82F forms should be provided to FMD on a weekly basis.

The supervisor / SME must keep records of refrigerant purchased, documenting the name of the seller, date of purchase, and the type and quantity of refrigerant purchased. A copy of this information shall be attached to the weekly report that they send to FMD.

The supervisor / SME shall keep records of refrigerants sent for reclamation documenting the names and addresses of the reclaimer, and the quantity of refrigerant sent to them for reclamation. A copy of this information must be attached to the weekly report that they send to FMD.

The supervisor / SME must provide documentation to the Office of Environmental Program Manager when new recycling or recovery equipment is purchased.

11.2 ENVIRONMENTAL PROGRAM MANAGER

The EV Program Manager shall submit a report to US EPA for appliances containing 50+# of refrigerant that leaks 125% or more of the full charge in a calendar year (due by March 1st). This report shall describe efforts to identify leaks and repair the appliance.

The EV program manager will work with FMD and FM&S to prepare retirement / retrofit plans and ensure dated signature of authorized officials. EV will forward copies of EPA notifications to the PW Core FM&S, FMD and PWO.

The EV Program Manager will provide equipment information to the US EPA Regional Office when Facilities has acquired (built, bought, or leased) recovery or recycling equipment when requested. They will also request EPA relieve retrofit / retirement obligations if, within 6 months of plan date, repairs are made, and units are brought below applicable leak rates.

EV will conduct annual internal compliance audits to ensure that all regulations are being met. They will work with FMD, FM&S and PW Core to develop and execute POA&Ms and resolve compliance deficiencies.

11.3 FACILITIES MANAGEMENT DIVISION

The FMD plays a vital role in ensuring Refrigerant Compliance Management at NSGL, acting as sentinel for the PWD, aggregating information from the HVAC shops and alerting PWD Core and EV about repair / replace / retire needs. The FMD Representative is responsible for:

- Keeping an installation-wide inventory of all HVAC appliances, equipment, or individual circuits normally containing a charge of 50+# refrigerant, including the type of refrigerant, and refrigerant capacity (in pounds) at full charge. They conduct a semi-annual self-audit of this inventory information, and provide inventory information to PWD EV and other departments as needed;
- Conducting semi-annual audits of HVAC Technician records to verify that the BMS process is being followed;
- Generating work orders, including preventative maintenance, repairs, refrigerant removal for appliance disposal, follow-up leak repair verification tests, and recurring leak inspections;
- Requesting copies of all HVAC&R equipment service records from contractors, PWD Shops, and tenants; maintaining these copies for 5 years;
- Using information from all CAA 40-82F-1 and any service records obtained from tenants or contracted technicians to calculate the annualized leak rate each time refrigerant is added to 50+# equipment; performing these calculations within one week of receiving data; maintaining records of these calculations for 5 years;
- Notifying HVAC Supervisors, EV, PW Core FM&S and PWO immediately of any leak that cannot be repaired within 20 days from leak discovery and that may exceed the allowable annualized leak rate;

- Working with EV to evaluate how to comply with EPA regulations if 50+ units leak above the allowable leak rate and cannot be repaired within 30 days of leak discovery;
- Reporting to EV all 50+ appliances that leaked 125% or more of their full charge in the previous calendar year, and are therefore considered to be “chronically leaking;”
- Supporting EV with information for notifications to EPA on individual appliances as well as on the annual report to EPA (by March 1st each year).

11.4 TENANT COMMANDS

Tenant commands that do not use NAVFAC HVAC technicians must comply with the same certification, repair / retrofit / retirement plan, disposal, and recordkeeping requirements as NAVFAC shops personnel (technicians, supervisors, SMEs). They should make these records available to PWD FMD and EV within the same timeframes as NAVFAC Shops personnel. Their records must also be available upon request and should be maintained for five years.

12 ADDITIONAL REQUIREMENTS

12.1 LABELING

Refrigerant containers of Class I or Class II ozone depleting substances shall be labeled. The warning statement must be clearly legible and conspicuous such that it is likely to be read by consumers prior to making their purchase.

The label must include the following statement:

WARNING: [Contains / Manufactured with] [Name CFC/HCFC], a substance that harms public health and environment by destroying ozone in the upper atmosphere.

The warning may be printed directly on a product or its outer packaging or on an alternative label, including the following conspicuous placements:

- Principal display panel or display panel area.
- Alternative labels, such as hang tags, stickers, or similar over labeling.
- Outer packaging.
- Supplemental printed materials, such as invoices, bills of lading, or package inserts.
- Promotional materials, such that the warning statement is visible before the time of purchase.

Additional information on labeling placement and requirements can be found in the Code of Federal Regulations at Title 40 Part 82 Subpart E (40 CFR 82.108 and 82.110).

12.2 SITE CONTACTS

Contacts for this Refrigerant Compliance Plan are identified in Table 4 below.

Table 12.1 Contacts

Description	Name
Environmental Program Manager	TBD
Alternate Environmental Contact	Donald Sytkowski Taylor Bozman

Description	Name
	Heather Richtfort
Maintenance Supervisors	East Shop - Robert Scroggin West Shop – Jeramy Qualls
ODS Subject Matter Expert	Shelby Vinson
Maintenance Manager	Nate Colburn

12.3 REVIEW FREQUENCY

This procedure shall be reviewed annually and updated as needed to meet applicable regulatory changes. Annual field audits are also recommended.

12.4 REFERENCES

- Recycling and Emissions Reduction, 40 CFR Part 82
- OPNAV 5090.1E Chapter 22-3.6 Management of Ozone-Depleting Substances

Revision History Table

Changes	Effective Date	Made by:
New Refrigerant Compliance Plan	19 March 2019	MECX
Plan update: Inserted information about penalties; expanded Repair section; expanded section about retirement or retrofitting; added section about mothballing systems; added Regulatory Timeline & Checklist for Retrofit & Retirement to Appendices; added BMS 15-2.2 Refrigerant Compliance Management to Appendices; Added flowchart of Decision Guide for Leaking Refrigeration Systems to Appendices; Created template for Retrofit / Retirement Plan and added to Appendices	Jun-Jul 2024	H. RICHTFORT
Updating Plan date; routing to IEPD for signature.	Jan 2025	H. RICHTFORT

APPENDIX A. ML-B- NAVFAC MID-LANT PROCESS ML-B-15.2.2 REFRIGERANT COMPLIANCE MANAGEMENT

Last Updated: 28-February-2020

Process Number and Title: Refrigerant Compliance Management

Process Owner Team, Lead: NAVFAC MIDLANT PW, Jenny Richards

Document History:

Rev	Date	Change Description
00	02/14/2014	Initial Version
01	02/28/2020	Incorporate Regulatory Changes • Covers all refrigerants and replacements, unless exempt • Requires records of how full charge, and revisions to same, are determined • Lowers allowable leak rates for 50+ # appliances • Shortens time to perform follow-up leak repair verification tests to 10 days • Requires disposal records for appliances 5 – 50# • Requires technicians record details of refrigerant disposal • Requires records of method used to verify leak repairs • Requires recurring leak inspections to verify repairs hold at least 1 year • Requires reports of 50+ # appliances that leak 125+ % full charge per year Increase record retention to 5 years to accommodate Title V permit requirements

Introduction:

This process applies to the service, repair, or disposal of all Heating, Ventilation and Air Conditioning (HVAC) appliances or other refrigerant containing equipment located onboard NAVFAC MIDLANT facilities. Examples include comfort cooling air conditioning, commercial refrigeration (such as walk in refrigerators/freezers), information technology server cooling centers and other industrial refrigeration systems, and small appliances such as window units and refrigerators.

The regulatory scope includes all records and work practices required by 40 CFR 82, Protection of Stratospheric Ozone. This includes all refrigerants and refrigerant substitutes except those exempted under 40 CFR 82.154(a):

- Ammonia in commercial or industrial process refrigeration or in absorption units;
- Carbon dioxide, nitrogen, or water in any application;
- Chlorine in industrial process refrigeration (processing of chlorine and chlorine compounds);
- Hydrocarbons in industrial process refrigeration (processing of hydrocarbons);
- Ethane (R-170) in equipment for very low temp refrigeration or non-mechanical heat transfer;
- Propane (R-290) and R-441A in retail food refrigerators/freezers (stand-alone units only), household refrigerators, freezers, and combination refrigerators and freezers, self-contained residential and light commercial room air-conditioning and heat pumps, and vending machines.
- Isobutane (R-600a) in retail food refrigerators/freezers (stand-alone units only), household refrigerators, freezers, and combination refrigerators and freezers, self-contained commercial ice machines, very low temp refrigeration equipment, and water coolers.

Since the BMS covers all HVAC and refrigeration equipment serviced by Public Works (PW), contractors, or other non-NAVFAC entities, responsibilities are outlined according to job function.

The process owner is NAVFAC MIDLANT PW and is the responsible party for ensuring conformance with the BMS. If BMS associated procedures are contracted by a PWD, the PWD is responsible for ensuring contractor conformance with the BMS.

Last Updated: 28-February-2020

Installation Environmental (EV) has authority to allow deviation from the BMS as needed to ensure regulatory compliance. This includes determining if any individual HVAC appliance or refrigerant is exempt from this BMS on a case-by-case basis.

Policies:

Refer to OPNAVINST 5090.1, Navy Environmental Readiness Program Manual, for current Department of Navy requirements on the handling of refrigerants.

General References:

40 CFR 82 work practice, labeling, record keeping, reporting, and technician and equipment certification requirements can be found at <http://www.epa.gov/ozone/strathome.html>.

Acronyms, Definitions, and Symbols:

Acronyms	Definitions
<	Less than
>	Greater than
#	Pound(s) refrigerant
50+ #	50 or more pounds of refrigerant
CAA	Clean Air Act
CFR	Code of Federal Regulations
Commercial Refrigeration	Is used in retail food and cold storage warehouse sectors
EPA	Environmental Protection Agency
EV	Environmental (Installation Environmental)
FMS	Facilities Management Specialist
FEAD	Facility Engineering Acquisition Department
FM&S	Facility Management & Sustainment Product Line (PW Core)
FMD	Facility Management Division
HVAC	Heating, Ventilation, Air Conditioning (and Refrigeration)
Industrial Process Refrigeration (IPR)	Complex customized appliance directly linked to processes used in chemical, pharmaceutical, petrochemical, and manufacturing.
ODS	Ozone-Depleting Substances; used in Maximo to indicate units with 50+ # refrigerant that are covered by EPA regulations.
PM	Preventive Maintenance
PW	Public Works
PWD	Public Works Department
PWO	Public Works Officer
Refrigerant	All refrigerants including substitutes but excluding exempt compounds

Last Updated: 28-February-2020

Process Table:

Process Step	Associated Procedure	Resources
1 Certifications	HVAC Technicians: • Be certified through an EPA-approved program; • Be knowledgeable of current EPA requirements associated with HVAC service and repair; • Utilize recovery and recycling equipment that meets EPA certification standards according to manufacture date and intended service, and is not altered in any way that would affect ability to meet certification standards. HVAC Technician Supervisors: • Keep copies of technician certifications at Installation where work occurs; • Keep copies of technician certifications at least 5 years after technician no longer operates as a technician.	
2 Provide EPA Requirements and Training	EV: • Provide training on EPA requirements and NAVFAC process.	-
3 Equipment Inventory	FMD: • Keep an Installation-wide inventory of all HVAC appliances or equipment normally containing a charge of 50+ # refrigerant. Inventory must include for each unit: - o equipment description; - o location; asset number; - o manufacturer; model; serial no.; - o refrigerant # capacity full charge (per circuit if independent); - o type of refrigerant; - o how full charge was determined: if using established range, include full range, midpoint, and how range was determined); - o how any revision to full charge was determined, along with revision date.	

Last Updated: 28-February-2020

Process Step	Associated Procedure	Resources
	- Flag "50+ #" units in Maximo by putting "ODS" in Assets module comment field; - Verify all required information is clear and easily understood with no miscellaneous or irrelevant data; - Print current installation-wide 50+ # inventory for use performing semi-annual audit. Keep copy readily available for inspections and audits. Be sure list includes normal # charge. Contact the FM&S Core for assistance if you have difficulty pulling report showing # normal charge out of Maximo. FEAD: - Collect an inventory as described above from all contractors and provide to the FMD to update the PWD's inventory in Maximo.	
4 Provide Inventory to EV	FMD: Provide current installation-wide 50+ # inventory to EV as updates are made and at any time upon EV request.	-
5 Maintain Equipment Inventory	FMD: Update the installation-wide 50+ # inventory as equipment is added or removed. Keep records of equipment removed including reason removed from inventory and date.	-
6 Provide Updated Equipment Inventory	FMD: Provide the updated equipment inventory to EV as equipment is added or removed.	
7 Generate Work Order	FMD: Generate work orders, including: preventive maintenance; repairs; refrigerant removal for appliance disposal; follow-up leak repair verification tests; and recurring leak inspections.	-
8 Respond to Work Order	HVAC Technician: - Respond to and complete work orders in accordance with EPA requirements; - Use recycling/recovery equipment that meets EPA standards;	-

Last Updated: 28-February-2020

Process Step	Associated Procedure	Resources
	• Operate recycling/recovery equipment per manufacturer instructions; • Do not intentionally release refrigerant.	
9 Complete CAA Forms	HVAC Technician, complete forms/keep 5 years: • CAA 40-82F-1 for 50+ # units; - o Provide duplicate copies to supervisor daily. - o Notify supervisor immediately if leak is not repaired. • CAA 40-82F-2 for all units. - o Track all refrigerants used. - o Track all refrigerants disposed, including disposition details: ▪ date of transfer; ▪ quantity and type of refrigerant transferred; ▪ the entity or person to whom it was transferred. - o When no refrigerant work is performed, log “no work” and dates covered. - o Calculate and record total # each type refrigerant recovered from disposed appliances each calendar month. - o Provide duplicate copies to supervisor weekly. • CAA 40-82F-3 for disposal of all appliances (including 5 – 50 #): - o Affix to appliance a weatherproof label stating “all refrigerant and oil that has not leaked has been removed i.a.w. 40 CFR 82”; - o Indicate refrigerant disposition (reclaimed or disposed) and add to Form CAA 40-82F-2; - o Provide duplicate copies to supervisor weekly. HVAC Technician Supervisor: • Send copy of CAA 40-82F-1 to FMD	<u>CAA 40-82F-1</u> <u>CAA 40-82F-2</u> <u>CAA 40-82F-3</u>

Last Updated: 28-February-2020

Process Step	Associated Procedure	Resources
	immediately upon receipt so that FMD may calculate leak rate. FMD: Calculate leak rate for any refrigerant additions to 50+ # equipment.	
10 Follow-Up Work Order	HVAC Technician: - Notify supervisor immediately if leak is not repaired. HVAC Technician Supervisor: - Request follow-up leak repair ticket from FMD. FMD: - Schedule follow-up leak repair to occur as soon as possible to ensure repair is made and verified within 30 days of leak discovery.	
11 Initial Leak Repair Verification Testing	HVAC Technician: - Perform INITIAL leak repair verification test prior to adding refrigerant to appliance. Document results on CAA 40-82F-1. Include method of verification; Potential methods for verification tests include, but are not limited to, the use of soap bubbles as appropriate, electronic or ultrasonic leak detectors, pressure or vacuum tests, fluorescent dye and black light, infrared or near infrared tests, and handheld gas detection devices. - Notify supervisor immediately of failed verification testing; - Work order remains open until successful follow-up verification test is completed.	CAA 40-82F-1
12 Follow-up Leak Repair Verification Testing.	HVAC Technician: - Perform FOLLOW-UP leak repair verification test after unit resumes normal operation but no later than 10 days after initial test or return to normal operation. Document results on CAA 40-82F-1. Include method of verification; - Notify supervisor immediately of failed verification testing;	CAA 40-82F-1

Last Updated: 28-February-2020

Process Step	Associated Procedure	Resources
	Work order remains open until successful follow-up verification test is completed.	
13 Follow-Up Work Order for Failed Leak Repair Verification Test	HVAC Technician: - Notify supervisor immediately of failed verification testing. HVAC Technician Supervisor: - Request follow-up repair ticket from FMD. FMD: - Schedule follow-up leak repair to occur as soon as possible to ensure repair is made and verified within 30 days of leak discovery.	
14 Follow-up Work Order for Recurring Equipment Leak Inspections Note: this is different than the leak repair verification testing.	FMD: - Schedule recurring equipment leak inspections for all 50+ # units that leaked above the allowable leak rate for ANY AMOUNT of time (see Process Step 18 for calculation formula). Schedule frequency as follows: 500+ # commercial and industrial process refrigeration: once every 3 months until 4 consecutive quarters after leak repair are < allowable leak rate; - Comfort cooling and other appliances not listed above: once per calendar year until equipment remains < allowable leak rate for one year after leak repair. HVAC Technician: - Perform inspection and document results on CAA 40-82F-1. This inspection is checking for any leaks on the appliance, not just the most recently repaired leak; - Include inspection date, location of each leak identified, certification that all visible and accessible parts of the appliance were inspected, and method used for verification testing; - Use leak inspection methods appropriate for the device; EPA acceptable methods include, but are not limited to, the use of soap bubbles as	CAA 40-82F-1

Last Updated: 28-February-2020

Process Step	Associated Procedure	Resources
	appropriate, electronic or ultrasonic leak detectors, pressure or vacuum tests, fluorescent dye and black light, infrared or near infrared tests, and handheld gas detection devices. Methods that determine whether the appliance is leaking refrigerant but not the location of a leak, such as standing pressure/vacuum decay tests, sight glass checks, viewing receiver levels, pressure checks, and charging charts, must be used in conjunction with methods that can determine the location of a leak. • Provide duplicate copy to supervisor daily; • Notify HVAC technician supervisor immediately of leaks discovered. HVAC Technician Supervisor: • Request follow-up ticket from FMD if leaks are discovered. FMD • Schedule follow-up leak repair to occur as soon as possible to ensure repair is made and verified within 30 days of leak discovery.	
15 Shop Service Record	HVAC Technician Supervisor: • Maintain copy of all CAA 40-82F-1, CAA 40-82F-2, and CAA 40-82F-3 forms for 5 years.	CAA 40-82F-1 CAA 40-82F-2 CAA 40-82F-3
16 Provide Shop Service Record	HVAC Technician Supervisor: • Send copies to FMD of all CAA 40-82F-1 on a weekly basis. Include those that have been updated to document follow-up verification or annual recurring testing.	CAA 40-82F-1
17 Maintain Service Records	FMD: • Request a copy of all service records from contractors, PWD Shops, and tenants. Maintain for 5 years. FEAD: • Collect the above information from all contractors and forward to FMD.	

Last Updated: 28-February-2020

Process Step	Associated Procedure	Resources
18 Calculate Annualized Leak Rate	FMD: - Use information from all Form CAA 40-82F-1 and any service records obtained from tenants or contracted technicians to calculate the annualized leak rate EACH time refrigerant is added to 50+ # equipment using formula below; $100 \times \frac{\left\{ \begin{array}{l} \# \text{ added past 365 days or since} \\ \text{last successful follow-up leak repair} \\ \text{verification test if < 1 year} \end{array} \right\}}{(\# \text{ normal charge})}$ - Perform leak rate calculations within one week of receiving data; - Maintain copies for 5 years.	
19 Provide Notification on Leaks > Allowable Annualized Rate	FMD: - Notify HVAC Technician Supervisor, EV, PW Core FM&S and PWO immediately of any leak that cannot be repaired within 20 days from leak discovery and that may exceed the allowable annualized leak rate: 30% industrial process refrigeration; 20% commercial refrigeration; 10% comfort cooling or other appliances not listed above. - Units that fail leak repair verification tests and are not secured are considered to still be leaking; PW with EV input must evaluate courses of action to comply with EPA regulations if 50+ # units leak above the allowable leak rate and cannot be repaired within 30 days of leak discovery. Options are: - secure unit pending repair; - within 30 days of leak discovery, request extension if necessary repair parts are on order; - within 30 days of leak discovery, prepare retrofit or retirement plan.	

Last Updated: 28-February-2020

Process Step	Associated Procedure	Resources
20 Extensions due to parts being on order	EV (upon notification from FMD): Request an extension to leak repair 30-day deadline if time between leak discovery and repair will exceed 30 days due to necessary repair parts being on order. EPA allows an extension request with up to 30 days after receipt of necessary repair parts to complete repairs (not to exceed 180 days from date of leak discovery); If date is to be extended, a new date and documentation of reason for change must be submitted to EPA within 30 days of identifying need to change date. FMD: • Provide EV information for request: - o Leak rate; - o Method used to determine full charge; - o Date appliance exceeded applicable leak rate; - o Location of leaks; - o Any repair work performed thus far and date completed; - o Estimate of when work will be completed. EV: • Prepare and submit extension request; • Keep dated copy.	
21 Retrofit or Retirement Plans	Prepare a retrofit or retirement plan if: • intent is to retrofit or retire appliance rather than repair leak; • leak cannot be identified and/or repaired; • appliance continues to leak above the allowable leak rate after repairs and verification tests. FMD: • Provide information required for plan: - o Appliance identification, location; - o Refrigerant type and full charge;	

Last Updated: 28-February-2020

Process Step	Associated Procedure	Resources
	- o If retrofitting: refrigerant type and full charge to be converted to; itemized procedure for converting appliance to different refrigerant, including changes required for compatibility with new refrigerant. - o Plans for recovered refrigerant disposition; - o Plans for appliance disposition (if retiring); - o Schedule for retrofit/retirement (must be complete within one year). EV: • Prepare retirement/retrofit plan, including dated signature of authorized official; • Request EPA relieve the retrofit or retirement obligation if, within 6 months of plan date, repairs are made and the unit is brought below applicable leak rate; • Keep plans onsite and available for inspection.	
22 50+ # appliances Chronically Leaking 125+ % in a Calendar Year	FMD (by Feb 1 each year): • Review leak repair records and identify all 50+ # appliances that leaked during the previous calendar year. This includes leaks that have been repaired; • Calculate total pounds leaked on each appliance over the calendar year using formula below. Include all leaks, even those that have been repaired. Treat each independent circuit as a separate appliance; $100 \times \frac{(\# \text{ added calendar year on all repairs})}{(\# \text{ normal charge})}$ • Report to EV all 50+ # appliances that leaked 125% or more of their full charge in the previous calendar year. Include description of efforts to identify leaks and repair appliances. EV (by March 1 each year): • Prepare and submit report to EPA.	-

Last Updated: 28-February-2020

Process Step	Associated Procedure	Resources
23 Copy of EPA Notifications	EV: Forward a copy of EPA notifications to the PW Core FM&S and PWO.	
24 Maintain a Master File	PW Core FM&S: - Maintain a master file that includes: - inventory of all 50+ # equipment; - all EPA Notifications; - copies of EV audits; - copies of FMD self-audits. - Perform annual review of all equipment containing 50+ # of refrigerant in Maximo: - Verify that all required information is clear and easily understood on the report from Maximo with no miscellaneous or irrelevant data; - Verify that all 50+ # units are correctly identified in Maximo as "ODS" in comments section of Assets module.	-
25 Compliance Audits	EV: Conduct annual compliance audits to ensure that all regulations are being met.	-
26 Self-Audits	FMD: - Conduct semi-annual audits of HVAC Technician records to verify that the BMS process is being followed. At minimum: - Identify additional equipment that has not been inventoried and add to inventory in Maximo. Make sure 50+ # units are flagged as "ODS" in comments section of Assets module; - Use CAA 40-82F-2 forms to identify discrepancies between refrigerant amounts used and what is placed on HVAC technician's truck; - Verify receipt of all CAA 40-82F-1 forms; verify leak rates calculated for all CAA 40-82F-1 received.	-

Last Updated: 28-February-2020

Process Step	Associated Procedure	Resources
27 Audit Results	FMD: - Forward self-audit findings to EV air manager, EV Core Air SME, PW Core FM&S, PWD Production Division Director or Maintenance Superintendent, affected PWD Shop Supervisor(s)", and PWO. EV: - Forward compliance audit findings to PWO, Deputy PWO, Production Division Director, FMD Director, EV Core Air SME, Installation Environmental Program Director (IEPD), and PW Core FM&S.	-
28 Provide Follow-Up Training	EV: Provide additional training on the NAVFAC process as necessary. The training will be provided at minimum biennially at each PWD.	-

APPENDIX B. “CAA 40-82F” FORMS

Last Updated: 28-February-2020

Form CAA 40-82F-1		Service Invoice (50+ # Appliances)				50+	
Technician Name		PWD		Service Date			
Building No. and Location (ex: roof, N side)		Work Order or Contract PO No. (MRI No. for refrigerant purchases)		Refrigerant No.			
Appliance Description		Asset No. (Model & S/N if unknown)		Pounds Normal Charge			
Service Description <i>For leaks include cause, location on appliance, repair description</i>					Pounds refrigerant (+/-) V/R/D ¹		
Date leak DISCOVERED		Date leak REPAIRED		Days leaked		Pounds refrigerant leaked	
Initial leak repair verification test ²	Date: _____ Result: _____	Method: _____		Follow-up leak repair verification ²	Date: _____ Result: _____	Method: _____	
Recurring leak inspection ³	Date: _____ Result: _____	Method: _____		Initial to certify all visible and accessible parts of appliance were inspected during recurring leak inspection			

AREA BELOW FOR FMD: Complete and notify Environmental (EV) immediately of leak >10 % that could last 20 days

Date Received		Completed by (Name)		Report to EV? (Y/N)		Reported to (Name)	
Annualized Leak Rate= 100* [____ # added past 365 days ⁴] ÷ [____ #normal charge] = [____ %]						Date Reported	

¹ (V)irgin, (R)ecovered/recycled, (D)isposal/unusable. Inspect any appliance requiring refrigerant addition for leaks.² Perform initial before refrigerant add, follow-up within **10** days of return to normal operation. Notify supervisor of failed tests.³ Annually until < leak rate 1 yr: inspect for any leaks, not just most recent repair; record leak location/notify supervisor.⁴ or since last successful follow-up verification test showing all identified leaks were repaired if < 1 yr.

HVAC tech keep original 5 years, copy supervisor daily; supervisor keep 5 years, copy FMD weekly; FMD keep 5 years.

Form CAA 40-82F-2		Service and Inventory Tracking - All Appliances					ALL	
Technician Name				PWD			Pounds (+/-) V/R/D	
Date	WO or Contract PO No. (MRI No. for refrigerant purchases)	Building No. and Location Details	Appliance & Service Description		Asset No. (Model & S/N if unknown)	Pounds Normal Charge	Type Refrigerant	If (D) log date & disposition (ex: tool room, VRS)

Pounds disposed this month									
Type refrig	Amount	Type refrig	Amount	Type refrig	Amount	Type refrig	Amount	Type refrig	Amount

(V)irgin, (R)ecovered / recycled, (D)unusable for disposal

Complete for all refrigerant (+) added to or (-) removed from inventory or HVAC units. Additionally, complete CAA-40-82F-1 if 50+ pound unit and CAA-40-82F-3 if appliance disposed. **Document if no refrigerant work that week.** Keep forms on truck 5 years, make available upon request for inspection, and provide copies to supervisor weekly. Supervisor keeps copies 5 years.

Last Updated: 28-February-2020

Form CAA 40-82F-3	Disposal Invoice - All Appliances		DISPOSAL - ALL
Technician Name		PWD	
Technician's Office Location (Building or Street Address)		Work Order No. (or Contract PO No.)	
Equipment Location (Building)		Additional location (room, roof, N side, etc.)	
Equipment Asset No.		Model No. (if Asset No. unknown)	
Appliance Description (industry or MAXIMO UNIFORMAT name)		Serial No. (if Asset No. unknown)	
Refrigerant Recovery Date		Refrigerant No. (Type) Removed (example: R-22)	
Refrigerant Amount Removed (indicate 'pounds' or 'ounces')		Refrigerant Disposition: [R] Recovered for re-use or [D] Recovered for disposal	
If [D], Refrigerant Disposal Date		If [D], what entity took possession of refrigerant? (example: tool room, VRS)	
Oil Type Removed		Oil Amount Removed	
Technician Signature (certifies that all refrigerant and oil that had not leaked was removed from appliance prior to disposal in accordance with 40 CFR 82 and that a weatherproof label has been affixed to the appliance stating the same)			

Technician shall keep original forms on work truck for 5 years, make available upon request for inspection, and provide copies to supervisor weekly. Supervisor shall maintain copies for 5 years.

APPENDIX C. RETROFIT / RETIREMENT PLAN FORM



HVAC Retrofit/Retirement Plan

This plan must be signed by an authorized company official, dated, accessible at the site of the appliance in paper copy or electronic format, and available for EPA inspection upon request.

HVAC Appliance Identification:

Location	Enter Building Number
Manufacturer	Enter Unit Manufacturer
Model	Enter Unit Model
Serial Number	Enter Unit Serial Number
Circuit Number	Enter Circuit number (if more than one)
Installation Date	Date Unit Installed
Current Refrigerant	Enter Refrigerant Type
Capacity at Full Charge (LBS)	Enter LBS
Leak Rate	Enter Leak Rate %
Date Leak Discovered (Date Refrigerant added to circuit that exceeds applicable leak rate threshold)	Enter date MM/DD/YY
Date of Initial Repair Attempt Repair required within 30 days of "discovery."	Enter date MM/DD/YY
Was Repair Attempted?	Yes/No
Date of Failed Initial Verification Test Perform Initial Verification test (IVT) after repair (before recharging) within 30 days of leak rate exceedance.	Enter date MM/DD/YY
Date of Failed Follow-up Verification Test Perform Follow-Up Verification Test (FVT) within 10 days of IVT or return to normal operation.	Enter date MM/DD/YY
One year past failed FVT:	Enter date MM/DD/YY
When was decision made to retrofit or retire	Enter date MM/DD/YY
One year past Retrofit / Retire Decision	Enter date MM/DD/YY
Description of repair actions, and dates of occurrences to date:	
Enter description here.	

Retrofit Information:

New Refrigerant	Enter Refrigerant Type
Capacity at Full Charge (LBS)	Enter LBS
Estimated Retrofit Completion Date Not to exceed one year past Retrofit Decision.	Enter Date MM/DD/YY

HVAC Retrofit/Retirement Plan

Itemized procedure for converting appliance to a different refrigerant, including changes required for compatibility with new substitute, if needed:

Enter description here.

Retirement Information:

New Appliance Manufacturer	Enter Manufacturer Name
New Appliance Model	Enter Model Name
New Refrigerant	Enter Refrigerant Type
Capacity at Full Charge (LBS)	Enter LBS
Estimated Installation Date	Enter Date MM/DD/YY

Plan for disposition of current appliance, if retired, and schedule, not to exceed on year for completion from failed FVT in Section 1 of this form:

Enter description here.

Other Information

Disposition Plan for recovered refrigerant (if retrofit or retire):

Enter description here.

Plan certified by:

Signature

Date

Printed Name

Title

APPENDIX D. TECHNICIAN CERTIFICATION

Certification Level	Allowed activities
Type I	Maintain, service, or repair small appliances with 5 pounds or less of refrigerant
Type II	Maintain, service, repair, or dispose of high- or very high-pressure appliances (an appliance using a refrigerant with a boiling point between -50° and 10°C); typically comfort cooling appliances with greater than 50 lbs of refrigerant charge. Type II technicians can also maintain, service, or repair MVAC-like appliances.
Type III	Maintain, service, repair, or dispose of low-pressure appliances (an appliance using a refrigerant with a boiling point above 10°C at atmospheric pressure); typically, industrial cooling systems such as large building chillers.
Type IV (Universal Technician)	Maintain, service, repair, or dispose of low and high-pressure equipment must be certified as Universal Technicians.

APPENDIX E. CLASS I REFRIGERANT PHASE OUT LIST

Class 1 Controlled Substances	Ozone Depleting Potential (ODP)
A. Group I:	
CFCl ₃ -Trichlorofluoromethane (CFC-II)	1.0
CF ₂ Cl ₂ -Dichlorodifluoromethane (CFC-12)	1.0
C ₂ F ₃ Cl ₃ -Trichlorotrifluoroethane (CFC-113)	0.8
C ₂ F ₄ Cl ₂ -Dichlorotetrafluoroethane (CFC-114)	1.0
C ₂ F ₅ Cl-Monochloropentafluoroethane (CFC-115)	0.6
All Isomers of the above chemicals	
B. Group II:	
CF ₂ ClBr-Bromochlorodifluoromethane (Halon-1211)	3.0
CF ₃ Br-Bromotrifluoromethane (Halon-1301)	10.0
C ₂ F ₄ Br ₂ -Dibromotetrafluoroethane (Halon-2402)	6.0
All Isomers of the above chemicals	
C. Group III:	
CF ₃ Cl-Chlorotrifluoromethane (CFC-13)	1.0
C ₂ FCl ₅ -(CFC-111)	1.0
C ₂ F ₂ Cl ₄ -(CFC-112)	1.0
C ₃ FCl ₇ -(CFC-211)	1.0
C ₃ F ₂ Cl ₆ -(CFC-212)	1.0
C ₃ F ₃ Cl ₅ -(CFC-213)	1.0
C ₃ F ₄ Cl ₄ -(CFC-214)	1.0
C ₃ F ₅ Cl ₃ -(CFC-215)	1.0
C ₃ F ₆ Cl ₂ -(CFC-216)	1.0
C ₃ F ₇ Cl-(CFC-217)	1.0
All Isomers of the above chemicals	
D. Group IV:	
CCl ₄ -Carbon Tetrachloride	1.1
E. Group V:	
C ₂ H ₃ Cl ₃ -1,1,1 Trichloroethane (Methyl chloroform)	0.1

Class 1 Controlled Substances	Ozone Depleting Potential (ODP)
Isomers of the above chemical except 1,1,2-trichloroethane	
F. Group VI:	
CH ₃ Br—Bromomethane (Methyl Bromide)	0.7
G. Group VII:	
CHFBR ₂	1.00
CHF ₂ Br (HBFC-2201)	0.74
CH ₂ FBr	0.73
C ₂ HFBR ₄	0.3-0.8
C ₂ HF ₂ Br ₃	0.5-1.8
C ₂ HF ₃ Br ₂	0.4-1.6
C ₂ HF ₄ Br	0.7-1.2
C ₂ H ₂ FBR ₃	0.1-1.1
C ₂ H ₂ F ₂ Br ₂	0.2-1.5
C ₂ H ₂ F ₃ Br	0.7-1.6
C ₂ H ₂ FBR ₂	0.1-1.7
C ₂ H ₃ F ₂ Br	0.2-1.1
C ₂ H ₄ FBr	0.07-0.1
C ₃ HFBR ₆	0.3-1.5
C ₃ HF ₂ Br ₅	0.2-1.9
C ₃ HF ₃ Br ₄	0.3-1.8
C ₃ HF ₄ Br ₃	0.5-2.2
C ₃ HF ₅ Br ₂	0.9-2.0
C ₃ HF ₆ Br	0.7-3.3
C ₃ H ₂ FBR ₅	0.1-1.9
C ₃ H ₂ F ₂ BR ₄	0.2-2.1
C ₃ H ₂ F ₃ Br ₃	0.2-5.6
C ₃ H ₂ F ₄ Br ₂	0.3-7.5
C ₃ H ₂ F ₅ BR	0.9-14
C ₃ H ₃ FBR ₄	0.08-1.9
C ₃ H ₃ F ₂ Br ₃	0.1-3.1

Class 1 Controlled Substances	Ozone Depleting Potential (ODP)
C ₃ H ₃ F ₃ Br ₂	0.1-2.5
C ₃ H ₃ F ₄ Br	0.3-4.4
C ₃ H ₄ FBr ₃	0.03-0.3
C ₃ H ₄ F ₂ Br ₂	0.1-1.0
C ₃ H ₄ F ₃ Br	0.07-0.8
C ₃ H ₅ FBr ₂	0.04-0.4
C ₃ H ₅ F ₂ Br	0.07-0.8
C ₃ H ₆ FB	0.02-0.7
H. Group VIII:	
CH ₂ BrCl (Chlorobromomethane)	0.12

- a) Source: Appendix A to Subpart A of Part 82—Class I Controlled Substances; [60 FR 24986, May 10, 1995, as amended at 68 FR 42892, July 18, 2003]
- b) Ozone Depletion Potential (ODP) of a compound is unitless and is the relative amount of degradation the compound can cause to the ozone layer. The value is relative to R-12 refrigerant, with an ODP of 1.

APPENDIX F. CLASS II REFRIGERANT PHASE OUT LIST

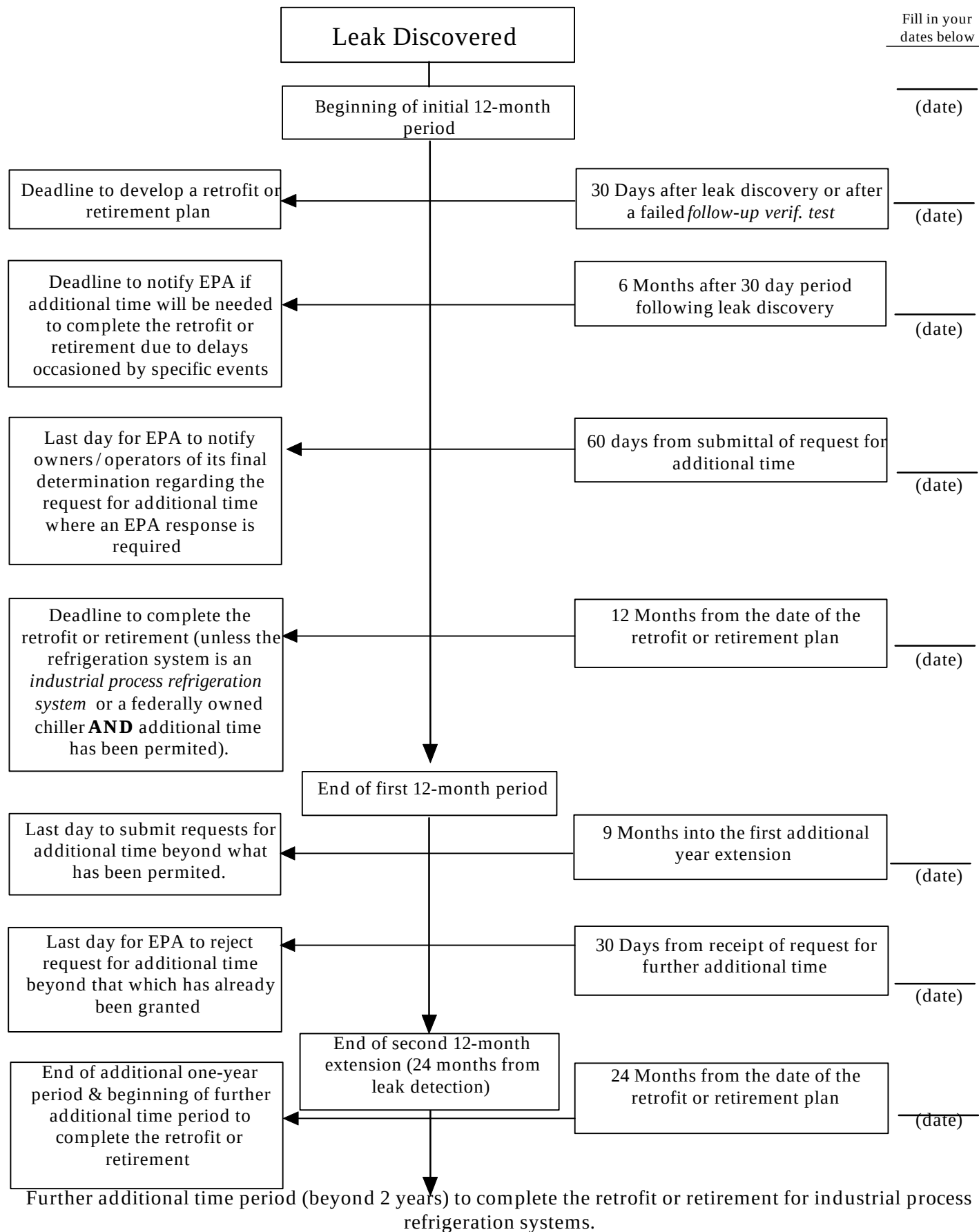
Controlled Class II Substance	Ozone Depleting Potential (ODP)
HCFC-21 (CH ₂ Cl ₂) Dichlorofluoromethane	0.04
HCFC-22 (CHF ₂ Cl) Monochlorodifluoromethane	0.055
HCFC-31 (CH ₂ Cl) Monochlorofluoromethane	0.02
HCFC-121 (C ₂ HFCl ₄) Tetrachlorofluoroethane	0.01-0.04
HCFC-122 (C ₂ HF ₂ Cl ₃) Trichlorodifluoroethane	0.02-0.08
HCFC-123 (C ₂ HF ₃ Cl ₂) Dichlorotrifluoroethane	0.02
HCFC-124 (C ₂ HF ₄ Cl) Monochlorotetrafluoroethane	0.022
HCFC-131 (C ₂ H ₂ FC ₃) Trichlorofluoroethane	0.007-0.05
HCFC-132 (C ₂ H ₂ F ₂ Cl ₂) Dichlorodifluoroethane	0.008-0.05
HCFC-133 (C ₂ H ₂ F ₃ Cl) Monochlorotrifluoroethane	0.02-0.06
HCFC-141 (C ₂ H ₃ FC ₂) Dichlorofluoroethane	0.005-0.07
HCFC-141b (CH ₃ CFCl ₂) Dichlorofluoroethane	0.11
HCFC-142 (C ₂ H ₃ F ₂ Cl) Chlorodifluoroethane	0.008-0.07
HCFC-142b (CH ₃ CF ₂ Cl) Monochlorodifluoroethane	0.065
HCFC-151 (C ₂ H ₄ Cl) Chlorofluoroethane	0.003-0.005
HCFC-221 (C ₃ HFCl ₆) Hexachlorofluoropropane	0.015-0.07
HCFC-222 (C ₃ HF ₂ Cl ₅) Pentachlorodifluoropropane	0.01-0.09
HCFC-223 (C ₃ HF ₃ Cl ₄) Tetrachlorotrifluoropropane	0.01-0.08
HCFC-224 (C ₃ HF ₄ Cl ₃) Trichlorotetrafluoropropane	0.01-0.09
HCFC-225 (C ₃ HF ₅ Cl ₂) Dichloropentafluoropropane	0.02-0.07
HCFC-225ca (CF ₃ CF ₂ CHCl ₂) Dichloropentafluoropropane	0.025
HCFC-225cb (CF ₂ ClCF ₂ CHClF) Dichloropentafluoropropane	0.033
HCFC-226 (C ₃ HF ₆ Cl) Monochlorohexafluoropropane	0.02-0.1
HCFC-231 (C ₃ H ₂ FC ₅) Pentachlorofluoropropane	0.05-0.09
HCFC-232 (C ₃ H ₂ F ₂ Cl ₄) Tetrachlorodifluoropropane	0.008-0.1
HCFC-233 (C ₃ H ₂ F ₃ Cl ₃) Trichlorotrifluoropropane	0.007-0.23
HCFC-234 (C ₃ H ₂ F ₄ Cl ₂) Dichlorotetrafluoropropane	0.01-0.28
HCFC-235 (C ₃ H ₂ F ₅ Cl) Monochloropentafluoropropane	0.03-0.52

Controlled Class II Substance	Ozone Depleting Potential (ODP)
HCFC-241 (C ₃ H ₃ FCI ₄) Tetrachlorofluoropropane	0.004-0.09
HCFC-242 (C ₃ H ₃ F ₂ Cl ₃) Trichlorodifluoropropane	0.005-0.13
HCFC-243 (C ₃ H ₃ F ₃ Cl ₂) Dichlorotrifluoropropane	0.007-0.12
HCFC-244 (C ₃ H ₃ F ₄ Cl) Monochlorotetrafluoropropane	0.009-0.14
HCFC-251 (C ₃ H ₄ FCI ₃) Monochlorotetrafluoropropane	0.001-0.01
HCFC-252 (C ₃ H ₄ F ₂ Cl ₂) Dichlorodifluoropropane	0.005-0.04
HCFC-253 (C ₃ H ₄ F ₃ Cl) Monochlorotrifluoropropane	0.003-0.03
HCFC-261 (C ₃ H ₅ FCI ₂) Dichlorofluoropropane	0.002-0.02
HCFC-262 (C ₃ H ₅ F ₂ Cl) Monochlorodifluoropropane	0.002-0.02
HCFC-271 (C ₃ H ₆ FCI) Monochlorofluoropropane	0.001-0.03

- a) According to Annex C of the Montreal Protocol, "Where a range of ODPs is indicated, the highest value in that range shall be used for the purposes of the Protocol. The ODPs listed as single value have been determined from calculations based on laboratory measurements. Those listed as a range are based on estimates and are less certain. The range pertains to an isomeric group. The upper value is the estimate of the ODP of the isomer with the highest ODP, and the lower value is the estimate of the ODP of the isomer with the lowest ODP.
- b) This table includes isomers of the substances above, regardless of whether the isomer is explicitly listed on its own.
- c) Ozone Depletion Potential (ODP) of a compound is unitless and is the relative amount of degradation the compound can cause to the ozone layer. The value is relative to R-12 refrigerant, with an ODP of 1.
- d) Source: Appendix F to Subpart A of Part 82—Listing of Ozone-Depleting Chemicals

**APPENDIX G. REGULATORY TIMELINE & CHECKLIST FOR LEAKING
REFRIGERATION SYSTEMS**

Regulatory Timeline & Checklist for Retrofit & Retirement Activities



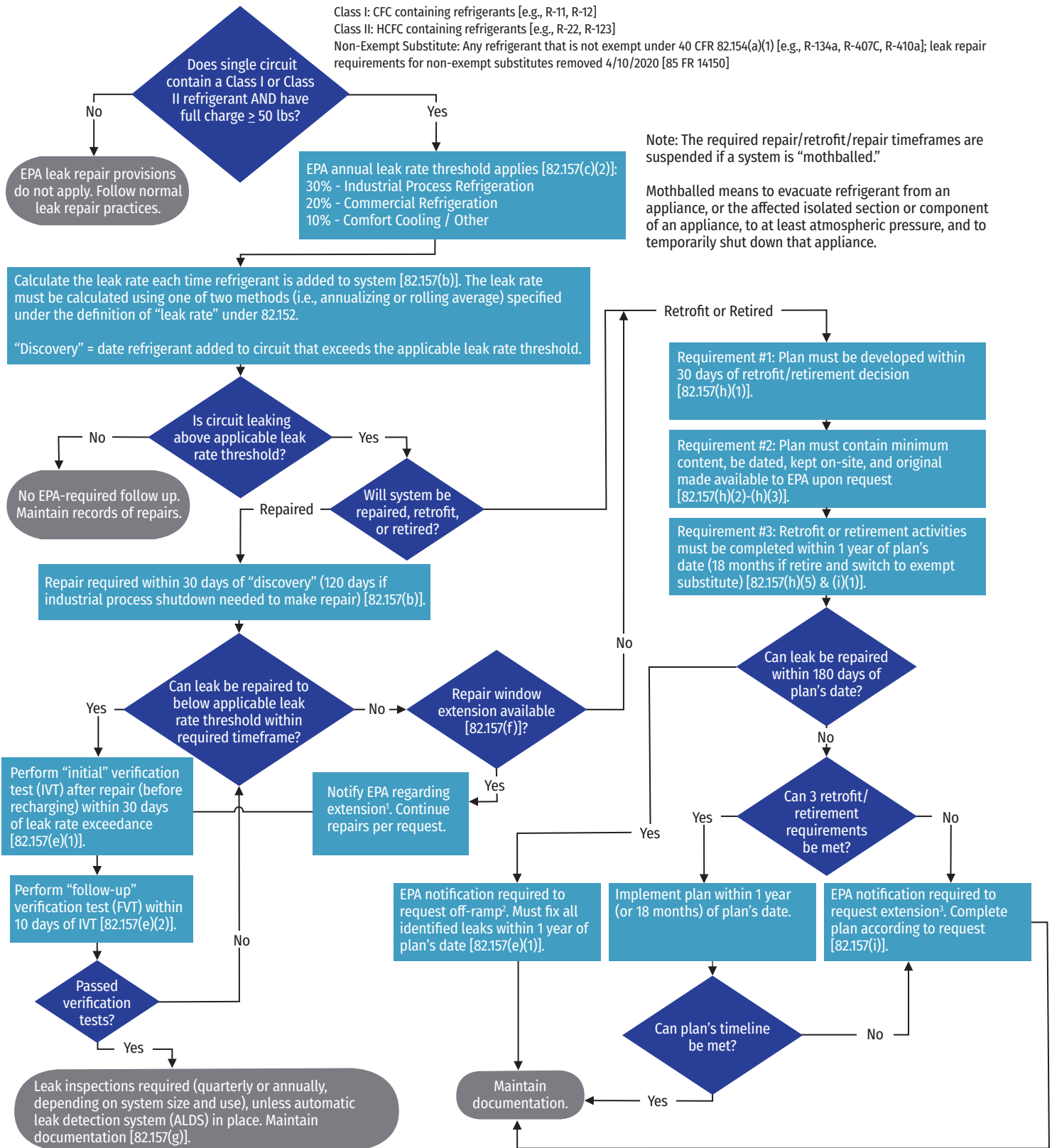
APPENDIX H. DECISION GUIDE FOR LEAKING REFRIGERATION SYSTEMS

DECISION GUIDE FOR LEAKING REFRIGERATION SYSTEMS

Class I: CFC containing refrigerants [e.g., R-11, R-12]

Class II: HCFC containing refrigerants [e.g., R-22, R-123]

Non-Exempt Substitute: Any refrigerant that is not exempt under 40 CFR 82.154(a)(1) [e.g., R-134a, R-407C, R-410a]; leak repair requirements for non-exempt substitutes removed 4/10/2020 [85 FR 14150]



¹Notify the Environmental Department on systems that cannot be repaired within the required timeframes. Extension requests, if applicable, must be submitted to EPA within 30 days of discovering leaks above the applicable leak rate threshold. Refer to 82.157(f)(4) for details regarding repair window extension requests.

²Retrofit/retirement plan off-ramp requests considered approved unless EPA disapproves within 60 days of receipt of request. Refer to 82.157(h)(5)(ii) for details regarding retrofit/retirement plan off-ramp requests.

³Retrofit/retirement plan extension requests considered approved unless EPA disapproves within 60 days of receipt of request. Refer to 82.157(i) for details regarding retrofit/retirement plan extension requests.

APPENDIX I. GLOSSARY

This glossary provides important terms used in this guidance.

Appliance	Any device that contains and uses a Class I or Class II substances as a refrigerant and is used for household, commercial, or industrial purposes, including any air conditioner, refrigerator, chiller or freezer. Essentially any sort of colling equipment that uses a Class I or Class II substance as a refrigerant is an “appliance.”
Class I refrigerant	Most Class I substances are usually chlorofluorocarbons (CFCs). A complete listing of class I substances can be found at 40 CFR Part 82, Subpart A, Appendix A.
Class II refrigerant	Class II substances are hydrochlorofluorocarbons (HCFCs). A complete listing of class II substances can be found at 40 CFR Part 82, Subpart A, Appendix B.
Commercial refrigeration	Refrigeration appliances utilized in the retail food and cold storage warehouse sectors. Retail food includes the refrigeration equipment found in supermarkets, convenience stores, restaurants, and other food service establishments. Cold storage includes the equipment used to store meat, produce, dairy products, and other perishable goods.
Critical component	A component without which an industrial process refrigeration system will not function, will be unsafe in its intended environment, and/or will be subject to failures that would cause the industrial process served by the system to be unsafe.
Custom-built	The system or any of its critical components cannot be purchased and/or installed without being uniquely designed, fabricated, and/or assembled to satisfy a specific set of industrial process conditions.
Follow-up verification test (FVT)	Those tests that involve checking the repairs within 30 days of returning the system to normal operating characteristics or conditions. Follow-up verification tests for equipment from which the refrigerant charge has been evacuated means a test conducted after the system or portion of the system has resumed operation at normal operating characteristics or conditions of temperature and pressure, except in cases where sound professional judgment dictates that these tests will be more meaningful if performed prior to the return to normal operating characteristics or conditions. A follow-up verification test for a system that has not been evacuated means a reverification test conducted after the initial verification test and usually within 30 days of normal operating conditions. Where a system is not evacuated, it is only necessary to conclude any required changes in pressure, temperature, or other conditions to return the system to normal operating characteristics or conditions.
Full charge	The amount of refrigerant required for normal operating characteristics or conditions of industrial process refrigeration systems and certain federally owned appliances as determined using one or a combination of the following four methods: • Using the system manufacturer’s determination of the correct full charge for the system; • Determining the full charge by appropriate calculations based on component sizes, density of refrigerant, volume of piping, and all relevant considerations;

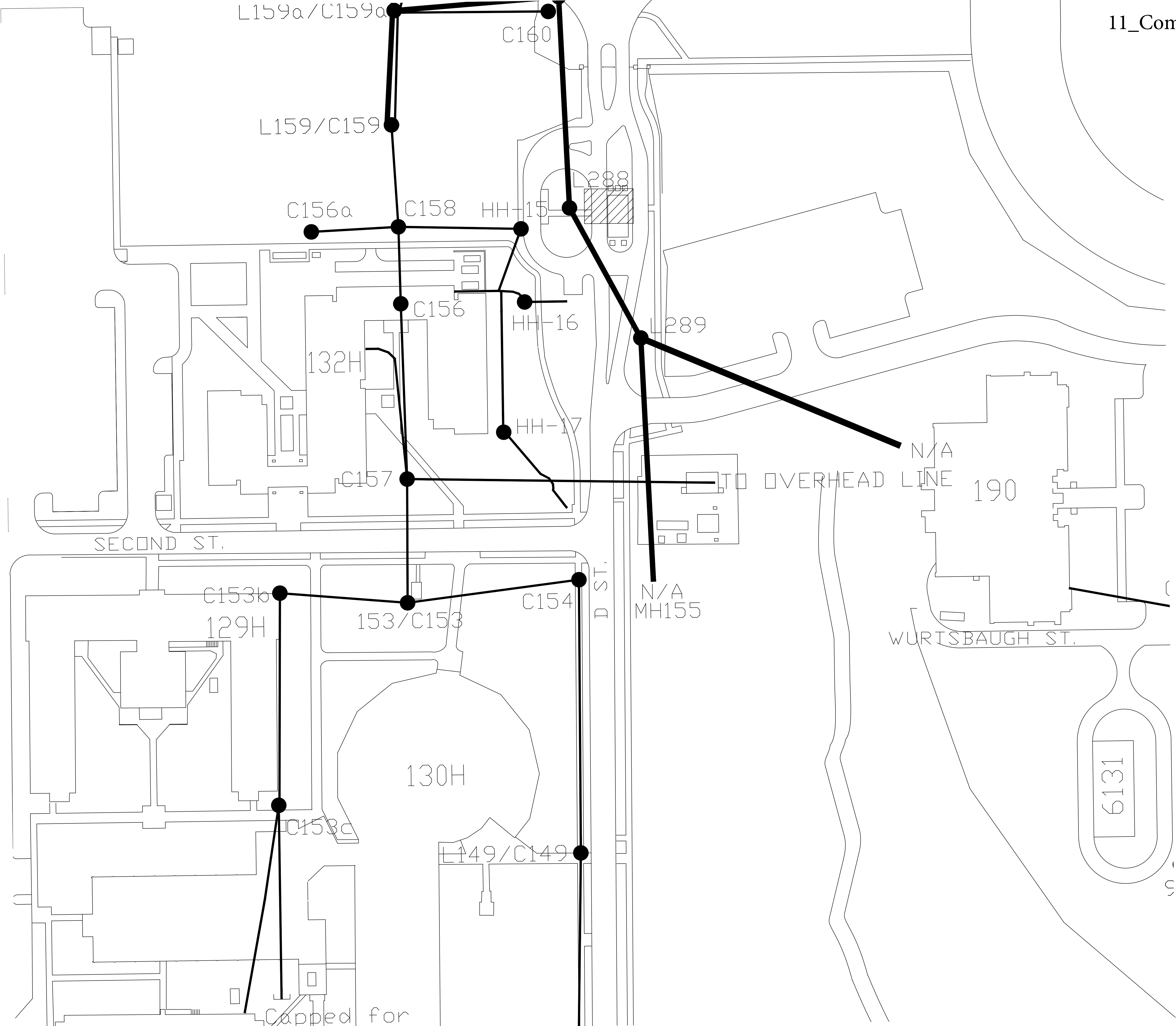
- Using actual measurements of the amount of refrigerant added or evacuated from an industrial process refrigeration system; and/or
- Using an established range based on the best available data, regarding the normal operating characteristics and conditions for the system, where the midpoint of the range will serve as the full charge and where records are maintained in accordance with 40 CFR 82.166(q).

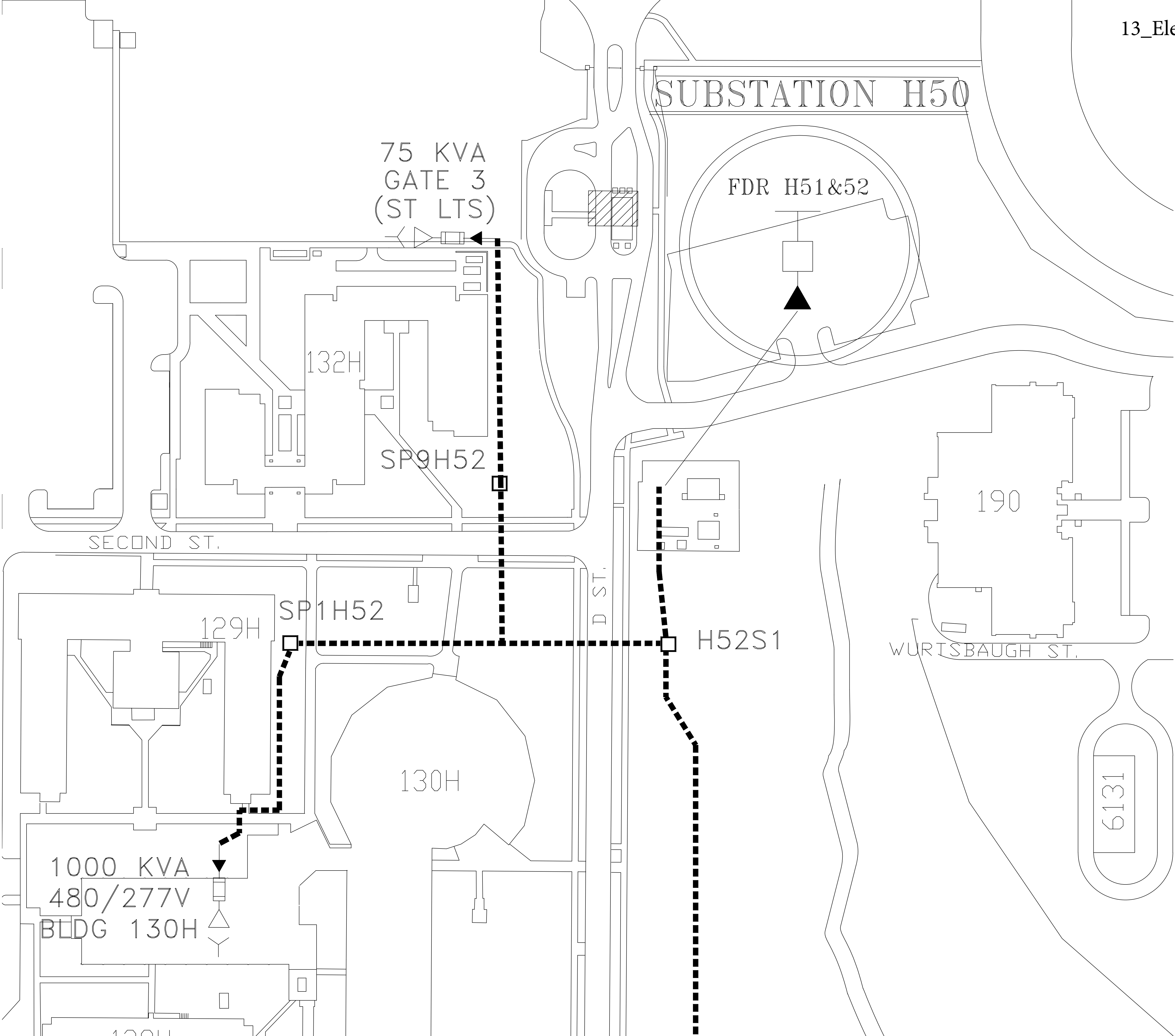
Industrial process refrigeration	Complex, customized systems used in the chemical, pharmaceutical, petrochemical, and manufacturing industries. These systems are directly linked to the industrial process. This sector also includes industrial ice machines, appliances used directly in the generation of electricity, and ice rinks. Where one appliance is used for both industrial process refrigeration and other applications, it will be considered an industrial process refrigeration system if 50 percent or more of its operating capacity is used for industrial process refrigeration.
Industrial process shutdown	An industrial process or facility temporarily ceases to operate or manufacture whatever is being produced at the facility.
Initial verification test (IVT)	Those leak tests that are conducted as soon as practicable after the repair is completed. If the system or isolated portion is evacuated, it means a test conducted prior to replacing of the full refrigerant charge and before the system or portion of the system has reached operation at normal operating characteristics or conditions of temperature and pressure. An initial verification test conducted without the evacuation of the refrigerant charge means a test conducted as soon as practicable after the completion of the repair work.
Normal operating characteristics or conditions	Temperatures, pressures, fluid flows, speeds, and other characteristics that would normally be expected for a given process load and ambient condition during operation. Normal operating characteristics or conditions are marked by the absence of atypical conditions affecting the operation of the refrigeration system.
Opening	Any service, maintenance, or repair on an appliance that would release class I or class II refrigerant from the appliance to the atmosphere unless the refrigerant were recovered previously from the appliance. Connecting and disconnecting hoses and gauges to and from the appliance to measure pressures within the appliance and to add refrigerant to or recover refrigerant from the appliance shall not be considered "opening."
Reclaim refrigerant	Reprocess refrigerant to at least the purity specified in appendix A to 40 CFR part 82, subpart F (based on ARI Standard 700-1993, Specifications for Fluorocarbon and Other Refrigerants) and verify this purity using the analytical methodology prescribed in appendix A. In general, reclamation involves the use of processes or procedures available only at a reprocessing or manufacturing facility.
Recover refrigerant	To remove refrigerant in any condition from an appliance without necessarily testing or processing it in any way.
Refrigerant circuit	The parts of an appliance that are normally connected to each other (or are separated only by internal valves) and are designed to contain refrigerant.

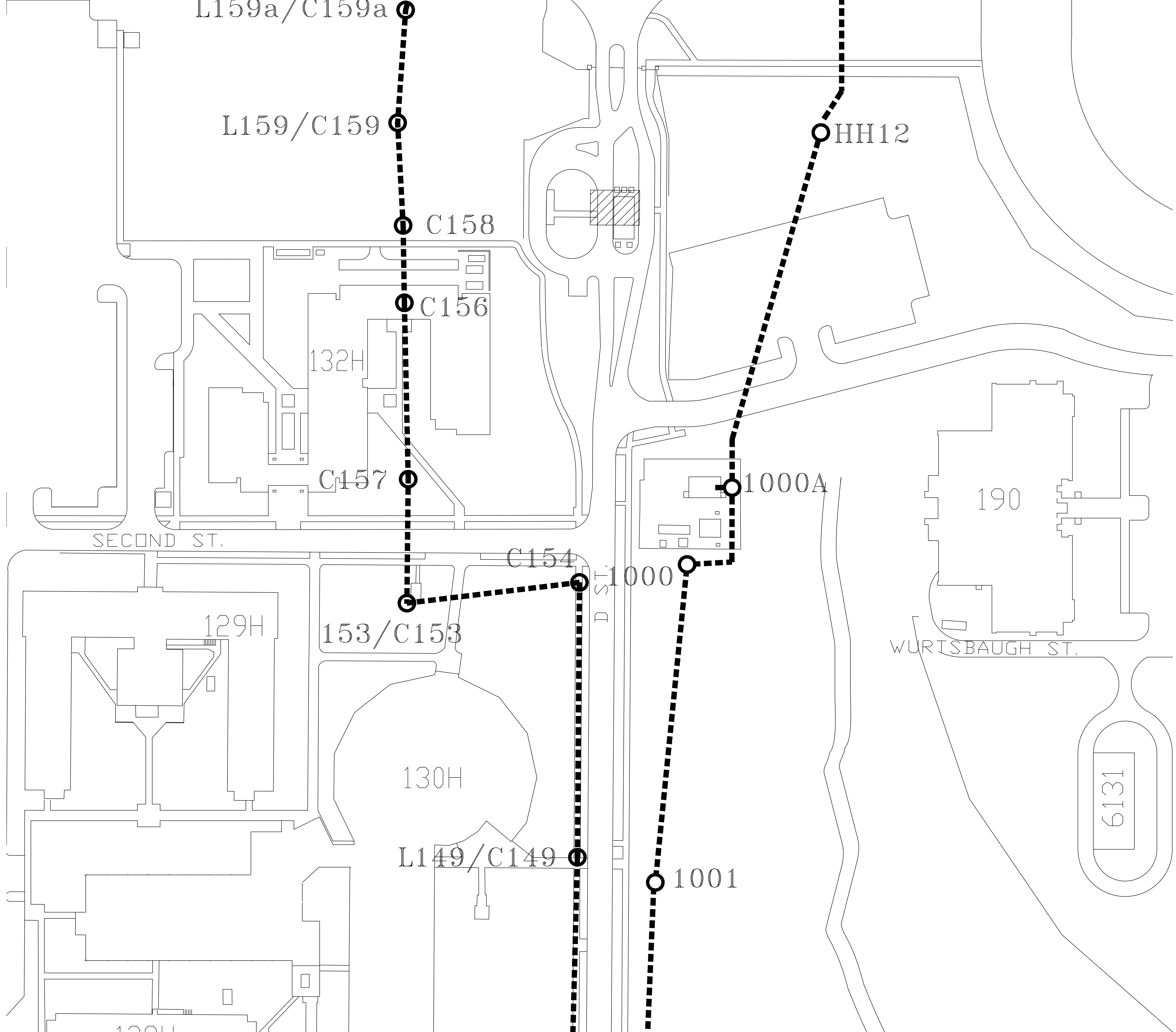
Sound professional judgement	A decision-making process that uses a combination of logic and operational experience, with methods of calculation that are practical, based on training, experience, and education.
Suitable replacement refrigerant	A refrigerant acceptable under section 612(c) of the Clean Air Act Amendments of 1990 and all regulations promulgated under that section, and compatible with other materials which it may come into contact. The SNAP-approved refrigerant must also be able to achieve the temperatures required for the affected industrial process in a technically feasible manner. NOTE: The Significant New Alternatives Policy (SNAP) program implements section 612 of the amended Clean Air Act of 1990. SNAP requires EPA to evaluate substitutes for ozone-depleting substances and disapprove the use of unacceptable substitutes. SNAP generates lists of acceptable and unacceptable substitutes for each of the major industrial use sectors.
System mothballing	Intentional shutdown of a refrigeration system undertaken for an extended period of time by the owners or operators of that facility, where the refrigerant has been evacuated from the appliance, or the affected isolated section of the appliance, at least to atmospheric pressure.
Technician	Any person who performs maintenance, service, or repair that could be reasonably expected to release class I or class II refrigerants from appliances, except for MVACs, into the atmosphere. Technician also means any person who performs disposal of appliances, except for small appliances, MVACs, and MVAC-like appliances, that could be reasonably expected to release class I or class II refrigerants from the appliances into the atmosphere. Performing maintenance, service, or repair, or disposal could be reasonably expected to release refrigerants only if the activity is reasonably expected to violate the integrity of the refrigerant circuit. Such activity includes attaching and detaching hoses and gauges to and from the appliance to add or remove refrigerant or to measure pressure and adding refrigerant to and removing refrigerant from the appliance. Activities such as painting the appliance, rewiring an external electrical circuit, replacing insulation on a length of pipe, or tightening nuts and bolts on the appliance are not reasonably expected to violate the integrity of the refrigerant circuit. Performing maintenance, service, repair, or disposal of appliances that have been evacuated pursuant to 40 CFR 82.156 could not be reasonably expected to release refrigerants from the appliance unless the maintenance, service, or repair consists of adding refrigerant to the appliance. Technician includes but is not limited to installers, contractor employees, in-house service personnel, and in some cases, owners or operators.

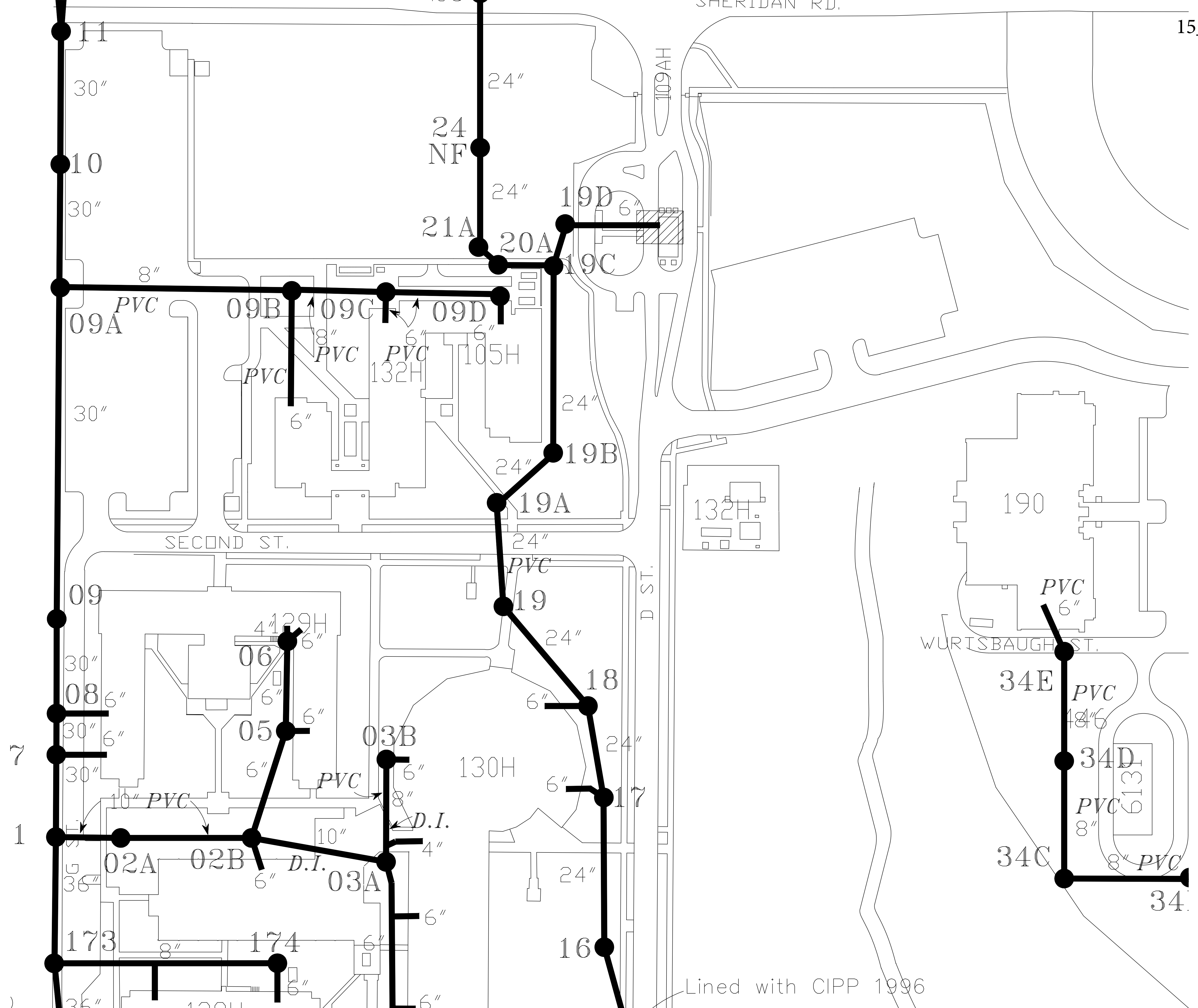
APPENDIX J. LIST OF COMMONLY USED ACRONYMS & ABBREVIATIONS

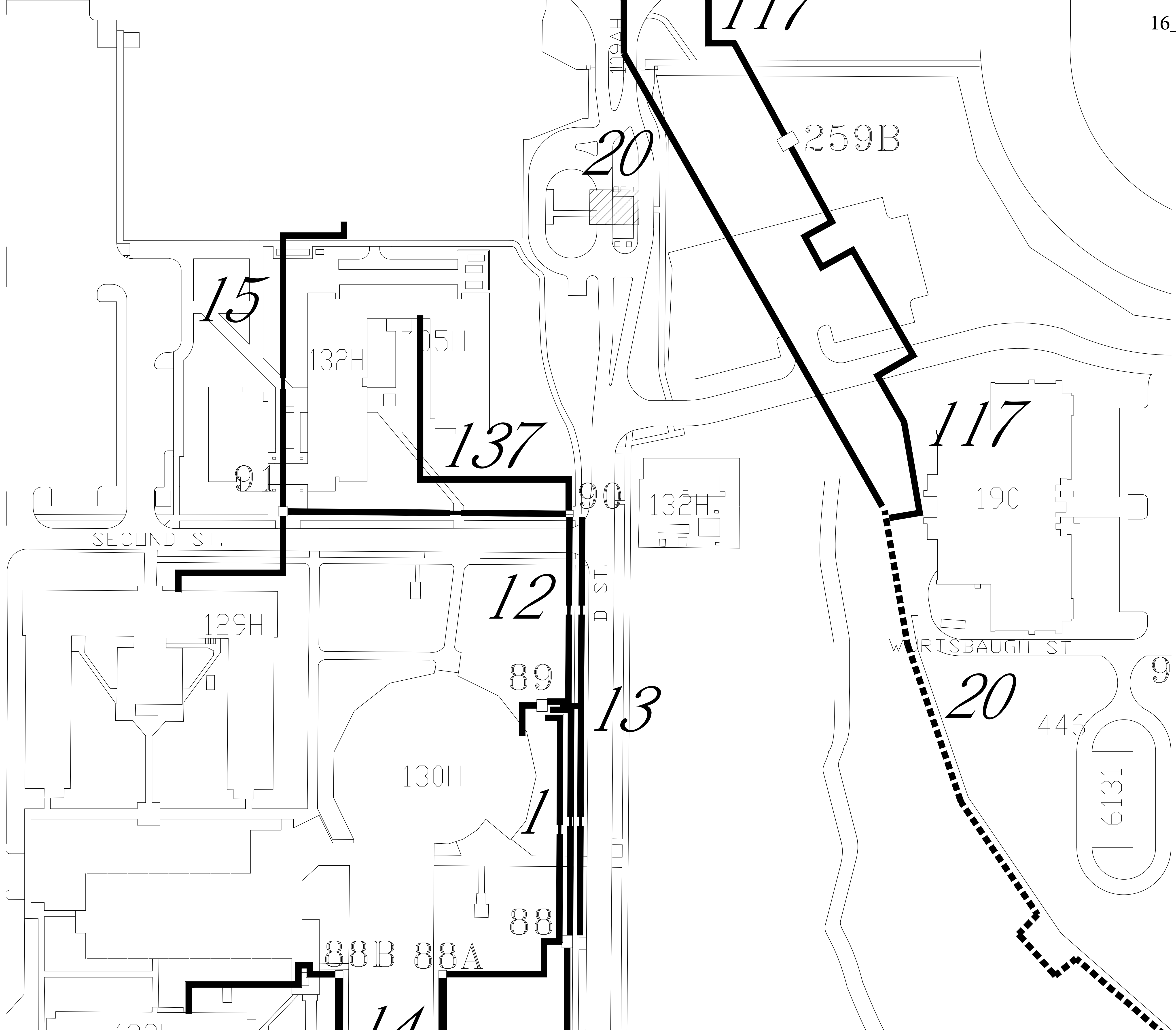
AC&R	Air Conditioning and Refrigeration
ASHRAE	American Society of Heating, Refrigeration, and Air Conditioning Engineers
° C	Degrees Celsius
CAA	Clean Air Act
CAS No.	Chemical Abstract Number
CFC	Chlorofluorocarbon
CFR	Code of Federal Regulations
CU	Classification Unit
DOD	U.S. Department of Defense
DON	U.S. Department of the Navy
DOT	US. Department of Transportation
DLA	Defense Logistics Agency
DRMO	Defense Reutilization and Marketing Office
EPA	U.S. Environmental Protection Agency
EV	Environmental Department (Installation Environmental)
FEAD	Facilities Engineering Acquisition Department
FMD	Facilities Management Division (PW Core)
FM&S	Facility Management & Sustainment Product Line (PW Core)
° F	Degrees Fahrenheit
FVT	Follow-up Verification Test
HCFC	Hydrochlorofluorocarbons
HFC	Hydrofluorocarbons
Hg	Mercury
HVAC	Heating, ventilation, and air conditioning
HVAC&R	Heating, ventilation, air conditioning, and refrigeration
IVT	Initial Verification Test
lbs	pounds
MEC	Mechanical
mm	millimeter
MVAC	Motor vehicle air conditioners
ODS	Ozone-depleting substances
ODP	Ozone-depleting potential
oz	ounce
PFC	Perfluorocarbon
psig	pounds per square inch gauge
SAE	Society of Automotive Engineers
SNAP	Significant New Alternatives Policy
SOP	Standard operating procedure
POA&M	Plan of Action and Milestones
PW	Public Works
PWD	Public Works Department
PWO	Public Works Officer
UL	Underwriters Laboratories
50+#	50 or more pounds

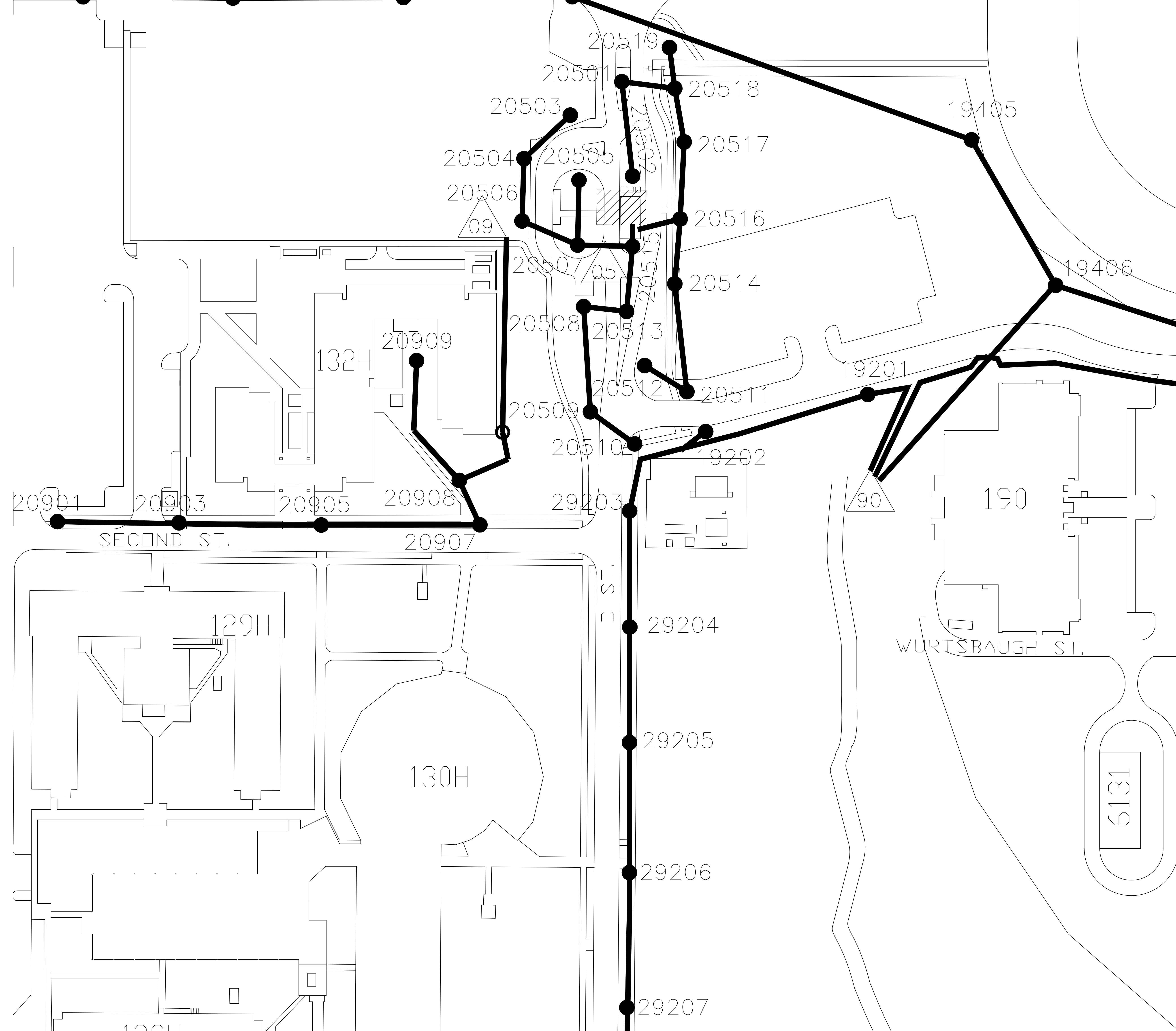














PWD Great Lakes Guidance Document
Asbestos Hazard Abatement Plan Format Requirements

The following asbestos abatement plan format outlines the minimum information required to be included in the plan to ensure compliance with Navy requirements and Federal, State and Local laws and regulations. Information in **blue** are additional requirements of the Unified Facilities Guide Specification (UFGS) 02 82 00, Asbestos Remediation, when Regulated Asbestos Containing Materials (RACM) are removed in quantities that exceed US EPA regulatory reporting thresholds OR when deemed applicable by the Government Designated Authority.

Asbestos Hazard Abatement Plan

Project Number

Project Name

Contract Number

Prepared For:

NAVAL STATION GREAT LAKES

NAVAL FACILITIES COMMAND MID-ATLANTIC

Date

Revision No. (if applicable)

Prepared By:

Contractor Name

Contractor Address

Contractor phone number

TABLE OF CONTENTS

1. INTRODUCTION

- Describe project scope of work and specify whether abatement is in support of a renovation or demolition project.
- Describe the Category I and II Non-Friable Asbestos Containing Material (ACM) and Regulated ACM (RACM) that will be impacted by the project. Specify type, condition, location, and quantity.
- Specify whether an Asbestos Project Notification Form will be submitted and to which regulatory agency(s). Include draft copy in Appendix for NAVFAC review and approval.
- All abatement projects must be preceded by a thorough asbestos inspection as required by 40 CFR Part 61 Subpart M. Incorporate by reference the inspection report in which the abatement plan is based upon. Include company name and report date.
- Identify the applicable Federal, State, and Local laws and regulations that will be adhered to.
- Identify the person (name and company) who prepared the plan. Provide qualifications in Appendix. [UFGS 02 82 00 requires this person be a Private Qualified Person \(PQP\)](#).
- Identify the Competent Person (name and company) who reviewed and approved the plan. Provide qualifications in Appendix.

2. ROLES/RESPONSIBILITIES

- Identify the name, company name, and roles/responsibilities of each person involved in the abatement project. Include Asbestos Abatement Contractor, Prime Contractor (if different from abatement contractor), Site Safety Manager, Quality Control Manager, Asbestos Abatement Workers, Asbestos Abatement Supervisors, Competent Person, [Private Qualified Person \(PQP\)](#), and the persons tasked with preparing the abatement plan, conducting oversight, and performing air monitoring. [UFGS 02 82 00 requires the last three functions be performed by a PQP](#). The information should be provided in a tabular format.
- Provide written qualifications in Appendix.

3. DESCRIPTION OF ASBESTOS WORK ACTIVITY(S)

3.1 OSHA Class Level of Work

- Identify the class level of work to be performed.

3.2 Abatement Method

- Engineering controls
- Work practices
- Materials (e.g. wetting agents, mastic removers, encapsulants, adhesives, etc.). Provide Safety Data Sheets in Appendix.
- Tools

3.3 Equipment to Be Used

- Describe the type, manufacturer, name and model of equipment to be used with statement that equipment is either contractor-owned or rented.
- If equipment is rented, provide copy of notification to rental company concerning intended use of the equipment and the possibility of asbestos contamination.
- Provide equipment spec sheets in Appendix.

3.4 Sequencing of Work

- Specify the order in which each asbestos work activity will be performed.

4. NOTIFICATION OF BUILDING OCCUPANTS AND OTHER EMPLOYERS AT THE WORKSITE

- Describe how building occupants and employers of other personnel working adjacent to asbestos work areas will be notified about the project and the measures taken to prevent the release of asbestos fibers and exposure to personnel working in the area.

5. DESCRIPTION OF REGULATED AREA AND CONTAINMENT SYSTEM**5.1 Regulated Area**

- Describe how each regulated area will be constructed, demarcated, and secured to restrict access. Describe the warning signs to be posted.
- Provide drawing in Appendix showing location, size, and details of each regulated area. Identify locations for viewing ports, critical barriers, physical and air distribution boundaries, clean and dirty areas, buffer zones, showers, change rooms, storage areas, waste load out area, waste storage location, negative air machines and local exhaust equipment.
- Identify location and quantity of impacted Category I and II Non-Friable ACM and RACM on drawings and indicate its condition.

5.2 Containment System

- If applicable, describe containment system to be used to ensure asbestos is not released outside the regulated area.
- Identify the required number of negative air machines (provide calculations).
- Describe how negative air pressure will be monitored and maintained.

6. DECONTAMINATION

- Describe decontamination system
- Describe decontamination procedures

7. AIR MONITORING**7.1 Personal Monitoring**

Revised 4/14/2022

- Initial determination. If applicable, provide a Negative Exposure Assessment if objective data is used to justify excluding the initial occupational exposure assessment. Describe the relevancy of such data to current job conditions.
- Continued exposure monitoring
- Identify who will perform the personal monitoring. Air sampling must be directed or performed by the Competent Person. [UFGS 02 82 00 requires this function be performed by a PQP.](#)
- Describe the sampling and analysis methods to be used.
- Describe when and how abatement workers will be notified of personal sampling results.

7.2 Environmental Monitoring

- Must demonstrate the absence of asbestos fiber migration outside the regulated area.
- Identify who will perform the environmental monitoring. [UFGS 02 82 00 requires this function be performed by a PQP.](#)
- Describe the sampling and analysis methods to be used.
- Describe background and area monitoring to be conducted. Where will samples be collected and how frequent? For what duration?
- Describe procedures to be followed should any of the sample results exceed permissible limits.

7.3 Clearance Monitoring

- Describe final cleanup, visual inspection, lock down and clearance testing procedures. What methods will be used for clearance sampling and analysis? What aggressive sampling techniques will be used? Where will samples be collected?
- Describe the measures to be taken if clearance testing fails.
- The post-abatement visual inspection and final clearance sampling must be directed or performed, and certified by the Competent Person. [UFGS 02 82 00 requires a PQP to perform and certify the visual inspection and final clearance sampling.](#)

7.3 Testing Laboratory and Analytical Methods

- Provide the name, address, and telephone number of each testing laboratory used for the analysis and reporting of airborne concentrations of asbestos fibers. The laboratory and asbestos abatement contractor cannot have an employee/employer relationship or financial relationship which could constitute a conflict of interest.
- Identify the analytical methods to be used.
- Provide proof in Appendix that laboratory is accredited by the American Industrial Hygiene Association (AIHA) and successfully participates in the Proficiency Analytical Testing (PAT) Program and that persons counting the samples have been judged proficient by current inclusion on the AIHA Asbestos Analysis Registry (AAR).

8. PERSONAL PROTECTION EQUIPMENT

- Describe respiratory protection equipment and protective clothing to be used.

Revised 4/14/2022

- If reusable clothing will be used, then describe decontamination methods to be used.
- [Provide Respiratory Protection Program in Appendix.](#)

9. HOUSEKEEPING PROCEDURES

- Describe procedures to prevent spread of asbestos contamination both in and beyond the regulated area.
- Describe how furnishings and building equipment will be relocated or covered and protected from asbestos fibers.

10. HYGIENE FACILITIES AND PRACTICES

- Describe eating, drinking, smoking and sanitary procedures

11. MEDICAL SURVEILLANCE

- Describe how employees are assessed for asbestos exposure and monitored for fitness to perform work tasks while wearing PPE including respiratory protection devices.

12. WASTE MANAGEMENT

- Describe how asbestos waste (including asbestos contaminated waste water) will be controlled, managed and disposed. Include procedures for packaging, labeling, storage, and transportation.
- Identify on-site temporary storage location of waste containers. If storage location is outside the perimeter of the construction site, then location must be pre-approved by the Naval Station Great Lakes Environmental Department.
- Identify waste transporter and waste landfill. Include in Appendix the Asbestos Waste Transporter Permit and Asbestos Waste Landfill Permit.
- Document your understanding that all asbestos waste manifests must be signed by the Naval Station Great Lakes Environmental Department prior to being transported off-site.
- Provide in Appendix an example Asbestos Waste Manifest to be used for this project.

13. SECURITY, FIRE, AND MEDICAL EMERGENCY RESPONSE PROCEDURES

13.1 Security Procedures

13.2 Fire Response Procedures

13.3 Medical Emergency Response Procedures

14. ABATEMENT CLOSEOUT REPORT

Identify the information that will be included in the abatement closeout report. The following is a list of the minimum required information.

- Summary of the type, location and amount of RACM and Category I and II Non-Friable ACM abated. The summary shall be consistent with the approved abatement plan and amendments. Otherwise, an explanation shall be provided.

Revised 4/14/2022

- Names, certifications and licenses of all Asbestos Abatement Workers and Supervisors, Competent Person [and Private Qualified Person \(Asbestos Building Inspector, Asbestos Project Designer, Asbestos Abatement Contractor/Supervisor and Air Sampling Professional\)](#).
- All personal, area, and clearance air monitoring results signed by the analyst, and chain of custody forms.
- Negative Exposure Assessments if historical or objective data is used in lieu of performing an initial exposure assessment.
- All daily pressure differential recordings for negative pressure enclosures.
- All contractor logs and containment sign-in/sign-out sheets.
- Competent Person clearance certifications (signed and dated).
- [All PQP daily monitoring logs and clearance certifications \(signed and dated\)](#).
- All notices and submissions to regulatory agencies documenting asbestos abatement activities.
- Asbestos waste manifests signed and dated by the generator (Naval Station Great Lakes Environmental Department), transporter and landfill agent.
- Asbestos waste turn-in documents or weight tickets prepared, signed and dated by a landfill agent.

15. COMPETENT PERSON SIGNATURE PAGE

- **The plan must be approved and signed by the Competent Person (CP).**
- **Include Name, Company Name and Address, Telephone Number, Asbestos Abatement Supervisor License Number, and Signature.**
- **Provide qualifications in Appendix.**

15. PQP SIGNATURE PAGE

- [The plan must be prepared, signed and sealed by the PQP](#)
- [Include Name, Company Name and Address, Telephone Number, Asbestos Project Designer License Number, and Signature](#)
- [Provide qualifications in Appendix.](#)

16. APPENDICES

The following additional information is required and would be best suited for appendices:

16.1 Detailed Drawings

16.2 Licenses, Permits, Notifications (to include Asbestos Abatement Contractor License, Liability Insurance and Bonding, Draft Asbestos Abatement Project Notification Form, Notification to Rental Company, Asbestos Waste Transporter Permit, Asbestos Waste Landfill Permit, and Example Asbestos Waste Manifest)

16.3 Written Qualifications (to include training certifications, licenses with endorsements, respirator fit tests, physicians written medical opinion, testing laboratory qualifications, Competent Person qualifications [and PQP](#)

Revised 4/14/2022

qualifications)

16.4 Transmittal and Approval sheets for the Accident Prevention Plan (AAP) and Activity Hazard Analysis (AHA). The APP and AHA must be reviewed and approved by the Government Designated Authority (GDA) per EM385-1-1 Appendix A prior to start of work. The Accident Prevention Plan is required to include both the Respiratory Protection Program and Site-Specific Safety Plan. These plans are rather large and as such should be incorporated by reference via the Transmittal and Approval sheets.

16.5 Equipment Spec Sheets

16.6 Safety Data Sheets

Revised 4/14/2022



PWD Great Lakes Guidance Document
Lead, Cadmium, Chromium Compliance Plan Format Requirements

The following Lead, Cadmium, Chromium Compliance Plan format outlines the minimum information required to be included in the plan to ensure compliance with Navy requirements and Federal, State and Local laws and regulations for non-residential and non-child-occupied facilities. Information in **blue** are additional requirements of UFGS 02 83 00, Lead Remediation, when deemed applicable by the Government Designated Authority.

Lead, Cadmium, Chromium Compliance Plan

Project Number
Project Name
Contract Number

Prepared For:
NAVAL STATION GREAT LAKES
NAVAL FACILITIES COMMAND MID-ATLANTIC

Date
Revision No. (if applicable)

Prepared By:
Contractor Name
Contractor Address
Contractor phone number

TABLE OF CONTENTS

1. INTRODUCTION

- Describe project scope of work and specify whether abatement is in support of a renovation or demolition project.
- Describe the lead-based paint (LBP), paint with lead (PWL), other material-containing lead (MCL), paint with cadmium and/or chromium (PWCC) and other material-containing cadmium and/or chromium (MCCC) that will be impacted by the project. Specify color, condition, location, and quantity.
- Incorporate by reference the lead, cadmium, chromium paint inspection report in which the compliance plan is based upon. Include company name and report date.
- Identify the applicable Federal, State, and Local laws and regulations that will be adhered to.
- Identify the person (name and company) who prepared the plan. Provide qualifications in Appendix.
- Identify the Competent Person (name and company) who reviewed and approved the plan. Provide qualifications in Appendix.

2. ROLES/RESPONSIBILITIES

- Identify the name, company name, and roles/responsibilities of each person involved in the abatement project. Include Lead Abatement Contractor, Prime Contractor (if different from abatement contractor), Site Safety Manager, Quality Control Manager, Lead Abatement Workers, Lead Abatement Supervisors, Competent Person, [Third Party Consultant \(when applicable\)](#), and the persons tasked with preparing the compliance plan, conducting oversight, and performing air monitoring. The information should be provided in a tabular format.
- Provide written qualifications in Appendix.

3. DESCRIPTION OF LEAD, CADMIUM, CHROMIUM WORK ACTIVITY(S)

3.1 Abatement Method

- Engineering controls
- Work practices
- Materials (e.g. wetting agents, encapsulants, adhesives, solvents, etc.). Provide Safety Data Sheets in Appendix.
- Tools

3.2 Equipment to Be Used

- Describe the type, manufacturer, name and model of equipment to be used with statement that equipment is either contractor-owned or rented.
- If equipment is rented, provide copy of notification to rental company concerning intended use of the equipment and the possibility of lead, cadmium, chromium contamination.
- Provide equipment spec sheets in Appendix.

3.3 Sequencing of Work

- Specify the order in which each lead, cadmium, chromium work activity will be performed.

Revised 4/14/2022

4. NOTIFICATION OF BUILDING OCCUPANTS AND OTHER EMPLOYERS AT THE WORKSITE

- Describe how building occupants and employers of other personnel working adjacent to lead, cadmium, chromium work areas will be notified about the project and the measures taken to prevent the release of lead, cadmium, chromium dust/debris and exposure to personnel working in the area.

5. DESCRIPTION OF LEAD, CADMIUM, CHROMIUM HAZARD CONTROL AREA AND CONTAINMENT SYSTEM**5.1 Lead, Cadmium, Chromium Hazard Control Area**

- Describe how each hazard control area will be constructed, demarcated, and secured to restrict access. Describe the warning signs to be posted.
- Provide drawing in Appendix showing location, size, and details of each hazard control area. Identify locations for viewing ports, critical barriers, physical and air distribution boundaries, clean and dirty areas, buffer zones, showers, change rooms, storage areas, waste load out area, waste storage location, and local exhaust equipment.
- Identify location and quantity of impacted LBP, PWL, MCL, PWCC, MCCC on drawings and indicate its condition.

5.2 Containment System

- If applicable, describe containment system to be used to ensure LBP, PWL, MCL, PWCC, MCCC is not released outside the hazard control area.
- Describe the local exhaust equipment to be used.

6. DECONTAMINATION

- Describe decontamination system
- Describe decontamination procedures

7. AIR MONITORING**7.1 Personal Monitoring**

- Initial determination. If applicable, provide an Occupational and Environmental Assessment Data Report if objective data is used to justify excluding the initial occupational exposure assessment. Describe the relevancy of such data to current job conditions.
- Continued exposure monitoring
- Identify who will perform the personal monitoring. [UFGS 02 83 00 requires that personal air sampling be directed or performed by the Competent Person.](#)
- Describe the sampling and analysis methods to be used.
- Describe when and how abatement workers will be notified of personal sampling results.

7.2 Environmental Monitoring

- Must demonstrate the absence of lead, cadmium, chromium dust migration outside the hazard control area.

Revised 4/14/2022

- Identify who will perform the environmental monitoring. [UFGS 02 83 00 requires that environmental air sampling be directed or performed by the Competent Person. Additionally, when applicable, a Third Party Consultant must collect baseline wipe and/or soil samples prior to start of work.](#)
- Describe the sampling and analysis methods to be used.
- Describe background and area monitoring to be conducted. Where will samples be collected? How frequent? For what duration?
- Describe procedures to be followed should any of the sample results exceed permissible limits.

7.3 Clearance Monitoring

- Describe final cleanup, visual inspection, and clearance testing procedures. What methods will be used for clearance sampling and analysis? What aggressive sampling techniques will be used? Will wipe samples be taken? Will soil samples be taken? Where will samples be collected?
- Describe the measures to be taken if clearance testing fails.
- The post-abatement visual inspection and final clearance sampling (wipe, soil and air) must be directed or performed, and certified by the Competent Person [and Third Party Consultant \(when applicable\).](#)

7.3 Testing Laboratory and Analytical Methods

- Provide the name, address, and telephone number of each testing laboratory used for the analysis and reporting of lead, cadmium, chromium concentrations in air and wipe samples. The laboratory and lead abatement contractor cannot have an employee/employer relationship or financial relationship which could constitute a conflict of interest.
- Identify the analytical methods to be used.
- Provide proof in Appendix that laboratory is successfully participating in the EPA National Lead Laboratory Accreditation Program (NNLAP) by being accredited by either the American Association for Laboratory Accreditation or the American Industrial Hygiene Association (AIHA) and is also successfully participating in the Environmental Lead Proficiency Analytical Testing (ELPAT) program to perform sample analysis.

8. PERSONAL PROTECTION EQUIPMENT

- Describe respiratory protection equipment and protective clothing to be used.
- If reusable clothing will be used, then describe decontamination methods to be used.
- [Provide Respiratory Protection Program in Appendix.](#)

9. HOUSEKEEPING PROCEDURES

- Describe procedures to prevent spread of lead, cadmium, chromium contamination both in and beyond the hazard control area.
- Describe how furnishings and building equipment will be relocated or covered and protected from lead, cadmium, chromium dust.

10. HYGIENE FACILITIES AND PRACTICES

Revised 4/14/2022

- Describe eating, drinking, smoking and sanitary procedures

11. ADMINISTRATIVE CONTROLS TO LIMIT EMPLOYEE EXPOSURE TO LEAD, CADMIUM, CHROMIUM

- Describe administrative controls to be employed, including employee rotation schedule, if engineering controls or PPE fail to eliminate exposures exceeding the PEL.

12. MEDICAL SURVEILLANCE

- Describe how employees are assessed for lead, cadmium, chromium exposure and monitored for fitness to perform work tasks while wearing PPE including respiratory protection devices.

13. WASTE MANAGEMENT

13.1 Waste Generation

- Provide estimated quantities of wastes to be generated, recycled or disposed of (include lead, cadmium, chromium contaminated waste water).
- Provide unit cost for hazardous waste disposal according to this plan.
- Identify any process that may alter or treat waste rendering a hazardous waste non-hazardous.
- Provide work plan and schedule for waste containment, temporary storage, off-site transportation, recycling and disposal.
 - All wastes shall be cleaned up and containerized daily.
 - Proper containment of waste includes using U.S. Dept. of Transportation approved waste containers (e.g. 55-gallon drums)
 - Proper marking/labeling of containers must be completed upon filling containers. Include the accumulation start date for hazardous wastes.
 - Identify on-site temporary (<90 days) storage location of waste containers. If storage location is outside the perimeter of the construction site, then location must be pre-approved by the Naval Station Great Lakes Environmental Department.

13.2 Characterization

- Describe how lead, cadmium, chromium wastes will be identified and classified as either hazardous or non-hazardous waste.

13.3 Transportation

- Provide names and qualifications of each contractor that will be transporting and storing wastes. Furnish written proof of EPA, State and local waste transporter permits, EPA Identification Number, and Transporter Number.

13.4 Disposal

- Provide name, physical address and phone number of recycling facility(s) and waste treatment, storage, and disposal facility(s). Provide a 24-hour point of contact. Include copy of facility permit(s) in Appendix. The facility(s) must be pre-approved by the Naval Station Great Lakes Environmental Department.

Revised 4/14/2022

- Document your understanding that all waste manifests must be signed by the Naval Station Great Lakes Environmental Department prior to being transported off-site.
- Provide in Appendix an example of the hazardous waste manifest to be used for this project. Include the Installation's EPA Generator Identification Number.

13.5 Recordkeeping

13.6 Names and qualifications (experience and training) of personnel who will be working on-site with hazardous wastes.

13.7 Waste handling equipment to be used in performing the work to include cleaning, volume reduction, and transport equipment.

13.8 Spill prevention, containment, and cleanup contingency measures including a health and safety plan.

14. SECURITY, FIRE, AND MEDICAL EMERGENCY RESPONSE PROCEDURES

14.1 Security Procedures

14.2 Fire Response Procedures

14.3 Medical Emergency Response Procedures

15. ABATEMENT CLOSEOUT REPORT

Identify the information that will be included in the abatement closeout report. The following is a list of the minimum required information.

- Summary of the type, location and amount of LBP, PWL, MCL, PWCCC and MCCC removed. The summary shall be consistent with the approved compliance plan and amendments. Otherwise, an explanation shall be provided.
- Names, certifications and licenses of all Lead Abatement Workers and Supervisors, Competent Person, Air Monitoring Professional and Third Party Consultant (when applicable).
- All personal, area, and clearance air monitoring results signed by the analyst, and chain of custody forms.
- All dust wipe sampling results signed by the analyst, and chain of custody forms.
- Occupational and Environmental Assessment Data Report if historical or objective data is used in lieu of performing an initial exposure assessment.
- All contractor logs and containment sign-in/sign-out sheets.
- Competent Person clearance certifications (signed and dated).
- When applicable, Third Party Consultant daily monitoring logs and clearance certifications (signed and dated).
- Hazardous waste manifests signed and dated by the generator (Naval Station Great Lakes Environmental Department), transporter and TSD facility agent.

Revised 4/14/2022

- Non-hazardous lead, cadmium, chromium waste turn-in documents or weight tickets prepared, signed and dated by a landfill agent.

16. COMPETENT PERSON SIGNATURE PAGE

- The plan must be approved and signed by the Competent Person (CP).
- Include Name, Company Name and Address, Telephone Number, Lead Abatement Supervisor License Number, and Signature.
- Provide qualifications in Appendix.

17. APPENDICES

The following additional information is required and would be best suited for appendices:

17.1 Detailed Drawings

17.2 Licenses, Permits, Notifications (to include Lead Abatement Contractor License, Liability Insurance and Bonding, Notification to Rental Company, Hazardous Waste Transporter Permit, Hazardous Waste Treatment, Storage and Disposal Facility Permit, and Example Hazardous Waste Manifest)

17.3 Written Qualifications (to include training certifications, licenses with endorsements, respirator fit tests, physicians written medical opinion, testing laboratory qualifications, Competent Person qualifications, and Third Party Consultant qualifications)

17.4 Transmittal and Approval sheets for the Accident Prevention Plan (AAP) and Activity Hazard Analysis (AHA). The APP and AHA must be reviewed and approved by the Government Designated Authority (GDA) per EM385-1-1 Appendix A prior to start of work. The Accident Prevention Plan is required to include the Respiratory Protection Program, Hazard Communication Program and Site-Specific Safety Plan. These plans are rather large and as such should be incorporated by reference via the Transmittal and Approval sheets.

17.7 Lead, Cadmium, Chromium Waste Management Plan (if not included in Waste Management section)

17.8 Equipment spec sheets

17.9 Safety Data Sheets